

MESSRS. :  **HYUNDAI**
MIPO DOCKYARD CO.,LTD.

SHIP No. : 4193

Control No.: B-AC-4193-3

ACCOMMODATION LADDER
& WINCH / MOTOR



JUNG-A MARINE CO.,LTD.

PLAN HISTORY

MESSRS.	HYUNDAI MIPO	KIND OF SHIP	2,800 TEU CLASS CONTAINER CARRIER
SHIP No.	4193/94	OWNER	TSAKOS
Control No.	B-AC-4193	CLASS	ABS

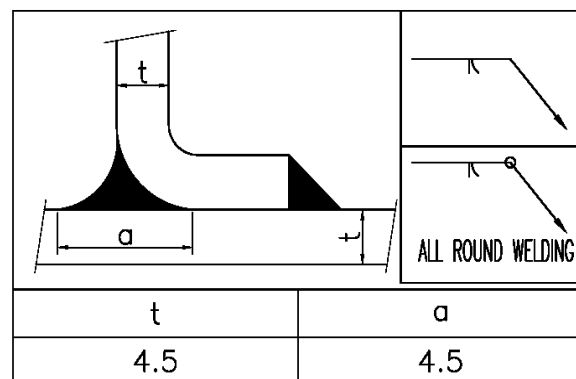
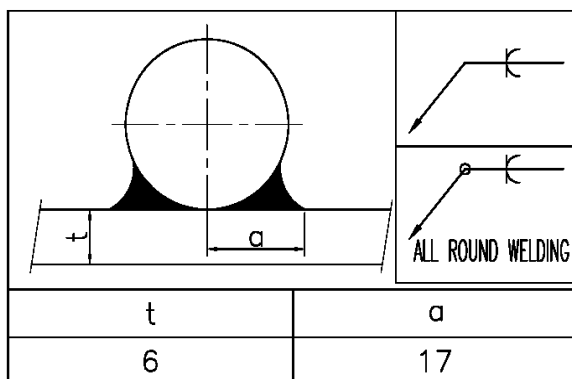
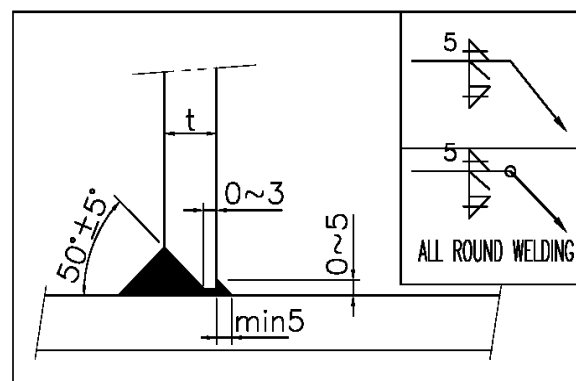
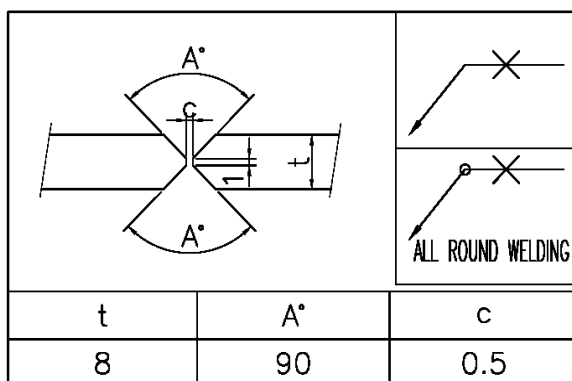
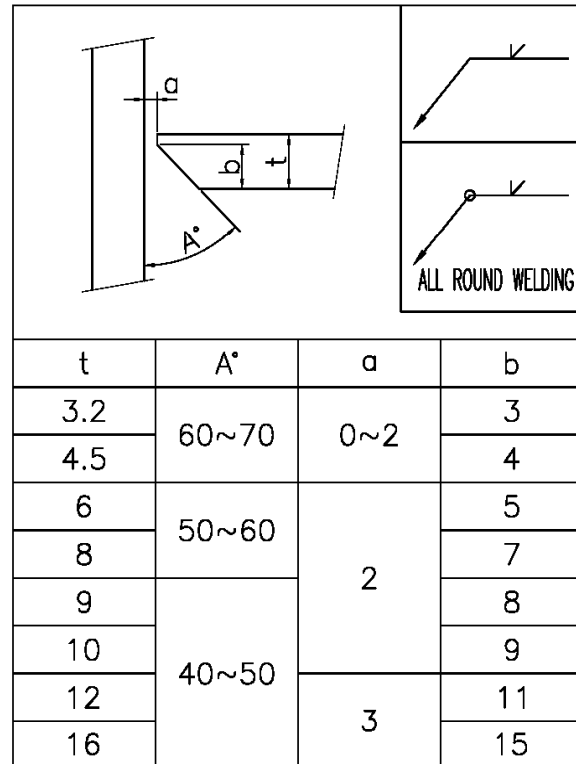
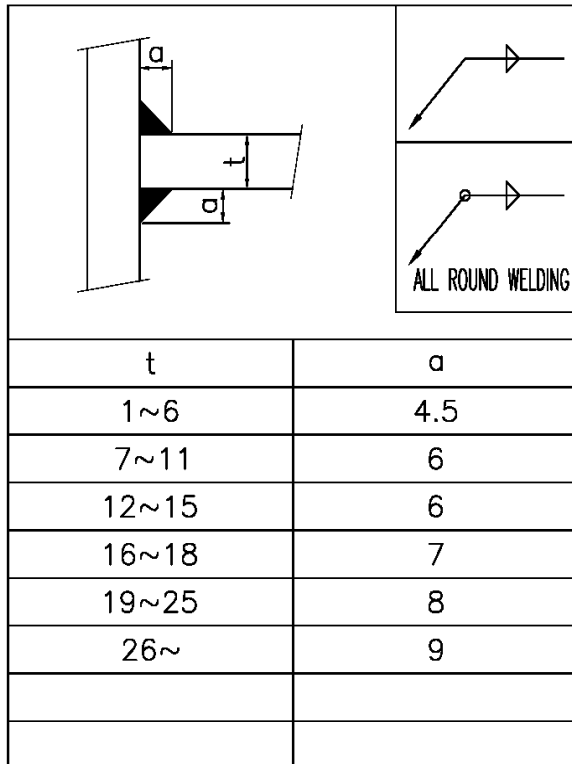
ITEM		ACCOMMODATION LADDER & WINCH / MOTOR		
REV.No.	DESCRIPTION	DATE	PAGE	APPROVED
△ 0	RECEIVED SPEC. FROM SHIPYARD	-	-	S.W.SHIN
	SUBMITTED FOR APPROVAL DWG. TO SHIPYARD	2021/11/10	-	
△ 1	RECEIVED APPROVAL FROM SHIPYARD	2022/5/27	-	
	ISSUED WORKSHOP DWG.(SEAT ONLY)	2022/6/3	-	
	ISSUED WORKSHOP DWG.	2022/6/9	-	
	SUBMITTED FOR WORKING DWG. TO SHIPYARD	2022/6/9	-	
	SUBMITTED FOR APPROVAL DWG. TO CLASS	2022/6/9	-	
△ 2	REVISED DWG FOR COMMENT FROM SHIPYARD	2022/11/11	6,18,19,19A	
	SUBMITTED FOR FINAL DWG. TO SHIPYARD	2022/12/9	-	

I N D E X

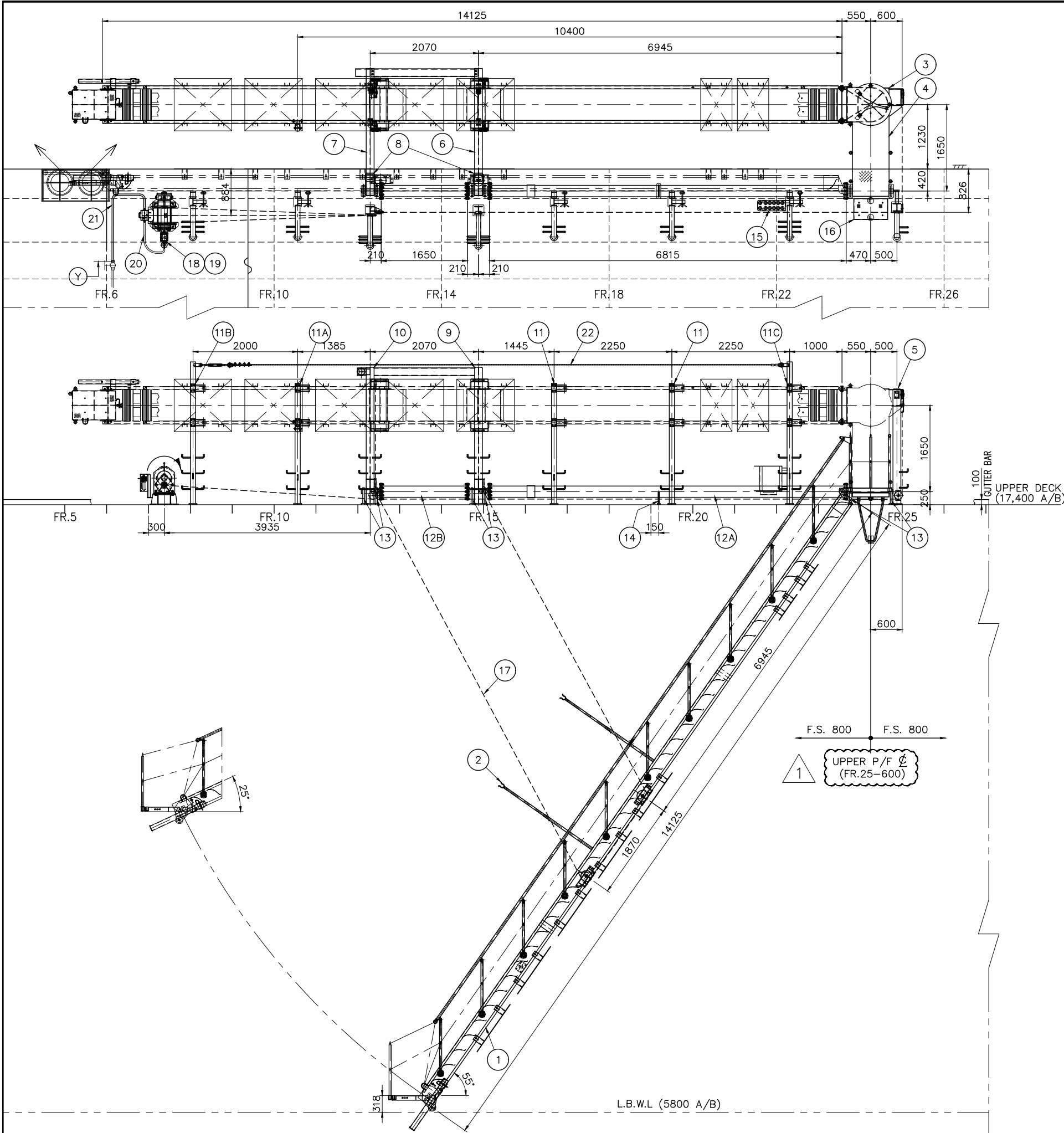
	PAGE	DESCRIPTION
	1	PLAN HISTORY
	2	INDEX
	3	GENERAL WELDING METHOD
	4	WEIGHT LIST
	5 ~ 6	ARRANGEMENT OF ACCOM. LADDER
	7 ~ 12	ACCOMMODATION LADDER PARTS
	13	LIFE NET
	14	P.P ROPE
	15	TURNTABLE
	16	UPPER P/F
	17	POST FOR UPPER P/F
	18	FWD. STOWING DAVIT
	19	AFT. STOWING DAVIT
	20	FWD. POST FOR DAVIT
	21	AFT. POST FOR DAVIT
	22	BOLSTER FOR DAVIT
	23~24A	REST
	25A,25B	SPINDLE PIPE
	26	SHAFT HOLDER
	27	SPINDLE REST
	28	STANCHION STAND
	29	CAUTION PLATE
	30	ACCESS P/F
	31	LOAD TEST METHOD
	32~34	WINCH & MOTOR PART
	35	OILER & FILTER
	36	PIPING PLAN
	37	LUBRICATION OIL CHART
	38	SPARE PARTS
	39	STRENGTH CALCULATION



GENERAL CUTTING & WELDING METHOD



		WEIGHT LIST		MESSRS.	HYUNDAI MIPO
				SHIP NO.	4193/94
NO.	PAGE	PART NAME	WE'T/SET	Q'TY/SHIP	WE'T/SHIP
1		ACCOMMODATION LADDER WITH SKID	895 Kg	2	1790 Kg
2		UPPER PLATFORM & TURN TABLE	350 Kg	2	700 Kg
2A		POST FOR UPPER P/F	85 Kg	2	170 Kg
3		STOWING DAVIT (FWD & AFT)	325 Kg	2	650 Kg
4		SUPPORTING ROD FOR UPPER P/F	15 Kg	2	30 Kg
5		BOLSTER FOR DAVIT	10 Kg	4	40 Kg
6		POST FOR DAVIT (FWD & AFT)	220 Kg	2	440 Kg
7		REST	85 Kg	10	850 Kg
8		LADDER SUPPORT	10 Kg	2	20 Kg
9		SPINDLE PIPE	210 Kg	2	420 Kg
10		SHAFT HOLDER	20 Kg	8	160 Kg
11		SPINDLE REST	15 Kg	2	30 Kg
12		STANCHION STAND	25 Kg	2	50 Kg
13		ACCESS P/F	20 Kg	2	40 Kg
14		AIR MOTOR/WINCH ASS'Y	250 Kg	2	500 Kg
15		LADDER FALL ROPE	40 Kg	2	80 Kg
16		ETC.	100 Kg	1	100 Kg
TOTAL WEIGHT : abt. 6070 Kg/SHIP					



NOTE

1. THIS DWG. SHOWS PORT SIDE
2. ALL STEEL PARTS TO BE GALVANIZED.
3. (Y) ITEMS TO BE PROVIDED BY SHIPYARD.
4. ALL GREASE NIPPLE : JIS B 1575 A-PT1/8"(MAT'L:BRASS)
5. DECK CAMBER : 50 / 2900
6. DOUBLING PAD (25t) TO BE YARD SUPPLY
EXCEPT DOUBLING PAD (10t) WITH WINCH SEAT & ACCESS P/F
7. Q'TY : 1 SET/SHIP

(FOR ONE SET)

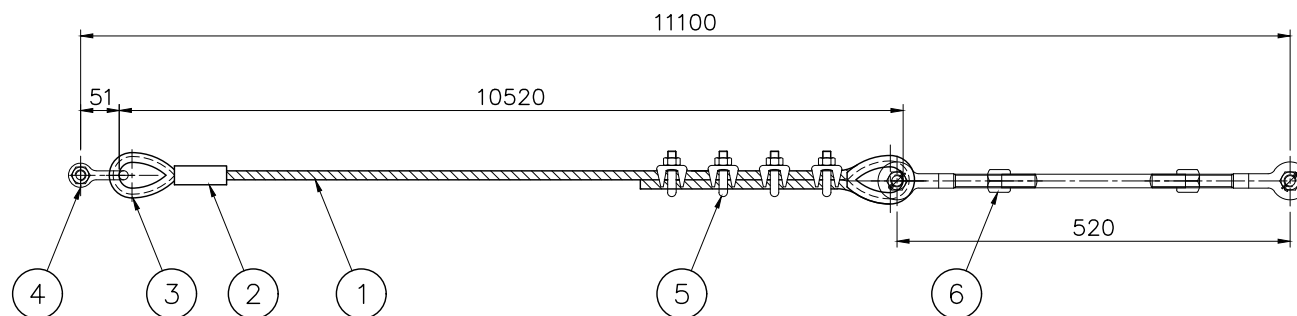
22	FALL ARREST	G.S.W.R	1	ø10(6x24)
21	AIR HOSE	RUBBER	1	25A x 4.5M
20	AIR HOSE	RUBBER	1	20A x 1.5M
19	AIR MOTOR	-	1	4.5HP
18	WINCH	-	1	OTW-1320
17	FALL ROPE	G.S.W.R	1	ø12(6x36)
16	ACCESS PLATFORM	SS400	1	L65x65x8t
15	STANCHION STAND	SS400	1	L50x50x6t
14	SPINDLE REST	SS400	1	25t
13	SHAFT HOLDER	SS400	6	32t
12	SPINDLE PIPE	SGP	2	200A
11	REST	SS400	5	H148x100x6/9t
10	AFT. POST FOR DAVIT	SS400	1	H150x150x7/10t
9	FWD. POST FOR DAVIT	SS400	1	H150x150x7/10t
8	BOLSTER FOR DAVIT	SS400	2	H100x100x6/8t
7	AFT. STOWING DAVIT	SS400	1	H194x150x6/9t
6	FWD. STOWING DAVIT	SS400	1	H194x150x6/9t
5	POST FOR UPPER P/F	SS400	1	H150x150x7/10t
4	UPPER PLATFORM	SS400	1	-
3	TURN TABLE	SS400	1	-
2	WIRE HANGER	AL ALLOY	-	ø43 PIPE
1	ACCOMMODATION LADDER	AL ALLOY	1	300Hx600B
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved	S.W.SHIN	Scale	1/90	Dwg No. AC-VET-ARRT-01
Checked	S.W.SHIN	Date	2021.10.25	Title
Designed	C.W.PARK	Ship No.	4193/94	ARRANGEMENT OF ACCOM. LADDER (PORT SIDE)
JUNG-A MARINE CO.,LTD.				



DETAIL OF FALL ARREST

(Q'TY : 2 SETS/SHIP)



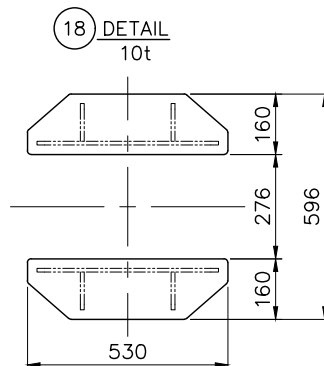
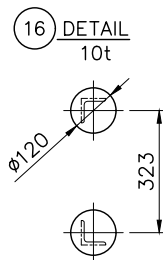
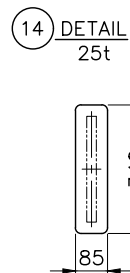
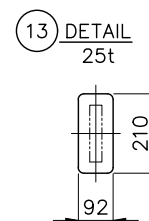
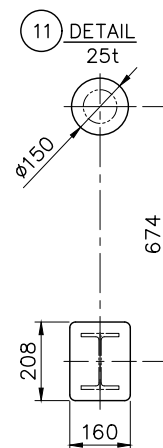
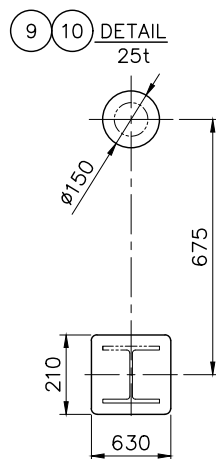
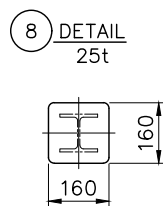
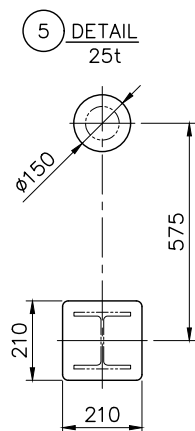
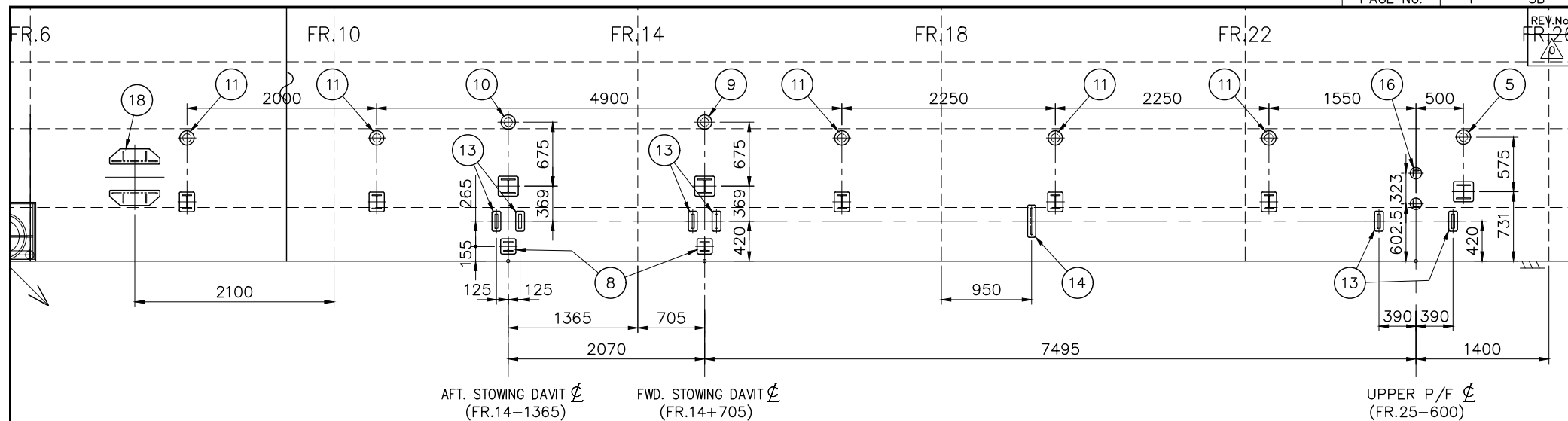
(FOR ONE SET)

6	TURNBUCKLE	SF	1	3/4"x9"
5	WIRE CLIP	SF	4	FOR ø10
4	SHACKLE	SF	1	BB-1/2"
3	THIMBLE	SS400	2	A10
2	METAL ALLOY	-	1	-
1	WIRE ROPE	G.S.W.R	1	ø10(6x24)
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved	S.W.SHIN	Scale	1/10	Dwg No. AC-STD-ROPE-01 Title FALL ARREST
Checked	S.W.SHIN	Date	2021.10.20	
Designed	C.W.PARK	Ship No.	4193/94	



JUNG-A MARINE CO., LTD.



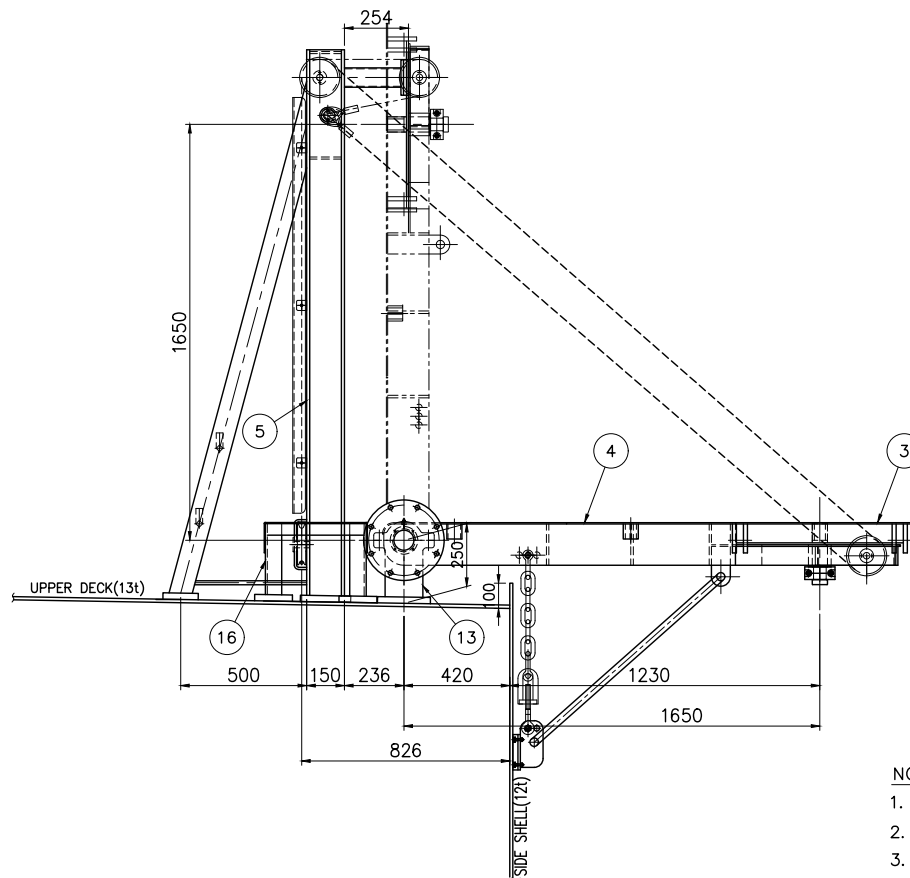
NOTE

1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE TO BE SYMMETRICAL.
2. ALL DOUBLING PAD(25t) TO BE ROUND OFF "R15".
(16 , 18) ---> 10t)

(FOR ONE SHIP)

18	WINCH SEAT	SS400	2	MAKER SUPPLY
16	ACCESS PLATFORM	SS400	2	MAKER SUPPLY
14	SPINDLE REST	SS400	2	YARD SUPPLY
13	SHAFT HOLDER	SS400	12	↑
11	REST	SS400	10/10	
10	AFT. POST FOR DAVIT	SS400	2/2	
9	FWD. POST FOR DAVIT	SS400	2/2	
8	BOLSTER FOR DAVIT	SS400	4	↓
5	POST FOR UPPER P/F	SS400	2/2	YARD SUPPLY
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved	S.W.SHIN	Scale	1/60-A4	Bwg No. AC-VET-DBAY-01
Checked	S.W.SHIN	Date	21.12.30	
Designed	C.W.PARK	Ship No.	4193/94	
 JUNG-A MARINE CO., LTD.				ARRANGEMENT OF DOUBLING PAD



NOTE

1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE TO BE SYMM.
2. DECK CAMBER : 50 / 2900
3. DOUBLING PLATE(25t) TO BE YARD SUPPLY.

Project Title.

ACCOMMODATION LADDER

Drawing Title.

"VIEW"-U

Dwg No.

AC-STD-VIEW-01

SCALE

1/30

SHIP NO.

4193/94

Approved

S.W.SHIN



JUNG-A MARINE co.,LTD.

DATE

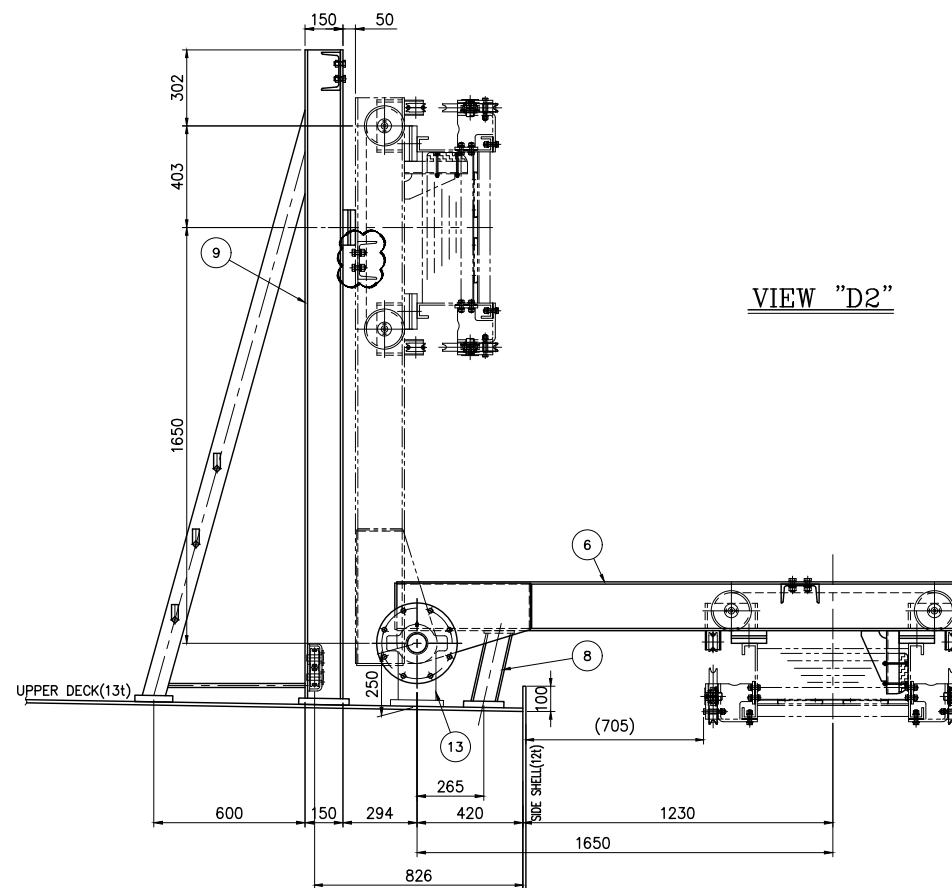
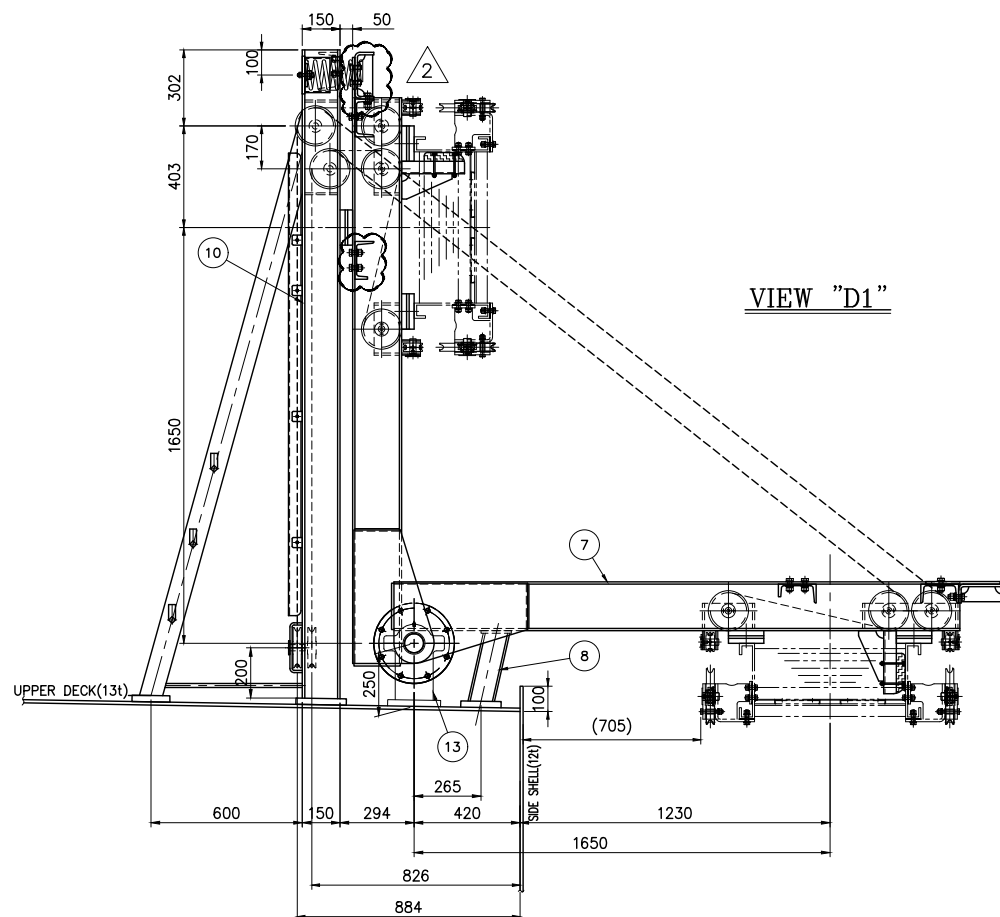
2021.10.25

Checked

S.W.SHIN

Designed

C.W.PARK



NOTE

1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE TO BE SYMM.
2. DECK CAMBER : 50 / 2900
3. DOUBLING PLATE(25t) TO BE YARD SUPPLY.

Project Title.

ACCOMMODATION LADDER

Drawing Title.

"VIEW"-D1 & D2

Dwg No.

AC-STD-VIEW-02

SCALE

1 / 30

SHIP NO.

4193/94

Approved

S.W.SHIN

DATE

2021.10.25

Checked

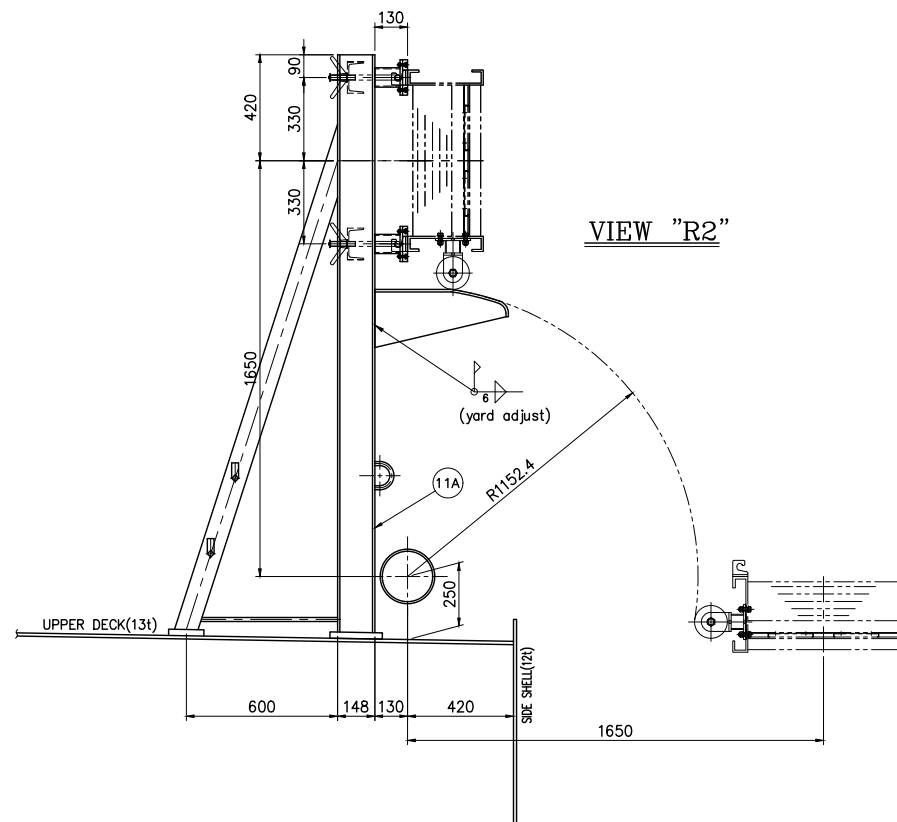
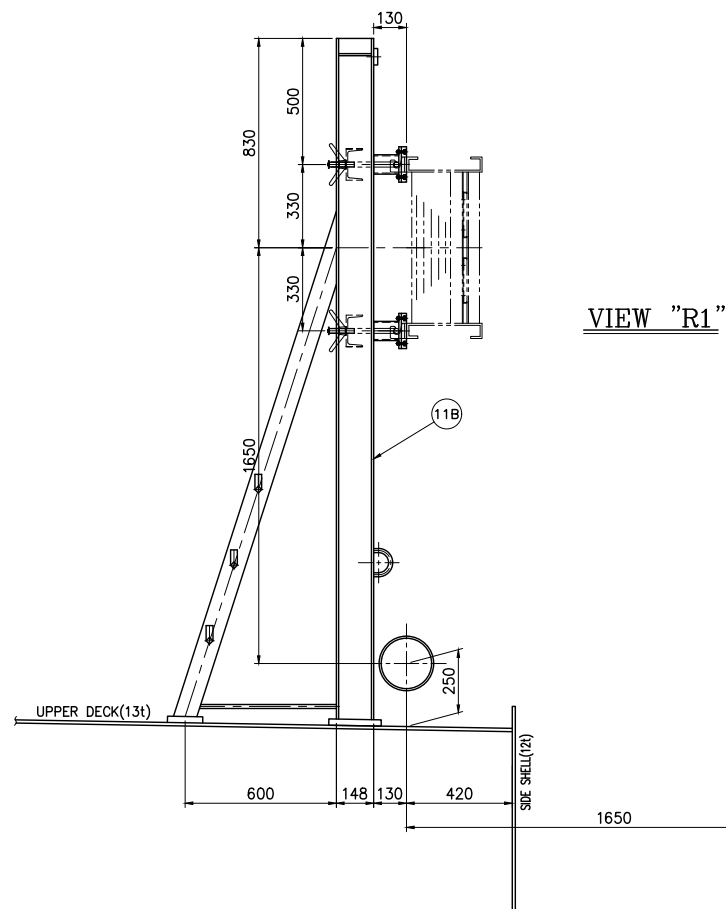
S.W.SHIN

Designed

C.W.PARK



JUNG-A MARINE co.,LTD.



NOTE

1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE TO BE SYMM.
2. DECK CAMBER : 50 / 2900
3. DOUBLING PLATE(25t) TO BE YARD SUPPLY.

Project Title.

ACCOMMODATION LADDER

Drawing Title.

"VIEW"-R & R1

Dwg No.

AC-STD-VIEW-03

SCALE

1/30

SHIP NO.

4193/94

Approved

S.W.SHIN

DATE

2021.10.25

Checked

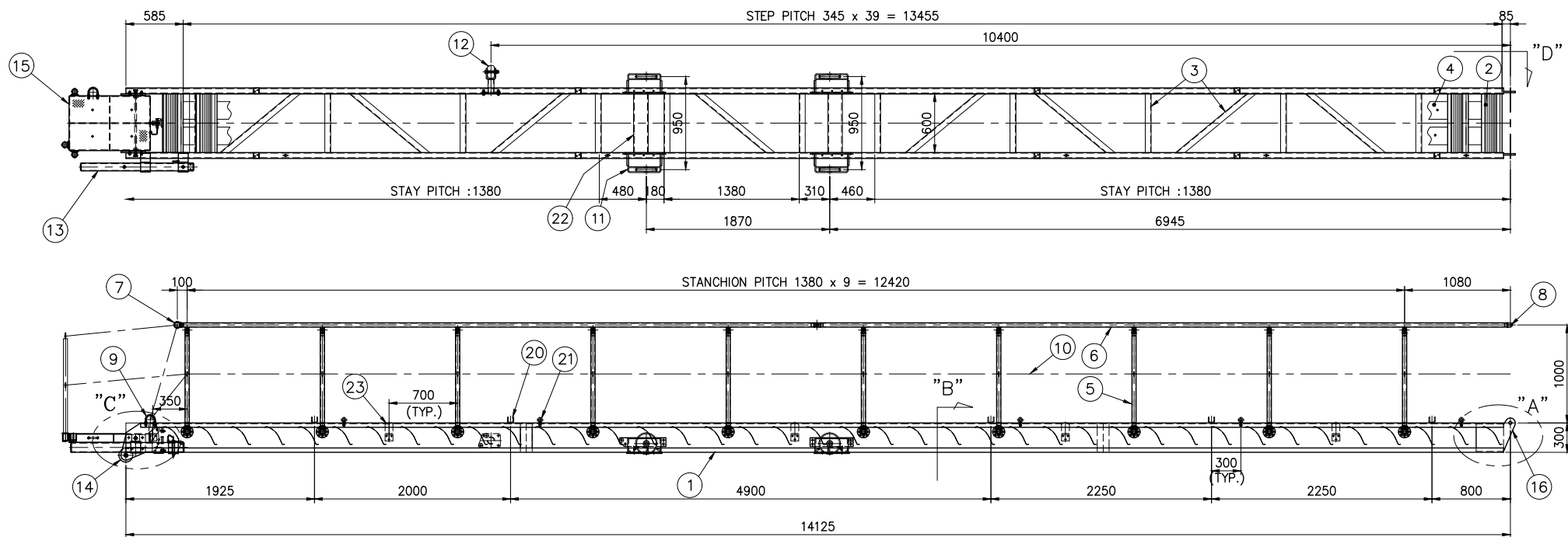
S.W.SHIN

Designed

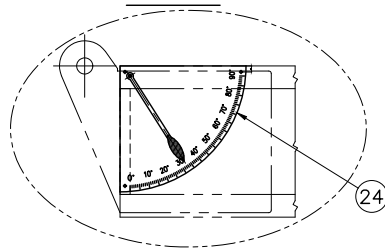
C.W.PARK



JUNG-A MARINE co.,LTD.



VIEW "D"



(FOR ONE SET)

NOTE

1. THIS DWG. SHOWS ST'D SIDE, PORT SIDE TO BE SYMM.
 2. ALL STEEL PARTS TO BE GALVANIZED.
 3. Q'TY : 2 SETS/SHIP

24	ANGLE INDICATOR	SUS316	1	1t
23	STANCHION GUIDE	AL ALLOY	5	6t
22	DOUBLING PLATE	AL ALLOY	2	[150x45x6/8t
21	LIFE NET EYE	SUS304	5	M12 EYE BOLT
20	LASHING HOOK	SUS304	10	8t
19	RIVET	AL ALLOY	—	—
18	STIFFENER	AL ALLOY	1	8t
17	BOLT/NUT/W/S.W	SUS304	16	M16x60L
16	HINGE PLATE	SS400	2	16t
15	LOWER PLATFORM	AL ALLOY	1	—
14	LOWER ROLLER	PE	1	ø140
13	FENDER	AL ALLOY	1	—
12	SIDE ROLLER	PE	1	ø130
11	SHEAVE BOX	SS400	4	—
10	MID RAIL	P.P ROPE	2	ø18(WITH VINYL COATED)
9	EYE RING	SUS316	2	U-BOLT

8	FORK EYE	AL ALLOY	2	—
7	EYE RING	AL ALLOY	2	ø10 R.BAR
6	HANDRAIL	AL ALLOY	2	ø43x3.5t
5	STANCHION	AL ALLOY	20	ø44.5x6t
4	BACK PLATE	AL ALLOY	2	180x15x5x2.5t
3	STAY	AL ALLOY	—	L50x60x6t
2	STEP	AL ALLOY	40	—
1	MAIN FRAME	AL ALLOY	2	H300x60x45x8/8t
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Project Title.

ACCOMMODATION LADDER

Drawing Title.

ACCOM. LADDER (300H)

Dwg No.

AC-VET-LAAY-01

SCALE

1/60

SHIP NO.

4193/94

DATE

2020.11.10

Checked

S.W.SHIN

Designed

C.W.PARK



JUNG-A MARINE

co.,LTD.

1 DETAIL

130

55

2-Ø5.5 HOLES
FOR SUS Ø4.8 RIVET

**ACCOMMODATION
LADDER (ST'BD)**

SHIP NO : 4193/94

CLASS : ABS

LENGTH : 14125 mm

WIDTH : 600 mm

HEIGHT : 300 mm

WEIGHT : 895 kg

MFG. NO : BAC4193S(P)

MIN.~MAX. ANGLE : 25 ~ 55°

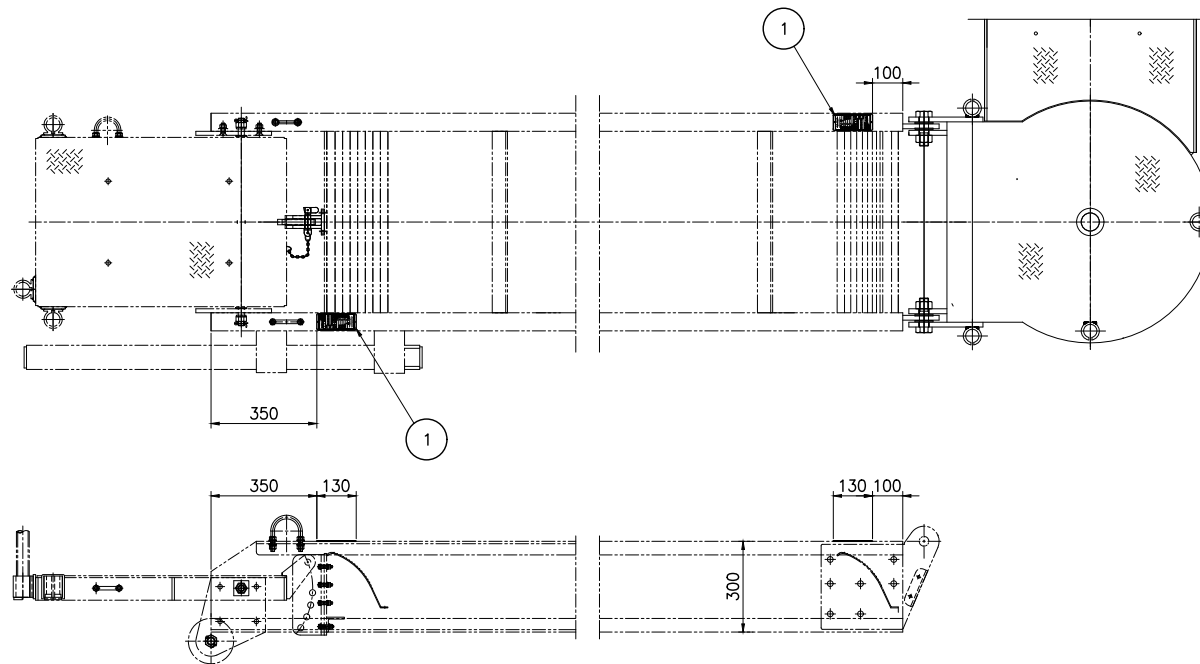
DESIGN LOAD : 3000 kg

MAX.OPERATIONAL LOAD(S.W.L):
1500 kg

MAX. LOWER P/F LOAD :
190 kg

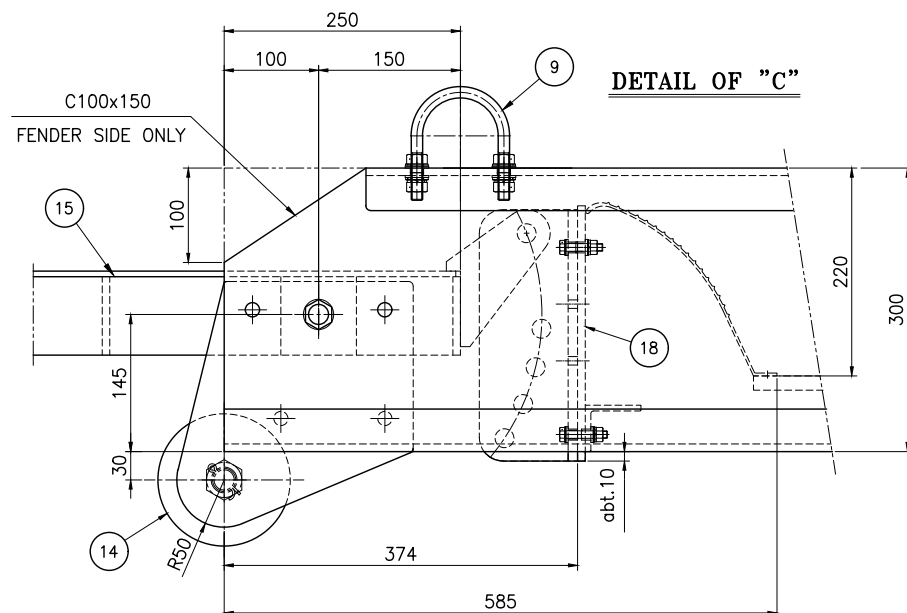
MFG. DATE : 20 . . .

 **JUNG-A MARINE**
BUSAN KOREA
TEL. +82-51-831-4147

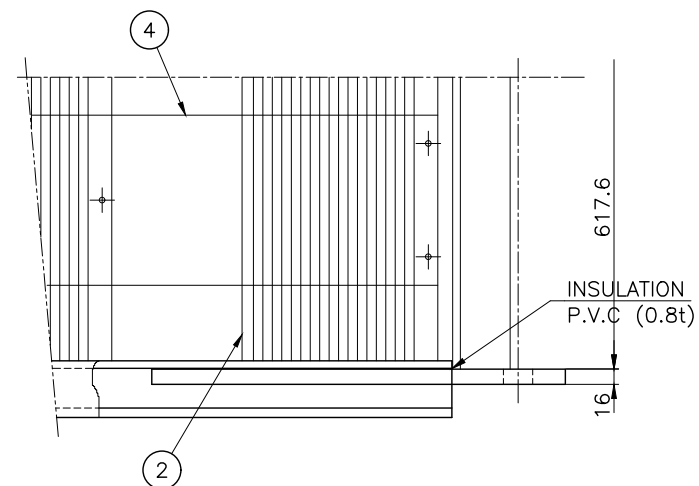
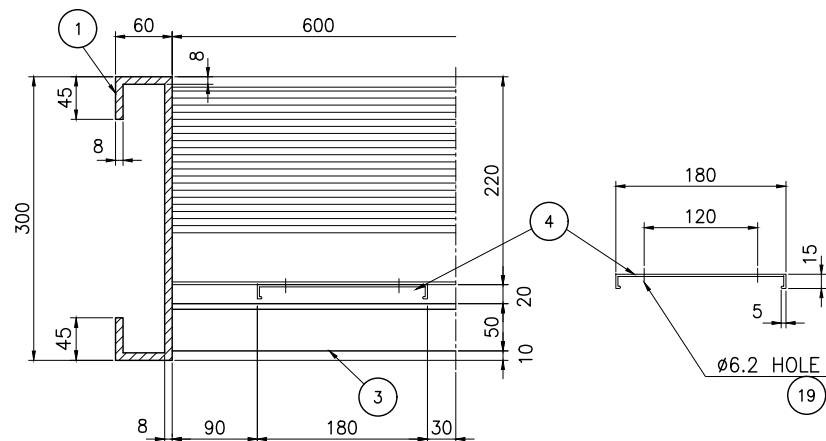


- * NOTE
1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE IS SYMMETRICAL.
2. Q'TY : 4EA/SHIP

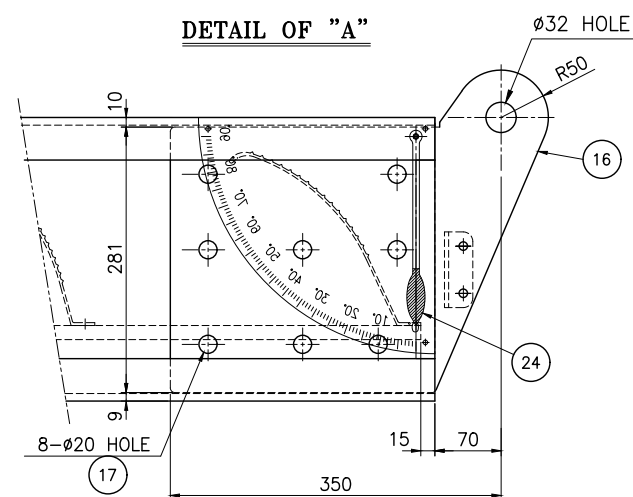
1	NAME PLATE	PL. 1t	SUS316	1	0.06KG
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS
Project Title. ACCOMMODATION LADDER					
Drawing Title. DEPLOYMENT OF NAME PLATE(FR-300)					
Dwg No. 18J010-LS-NPDP-300			SCALE 1/ 1	SHIP NO. 4193/94	
 JUNG-A MARINE co.,LTD.			DATE 2021.11.10	Approved S.W.SHIN	
				Checked S.W.SHIN	
				Designed C.W.PARK	



SEC' OF "B"



DETAIL OF "A"



Project Title.

ACCOMMODATION LADDER

Drawing Title.

LADDER DETAIL (FR-300)

Dwg No.

18J010-LS-DTLA-300

SCALE

1/8

SHIP NO.

4193/94

Approved

S.W.SHIN

DATE

2021.11.10

Checked

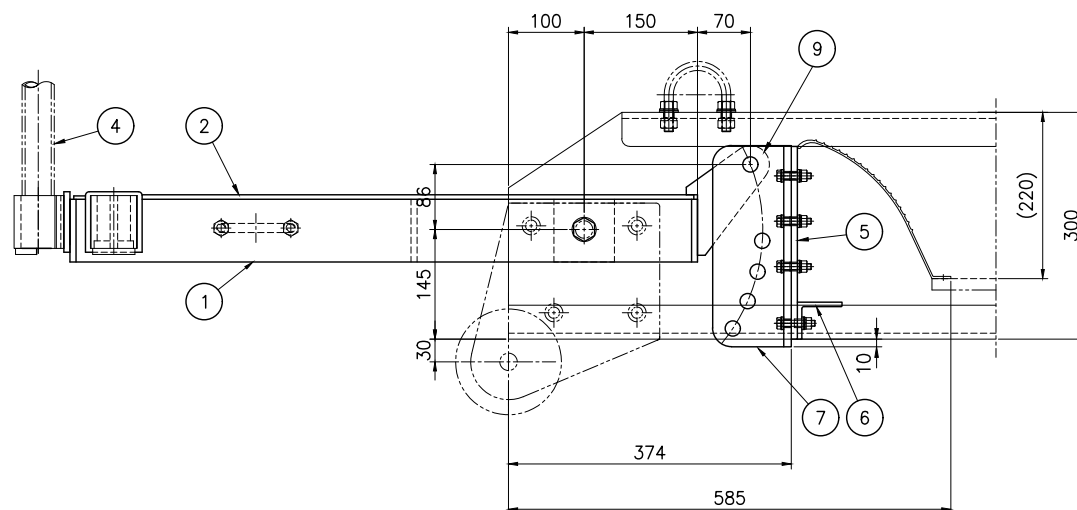
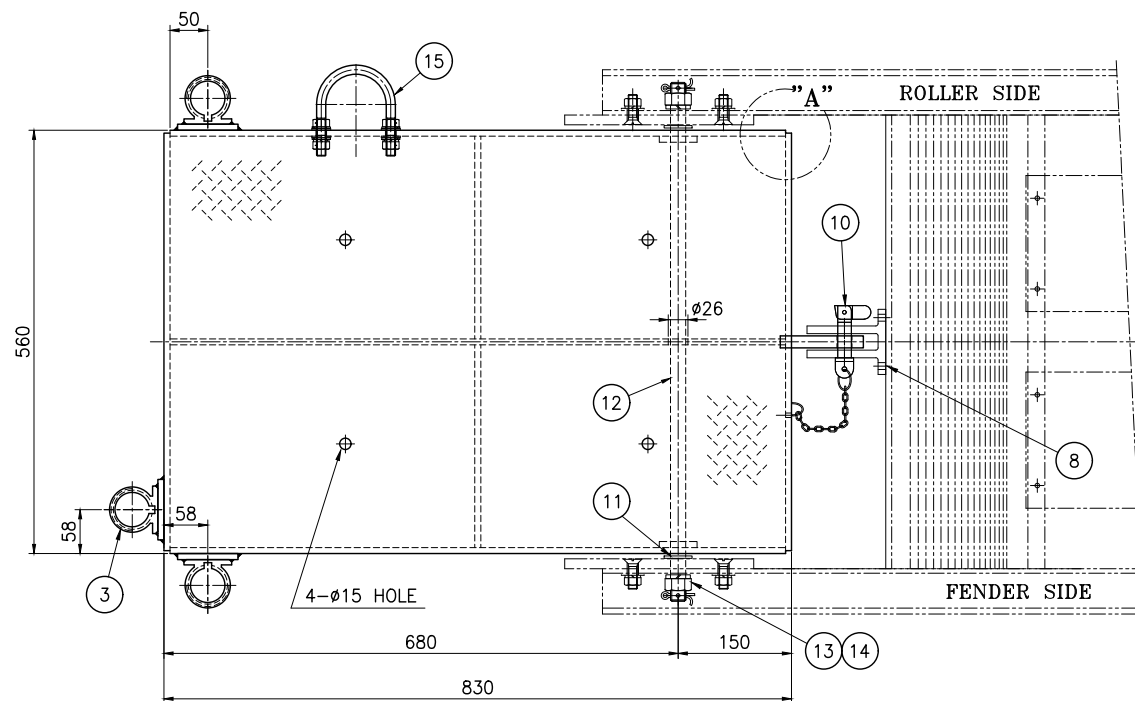
S.W.SHIN

Designed

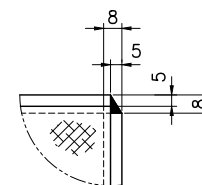
C.W.PARK



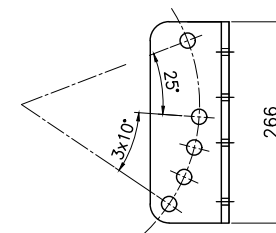
JUNG-A MARINE co.,LTD.



DETAIL OF "A"



DETAIL OF ⑦



NOTE

1. THIS DWG SHOWS ST'BD SIDE.
2. ONLY 2 STANCHIONS CAN BE USED ALTERNATIVELY AMONG 3 SOCKETS FOR ASSURING A PASSAGE.
3. WEIGHT : ABT. 26KG/SET
4. Q'TY : 2 SETS/SHIP(1SET SYMMETRY)

(FOR ONE SET)

15	U-BOLT/N/W/PVC W.	65A,M12	SUS316	1	65A/M12
14	SPLIT PIN	Ø4x32L	SUS304	2	
13	NUT/S.W	M20	SUS316	2	
12	SHAFT	Ø20x695L	SUS316	1	Ø20
11	WASHER	M20	SUS316	2	
10	TOGGLE PIN	M8x40/45L	SUS316	1	Ø18
9	EYE PLATE	16t	AL ALLOY	1	16t
8	BOLT/N/S.W/PVC.W	M8x40/45L	SUS316	6/2	M8
7	ANGLE SOCKET	10t	AL ALLOY	1	
6	STIFFENER	UA.50x60x6t	AL ALLOY	1	
5	BRACKET	8tx290	AL ALLOY	1	
4	PORTABLE STANCHION	Ø44.5x6t PIPE	AL ALLOY	2	
3	STANCHION SOCKET	Ø45.5xØ60x70L	AL ALLOY	3	
2	TOP PLATE	6t CH'KED PL.	AL ALLOY	1	
1	MAIN FRAME	8tx83	AL ALLOY	1	
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

Project Title.

ACCOMMODATION LADDER

Drawing Title.

LOWER PLATFORM_600B(FRAME-300)

Dwg No.

18J010-LS-LPAY-300

SCALE

1/ 10

SHIP NO.

4193/94

Approved

S.W.SHIN

DATE

2021.11.10

Checked

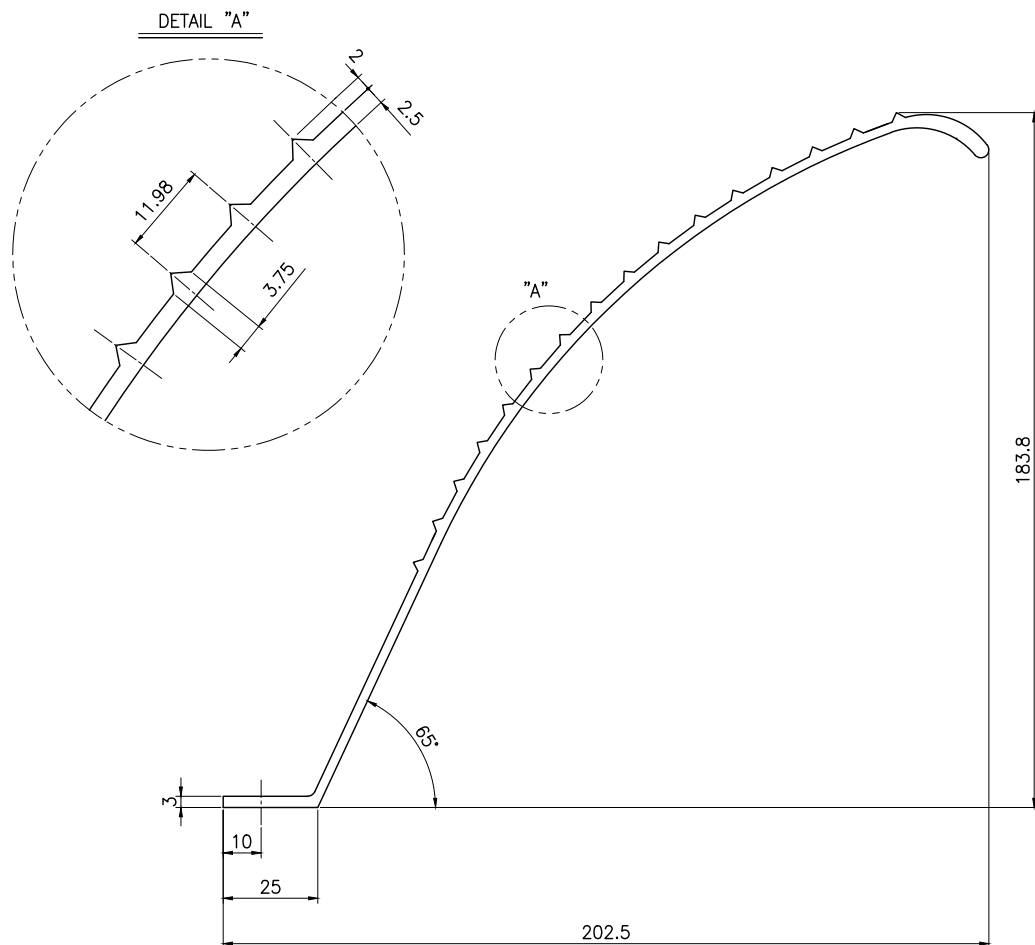
S.W.SHIN

Designed

C.W.PARK



JUNG-A MARINE co.,LTD.



* NOTE

1. Q'TY : 80EA/SHIP

2. WEIGHT : 1.39 KG/600mm

1	STEP	-	AL ALLOY	-	
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

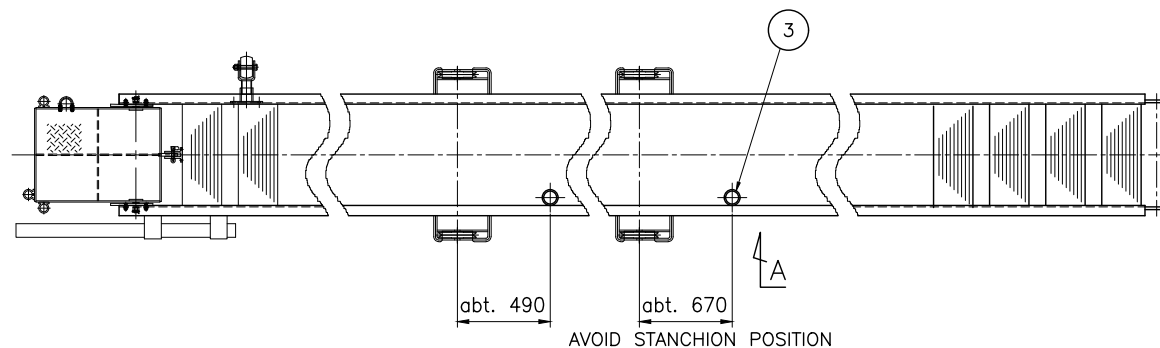
Project Title. **ACCOMMODATION/PILOT SLANT LADDER**

Drawing Title. **LADDER-ALUMINIUM STEP**

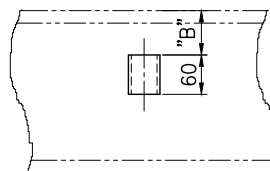
Dwg. No.	18J010-LS-SESC-01	SCALE	SHIP NO.	4193/94
		1/2	Approved	S.W.SHIN
		DATE	Checked	S.W.SHIN
		2021.11.10	Designed	C.W.PARK



JUNG-A MARINE co.,LTD.



VIEW : A



FRAME	"B"	REMARK
250H	75	
300H	90	←
350H	100	
400H,450H,500H	130	
600H,620	150	

* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. Q'TY : 2SETS/SHIP
3. WEIGHT : 3.5 KG/SET

(FOR ONE SET)

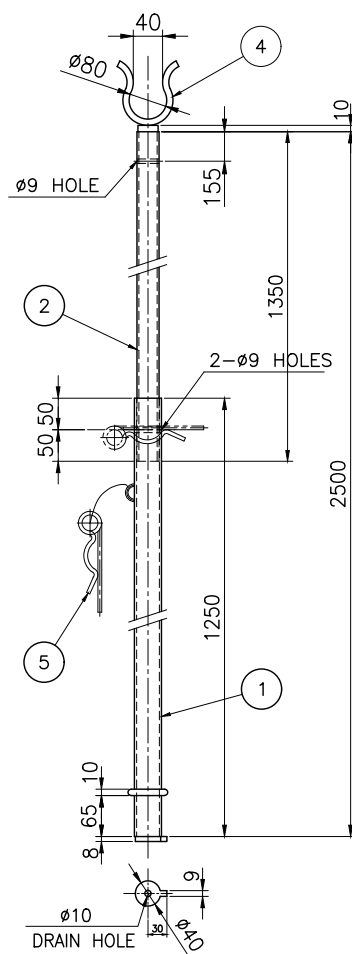
5	PIN W/WIRE	ø6	SUS304	1	
4	RING	ø10 R.BAR	AL ALLOY	1	
3	SOCKET	ø60xø44x60L	AL ALLOY	2	
2	ASSIST PIPE	PIPE ø34x3t	AL ALLOY	1	
1	MAIN PIPE	PIPE ø43x3.5t	AL ALLOY	1	
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

Project Title. **ACCOMMODATION/PILOT SLANT LADDER**Drawing Title. **ASSEMBLY OF WIRE HANGER**

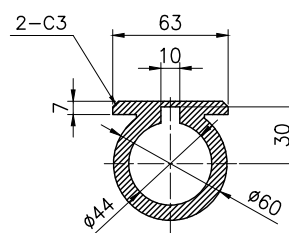
Dwg No.	18J010-LS-HWAY-01	SCALE	SHIP NO.	4193/94
		1/12	Approved	S.W.SHIN
		DATE	Checked	S.W.SHIN
		2021.11.10	Designed	C.W.PARK

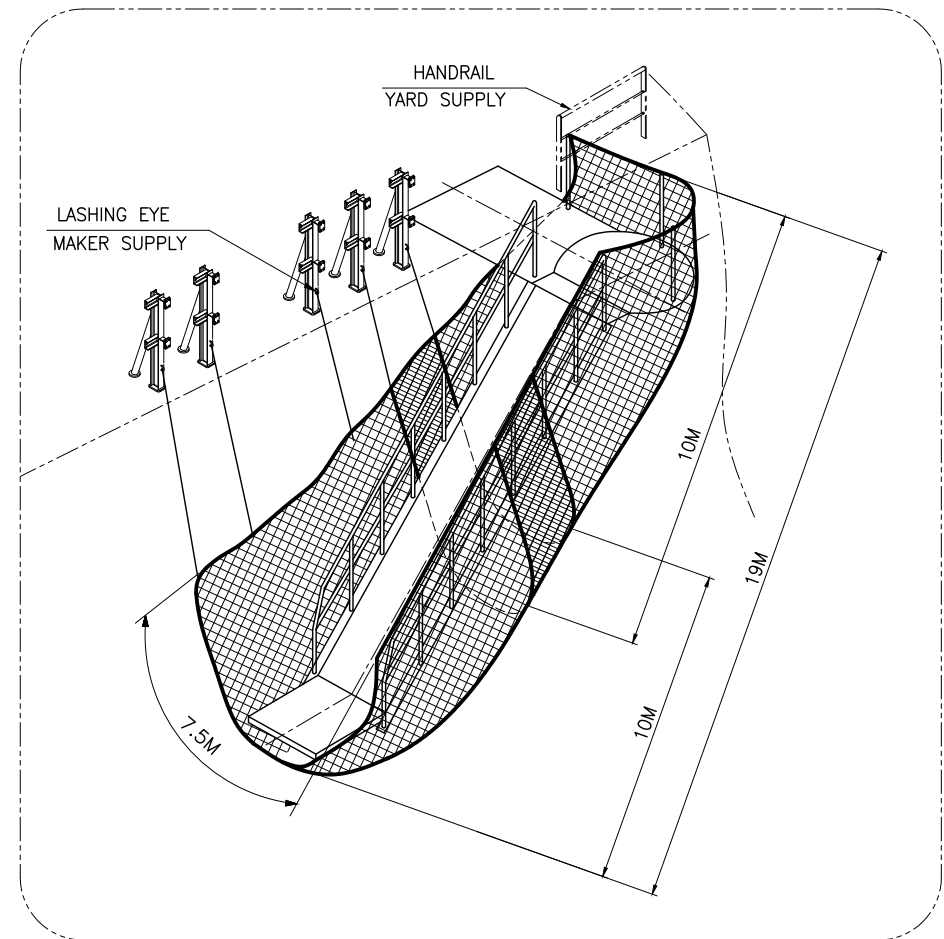
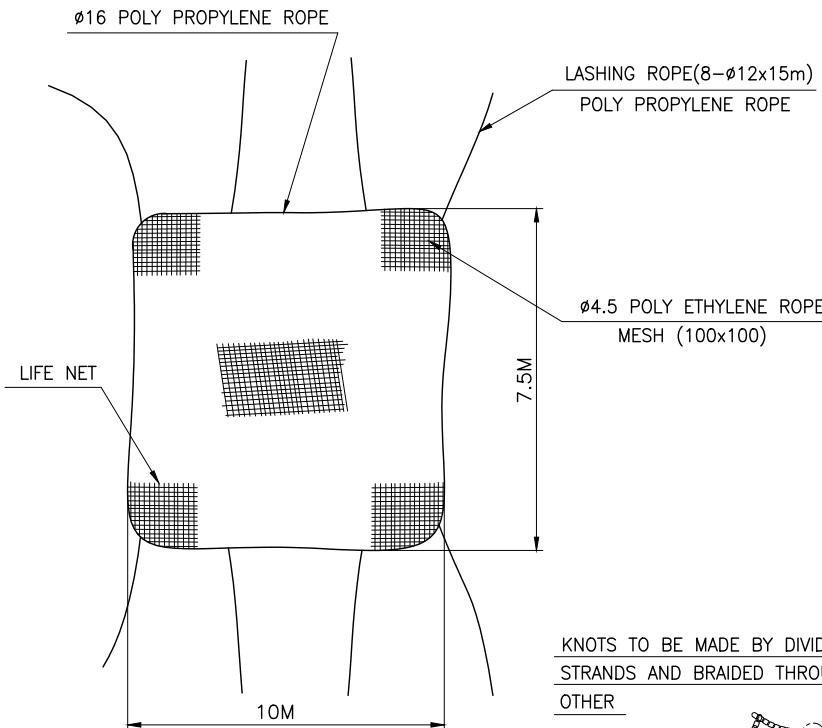


JUNG-A MARINE co.,LTD.



3 DETAIL

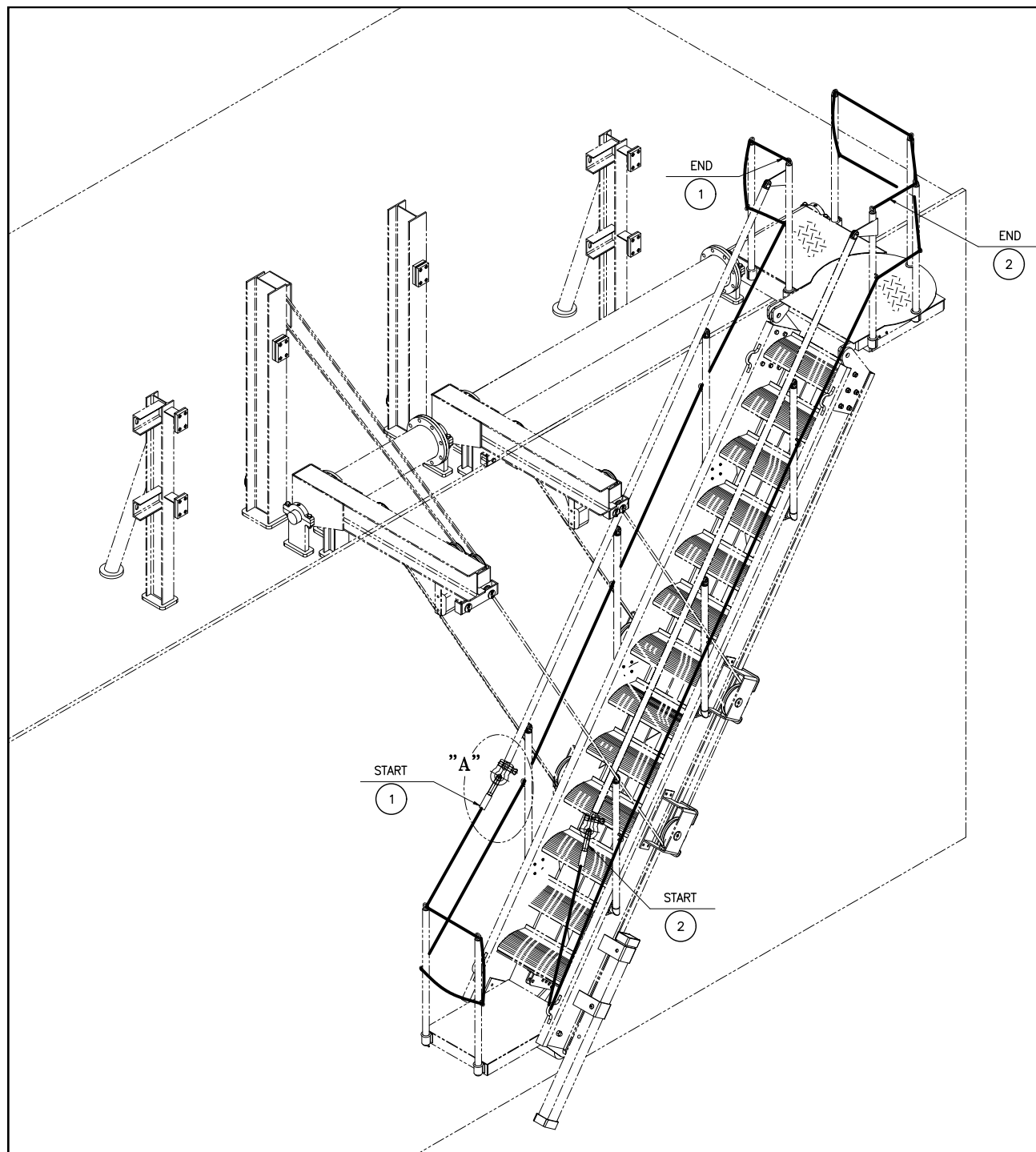




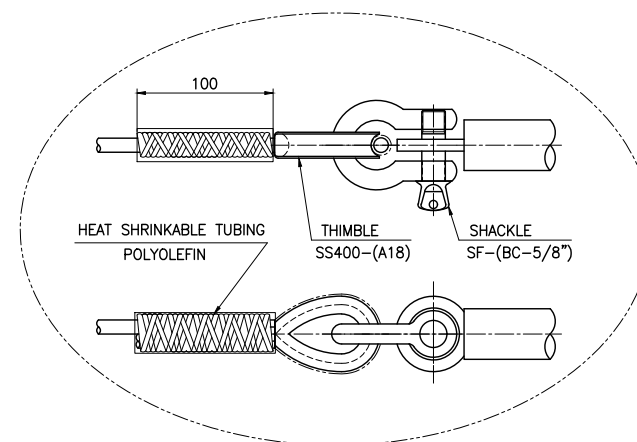
* NOTE

1. Q'TY : 2EA(7.5Mx10M)/SHIP
2. WEIGHT : abt. 29Kg/EA (7.5Mx10M)
3. NETTING : 136 KG (4.5 P.E.ROPE)
BREAKING LOAD : 1800KG (12 P.P.ROPE)
BREAKING LOAD : 3000KG (16 P.P.ROPE)

NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS
Product Title.					
ACCOMMODATION LADDER					
Drawing Title.					
LIFE NET (VET-PIPE SUPPORT)					
Dwg No.		SCALE	PROJECTION		
18J010-LS-LNVT-01		N/S			
			Designed by	Date	
STANDARD			S.H.HONG	2018.06.30	



DETAIL "A"



* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. WEIGHT : 0.156 KG/M
3. BREAKING LOAD : 3,300 KG (Ø18 P.P. ROPE)
4. SHACKLE(5/8") : SWL 1,500KG

(FOR ONE SHIP)

2	ROPE(WITH VINYL COATED)	Ø18 x 24M	P.P.	2	
1	ROPE(WITH VINYL COATED)	Ø18 x 23M	P.P.	2	
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

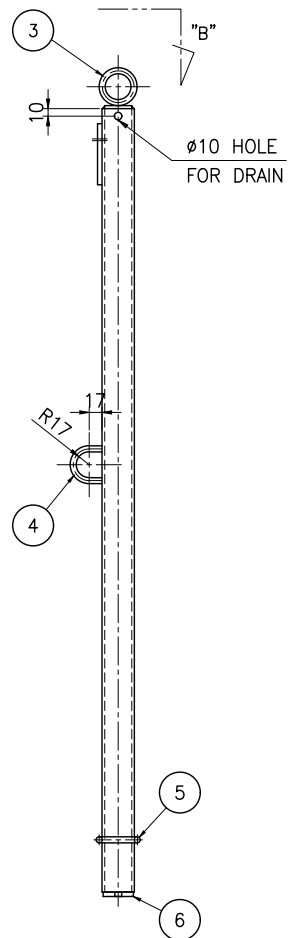
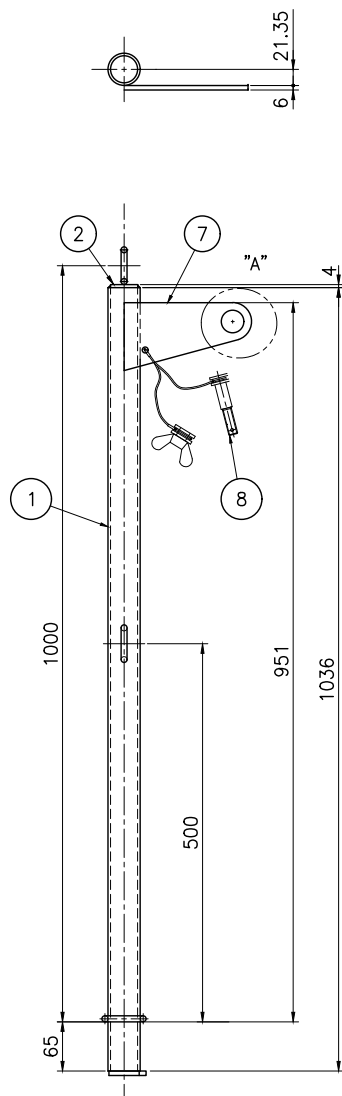
Product Title.

ACCOMMODATION LADDER

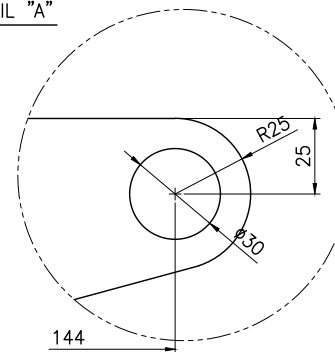
Drawing Title.

DEPLOYMENT OF MID RAIL

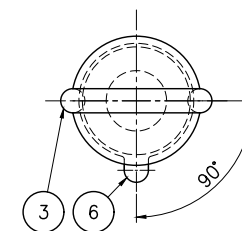
Dwg No.	SCALE	PROJECTION	
18J010-LS-PPDP-V101	N/S		
		Designed by	Date
		S.H.HONG	2018.06.30



DETAIL "A"



SECTION "B"



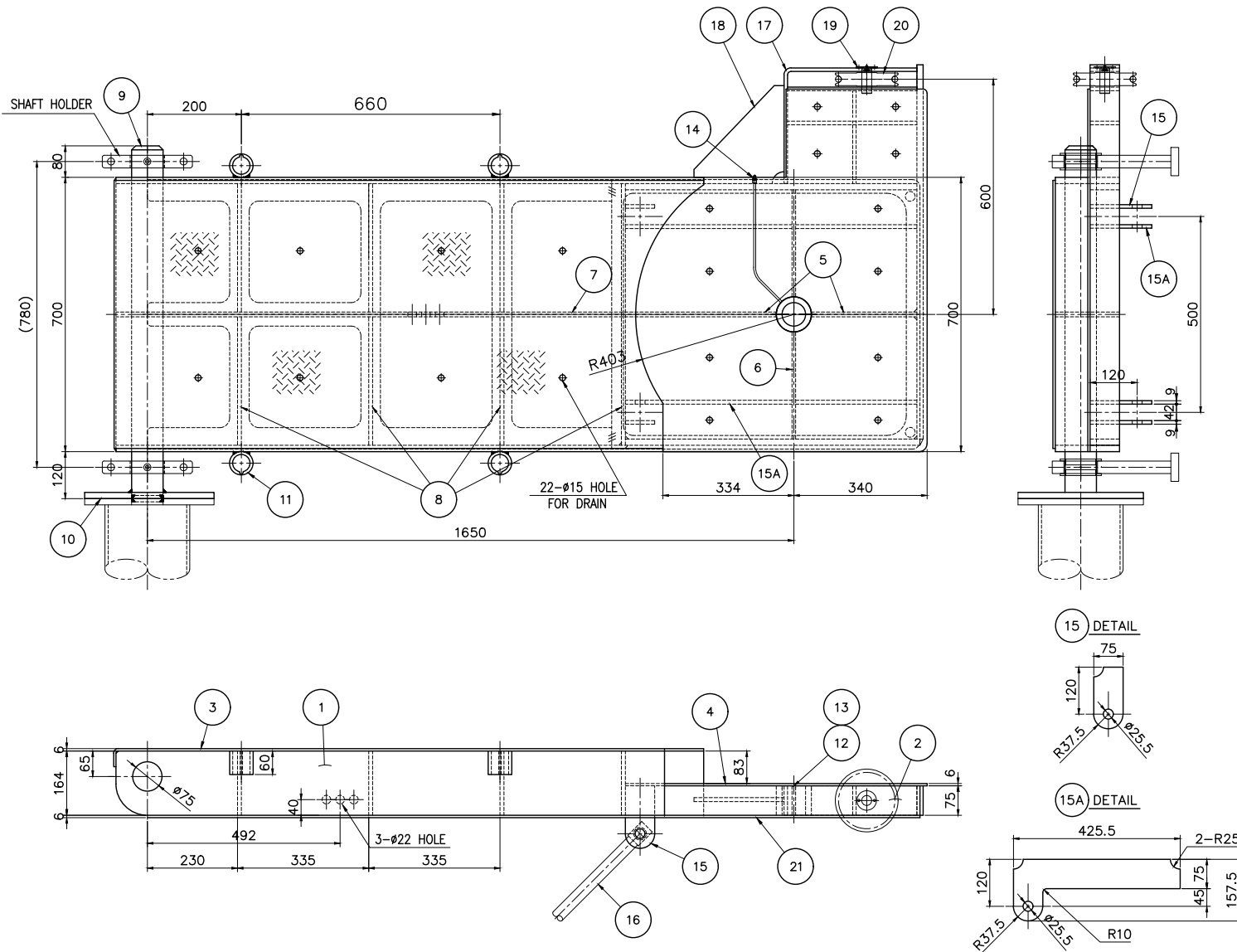
* NOTE

1. STEEL PARTS TO BE GALVANIZED.
2. Q'TY : 4SETS/SHIP (2SETS SYMMETRY)
3. WEIGHT : 1.05 KG/EA

(FOR ONE SET)

8	JOINT PIN	Ø12x35L	SUS304	1	
7	WING PLATE	PL. 6t	SS400	1	
6	KEY PLATE	9tx39x8	SS400	1	
5	STOPPER RING	R.BAR Ø8	SS400	1	
4	EYE RING	R.BAR Ø8	SS400	1	
3	TOP RING	R.BAR Ø8	SS400	1	
2	END PLATE	PL. 4t	SS400	1	
1	STANCHION	32A PIPE	SGP	1	JAC-SST-DTTST-01
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

Project Title.		ACCOMMODATION/PILOT SLANT LADDER			
Drawing Title.		TURN TABLE-WING STANCHION			
Dwg No.	18J010-SS-VTT8DE-10	SCALE 1/10	SHIP NO.	4193/94	
			Approved	S.W.SHIN	
JUNG-A MARINE co.,LTD.	2021.11.10	DATE	Checked	S.W.SHIN	
			Designed	C.W.PARK	



* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

(FOR ONE SET)

NO.	DESCRIPTION	MAT'L	Q'TY	REMARKS
21	UNDER PLATE	SS400	—	F.BAR 6tx65
20	SHEAVE	SS400	1	ø160
19	SHEAVE PIN	SUS304	1	R.BAR ø25
18	BRACKET	SS400	1	PL. 10t
17	SHEAVE BOX	SS400	1	F.B. 9tx75
16	SUPPORTING ROD	SS400	1	R.BAR ø32
15A	EYE PLATE	SS400	2	F.B. 10tx75
15	EYE PLATE	SS400	2	F.B. 9tx75
14	GREASE NIPPLE	BS	1	A-PT 1/8"
13	BUSH	BS	1	—
12	BOSS	SS400	1	—
11	STANCHION SOCKET	SS400	4	R.BAR ø60xø44
10	FLANGE	SS400	1	—
9	SHAFT	SS400	1	R.BAR ø75
8	STIFFENER	SS400	1/4	PL. 10t
7	STIFFENER	SS400	1	PL. 12t
6	STIFFENER	SS400	2	F.B. 9tx75/PL.10t
5	STIFFENER	SS400	1/1	F.B. 12tx75
4	LOWER PLATE	SS400	1	PL. 6t
3	TOP PLATE	SS400	1	PL. CH'KED 6t
2	SIDE FRAME	SS400	3	PL. 16t
1	MAIN PLATE	SS400	1/1	PL. 16t

Project Title.

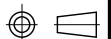
ACCOMMODATION LADDER

Drawing Title.

UPPER PLATFORM

Dwg No.
18J010-SS-VUP1600-01SCALE
1/16

PROJECTION



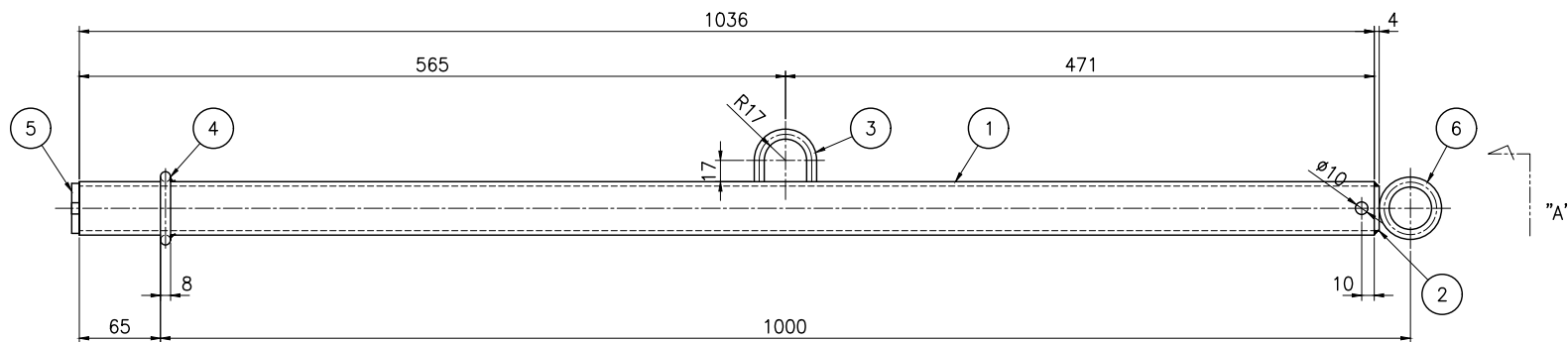
JUNG-A MARINE CO.,LTD.

Designed by

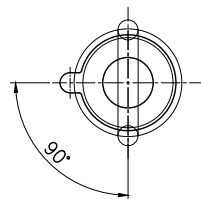
S.H.HONG

Date

2019.01.31



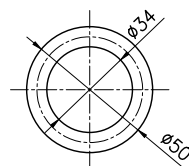
VIEW "A"



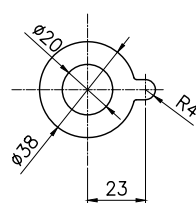
* NOTE

1. STEEL PARTS TO BE GALVANIZED.
2. Q'TY : 12SETS/SHIP (6SETS SYMMETRY)
2. WEIGHT : 1.05 KG/EA

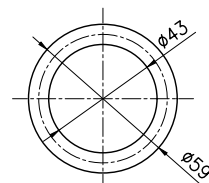
6 EYE



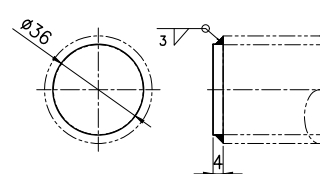
5 KEY PLATE



4 STOPPER RING



2 END PLATE



6	EYE(TOP RAIL)	R.BAR $\phi 8$	SS400	1	-
5	KEY PLATE	PL. 6t	SS400	1	-
4	STOPPER RING	R.BAR $\phi 8$	SS400	1	-
3	EYE(MID RAIL)	R.BAR $\phi 8$	SS400	1	-
2	END PLATE	PL. 4t	SS400	1	-
1	STANCHION	32A PIPE	SGP	1	-
NO.	DESCRIPTION	SPECIFICATION	MAT'L	Q'TY	REMARKS

Project Title.

ACCOMMODATION/PILOT SLANT LADDER

Drawing Title.

UPPER P/F(GEN) - STANCHION (2COURSES)

Dwg No.

18J010-SS-VUPDE-09

SCALE

1/6

SHIP NO.

4193/94

Approved S.W.SHIN

DATE

2021.11.10

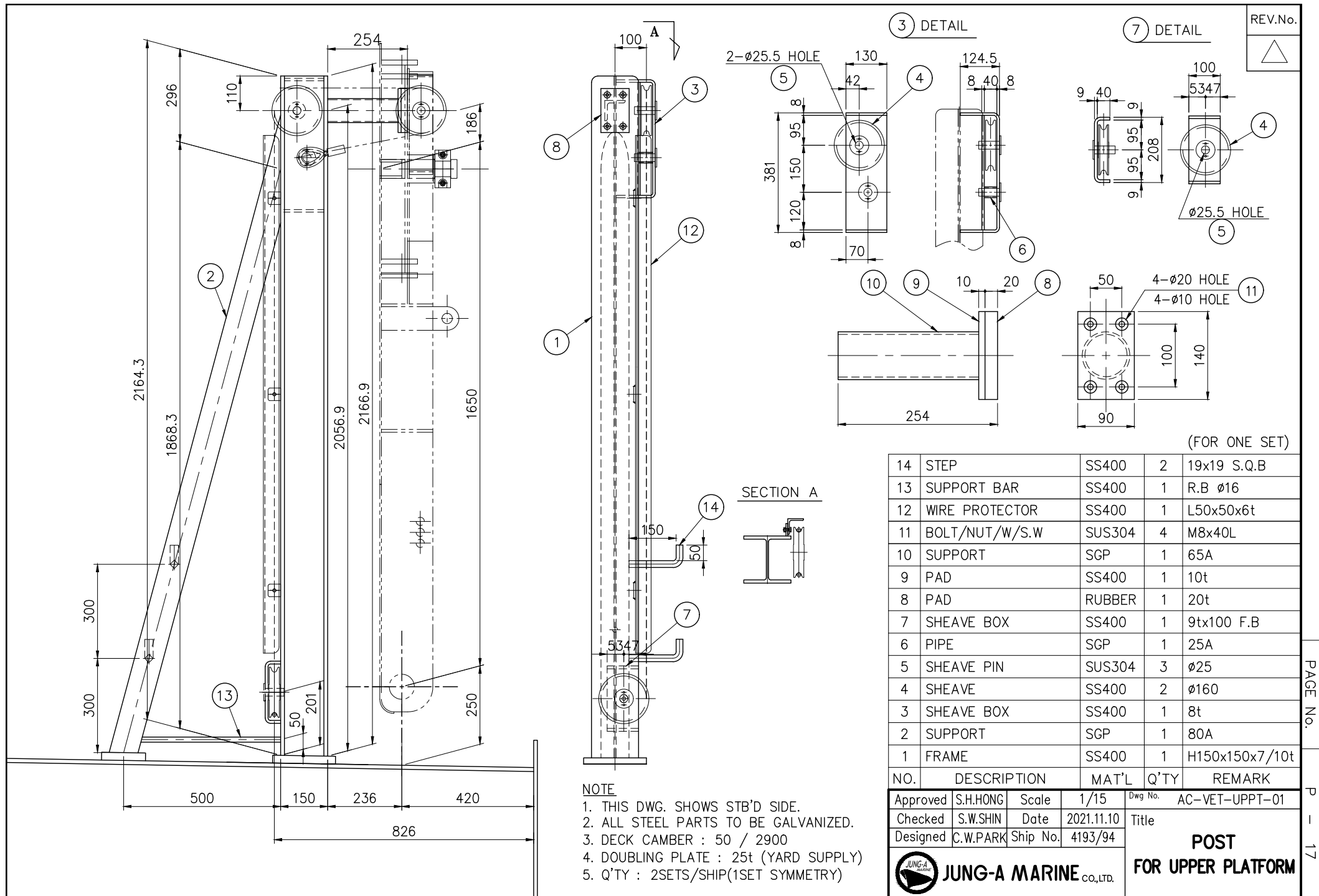
Checked

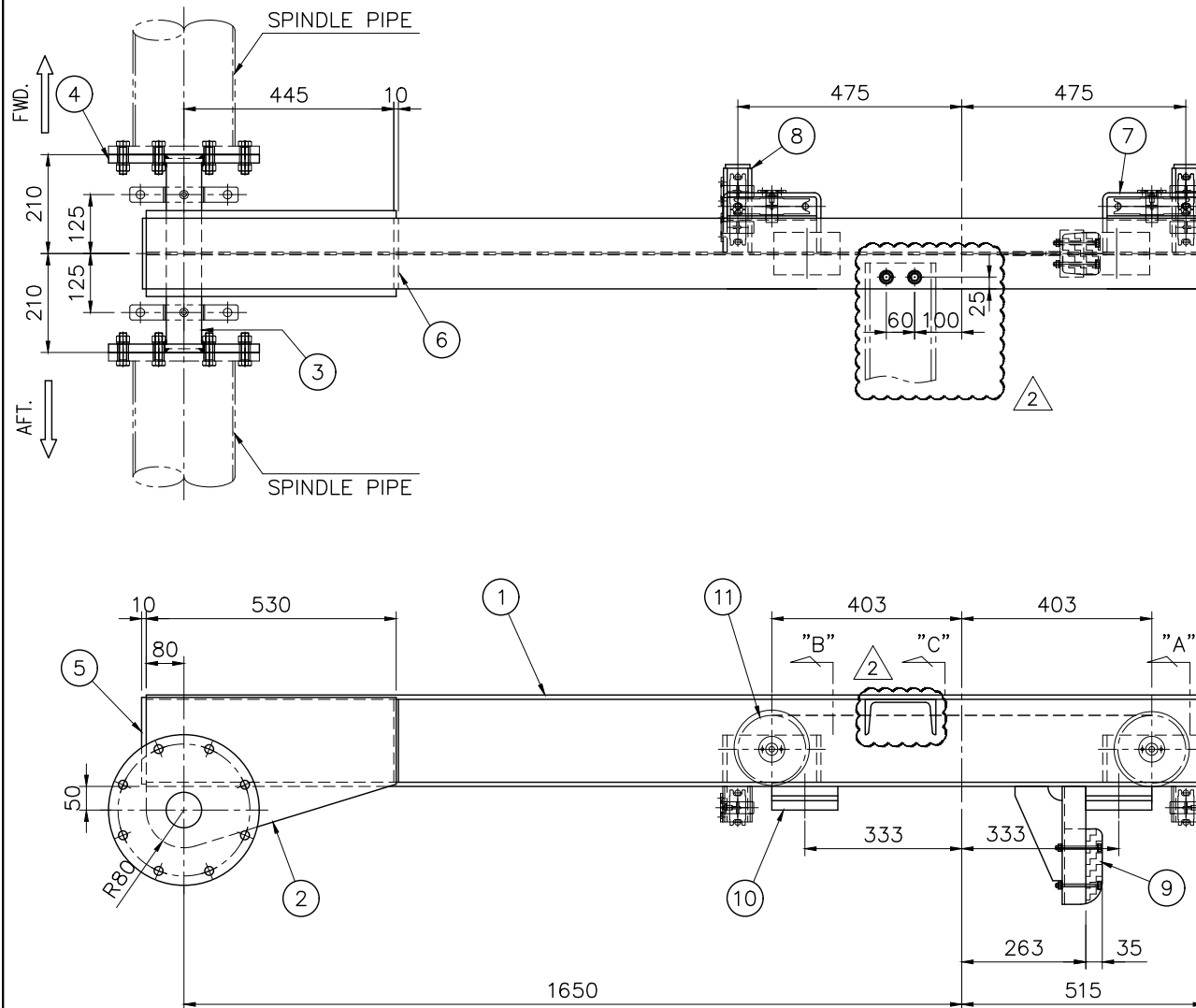
S.W.SHIN

Designed C.W.PARK



JUNG-A MARINE co.,LTD.

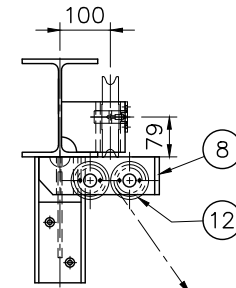




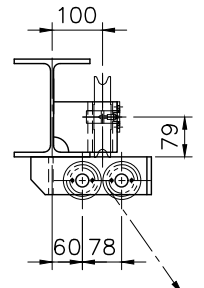
NOTE :

1. THIS DWG. SHOWS STB'D SIDE, PORT SIDE IS SYMMETRY.
2. SS400 PARTS TO BE GALVANIZED.
3. Q'TY: 2SETS/SHIP (1 SET SYMM.)

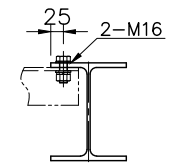
"A" VIEW



"B" VIEW



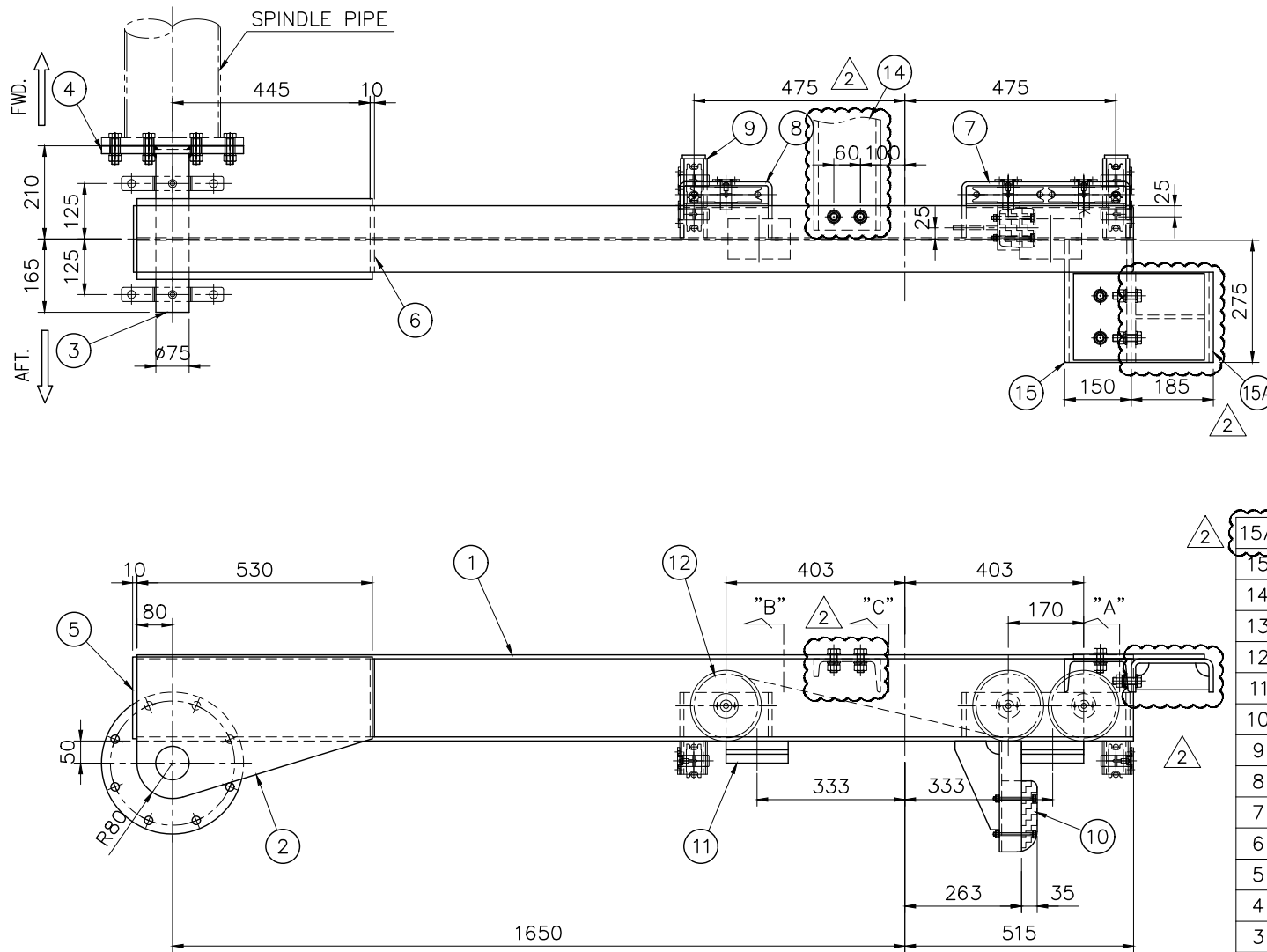
"C" VIEW



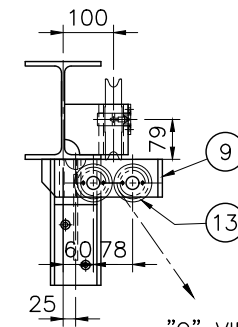
(FOR ONE SET)

NO.	DESCRIPTION	MAT'L	Q'TY	REMARK
12	ROLLER	SS400	4	ø75
11	SHEAVE	SS400	2	ø160
10	PAD	RUBBER/SS400	4/2	(20t+10t+20t)x90x140
9	SUPPORT	SS400	1	[100x50x5/7.5t
8	GUIDE ROLLER BOX		2	9tx75 F.B
7	SHEAVE BOX		2	9tx100 F.B
6	STIFFENER		2	10t
5	END PLATE		1	10t
4	FLANGE		2	—
3	SHAFT		1	ø75 R.B
2	HINGE PLATE		2	16t
1	FRAME	SS400	1	H194x150x6/9t

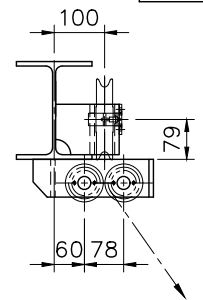
Approved	S.H.HONG	Scale	1/15-A4	Dwg No.	AC-STD-SDDT-01
Checked	S.W.SHIN	Date	2021.12.29	Title	
Designed	C.W.PARK	Ship No.	4193/94		
				FWD.STOWING DAVIT	
				⑥	



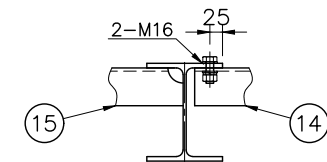
"A" VIEW



"B" VIEW



"C" VIEW



(FOR ONE SET)

15A	SPRING PAD	SS400	1	10t
15	SPRING PAD	SS400	1	[150x75x6.5/10t
14	CONNECTOR	SS400	1	[150x75x6.5/10t
13	ROLLER	SS400	4	ø75
12	SHEAVE	SS400	3	ø160
11	PAD	RUBBER/SS400	4/2	(20t+10t+20t)x90x140
10	SUPPORT	SS400	1	[100x50x5/7.5t
9	GUIDE ROLLER BOX		2	9tx75 F.B
8	SHEAVE BOX 2		1	9tx100 F.B
7	SHEAVE BOX 1		1	9tx100 F.B
6	STIFFENER		2	10t
5	END PLATE		1	10t
4	FLANGE		1	—
3	SHAFT		1	ø75 R.B
2	HINGE PLATE		2	16t
1	FRAME	SS400	1	H194x150x6/9t
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved S.H.HONG Scale 1/15-A4 Dwg No. AC-STD-SDDT-02

Checked S.W.SHIN Date 2021.12.29 Title

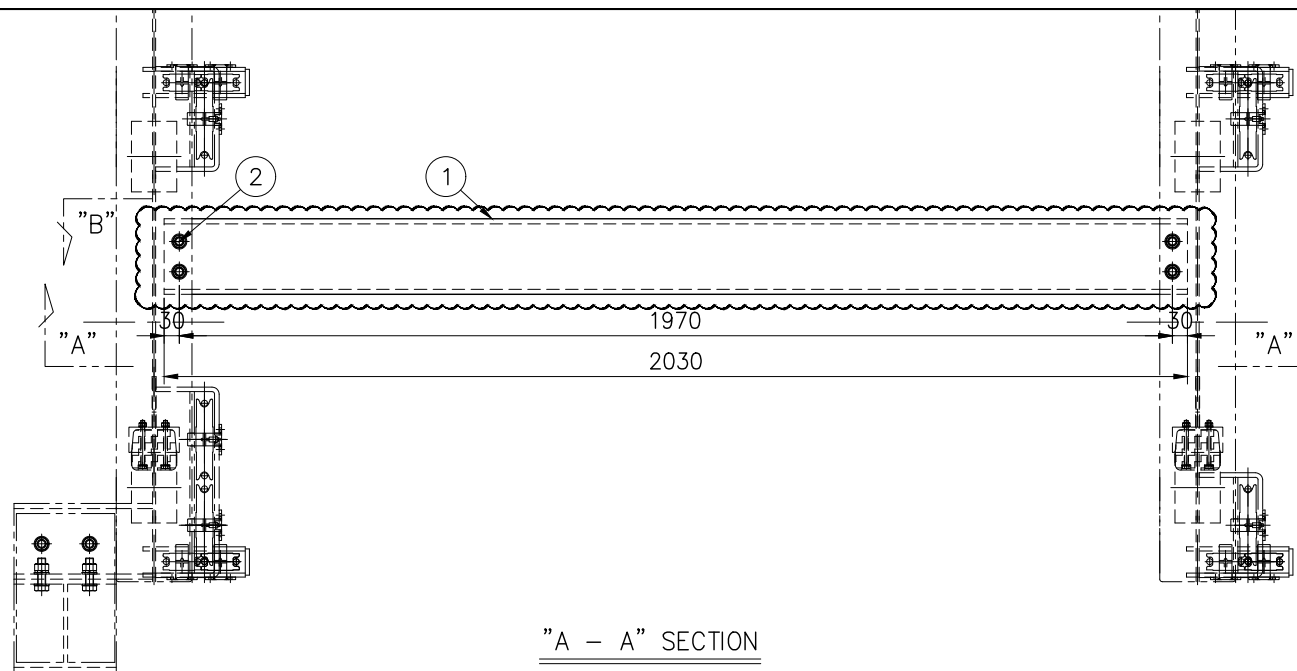
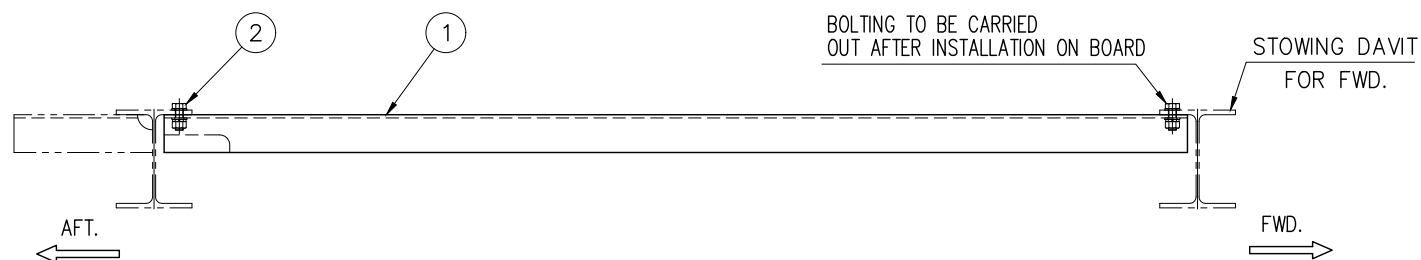
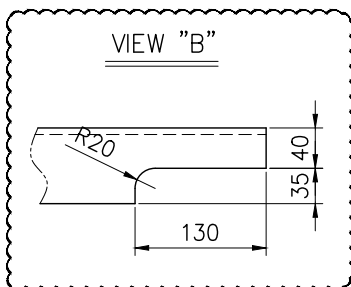
Designed C.W.PARK Ship No. 4193/94

JUNG-A MARINE CO., LTD.
AFT.STOWING DAVIT

7

NOTE :

1. THIS DWG. SHOWS STB'D SIDE, PORT SIDE IS SYMMETRY.
2. SS400 PARTS TO BE GALVANIZED.
3. Q'TY: 2SETS/SHIP (1 SET SYMM.)

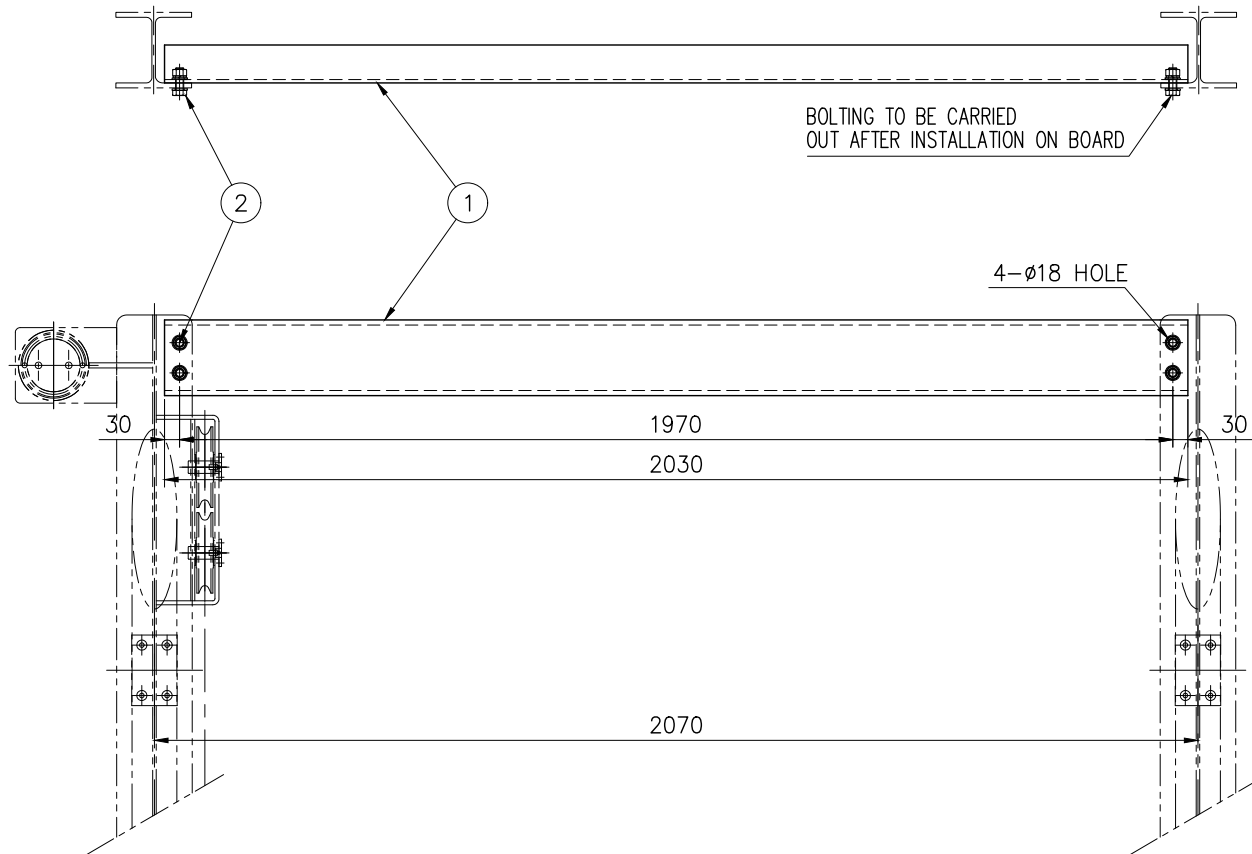
"A - A" SECTIONVIEW "B"

Q'TY : 2SETS/SHIP

2	BOLT/N/2W/S.W	SUS304	4	M16,45L
1	CONNECTION BEAM	SS400	1	[150x75x6.5/10t
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

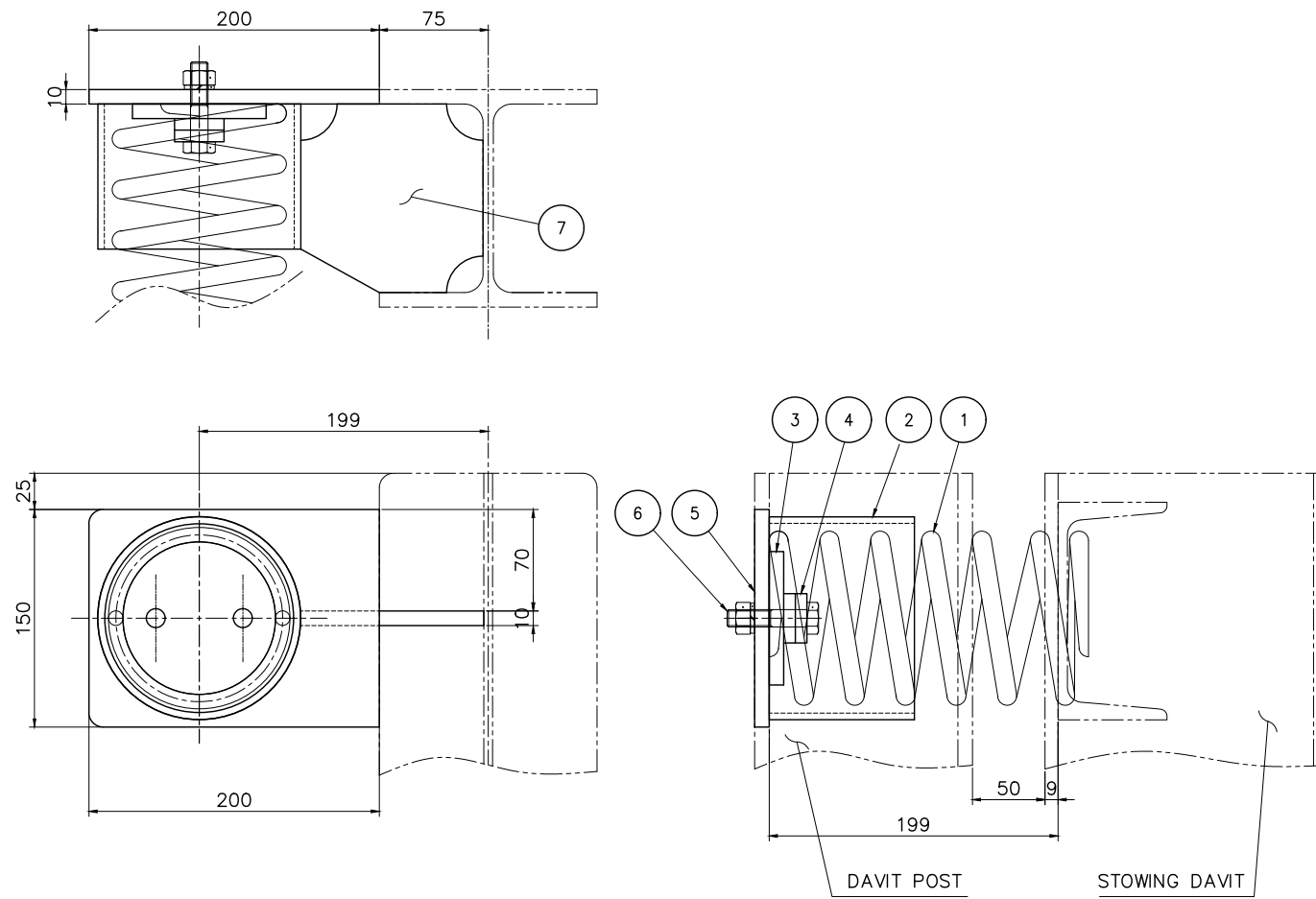
Approved	S.H.HONG	Scale	1/15	A4	Dwg No.	AC-VER-SDCB-01
Checked	S.W.SHIN	Date	2021.12.29		Title	
Designed	C.W.PARK	Ship No.	4193/94			

**JUNG-A MARINE** CO.,LTD.**CONNECTION BEAM
FOR DAVIT**



Q'TY : 2SETS/SHIP

2	BOLT/N/2W/S.W	SUS304	4	M16,45L
1	CONNECTION BEAM	SS400	1	[150x75x6.5/10t
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK
Approved	S.H.HONG	Scale	1/15	A4 Dwg No. AC-VEP-SDCB-01
Checked	S.W.SHIN	Date	2021.12.29	Title
Designed	C.W.PARK	Ship No.	4193/94	
 JUNG-A MARINE CO.,LTD.				CONNECTION BEAM FOR DAVIT POST

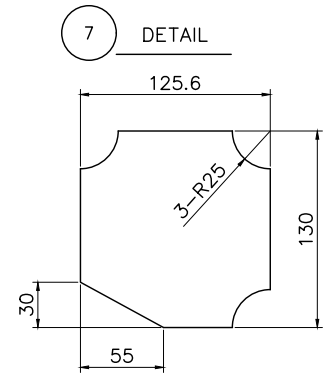
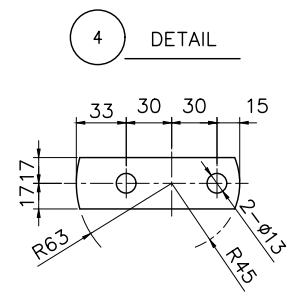
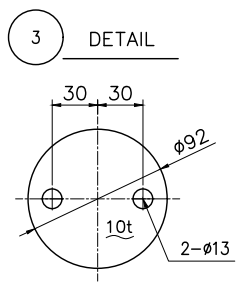


* NOTE

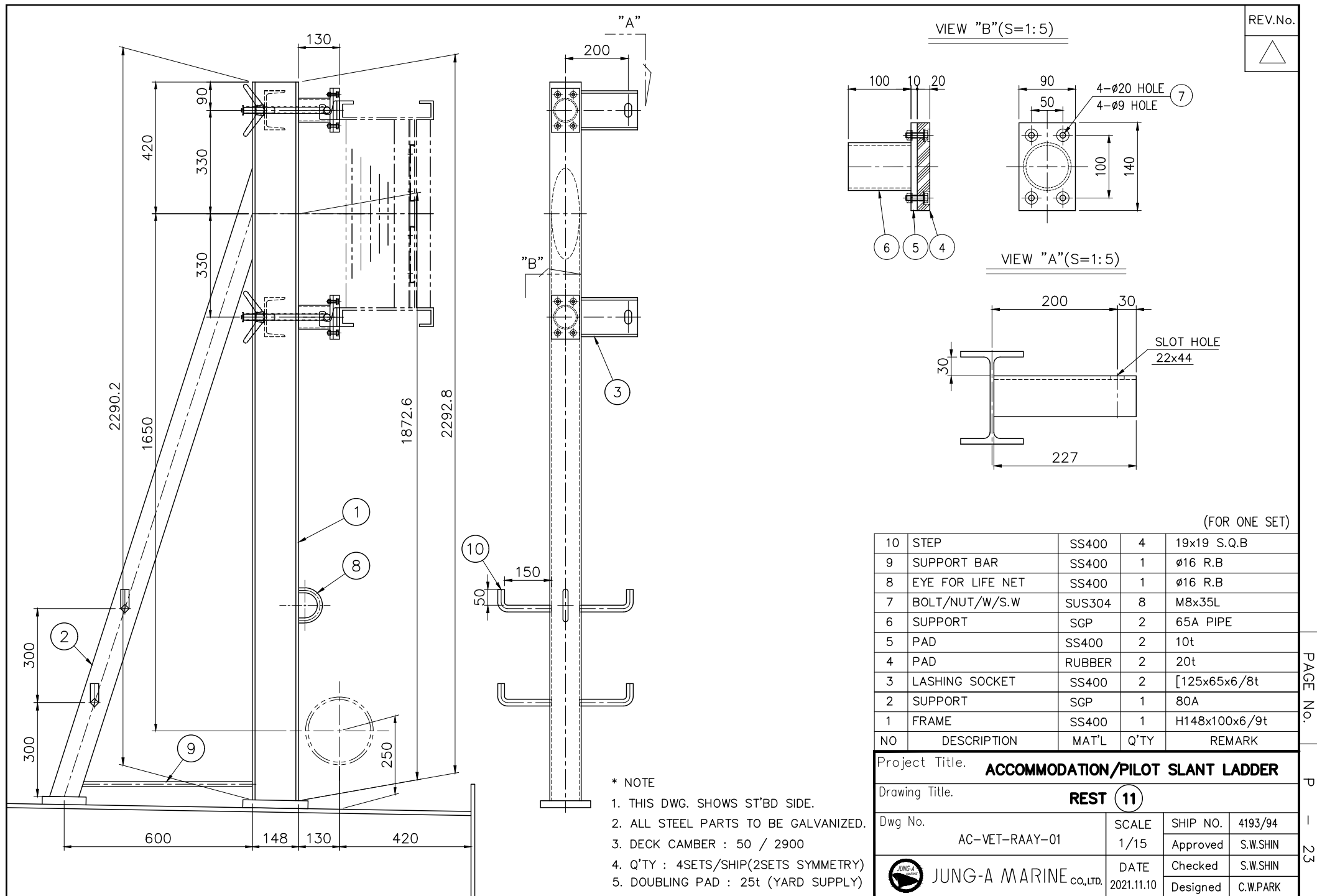
1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

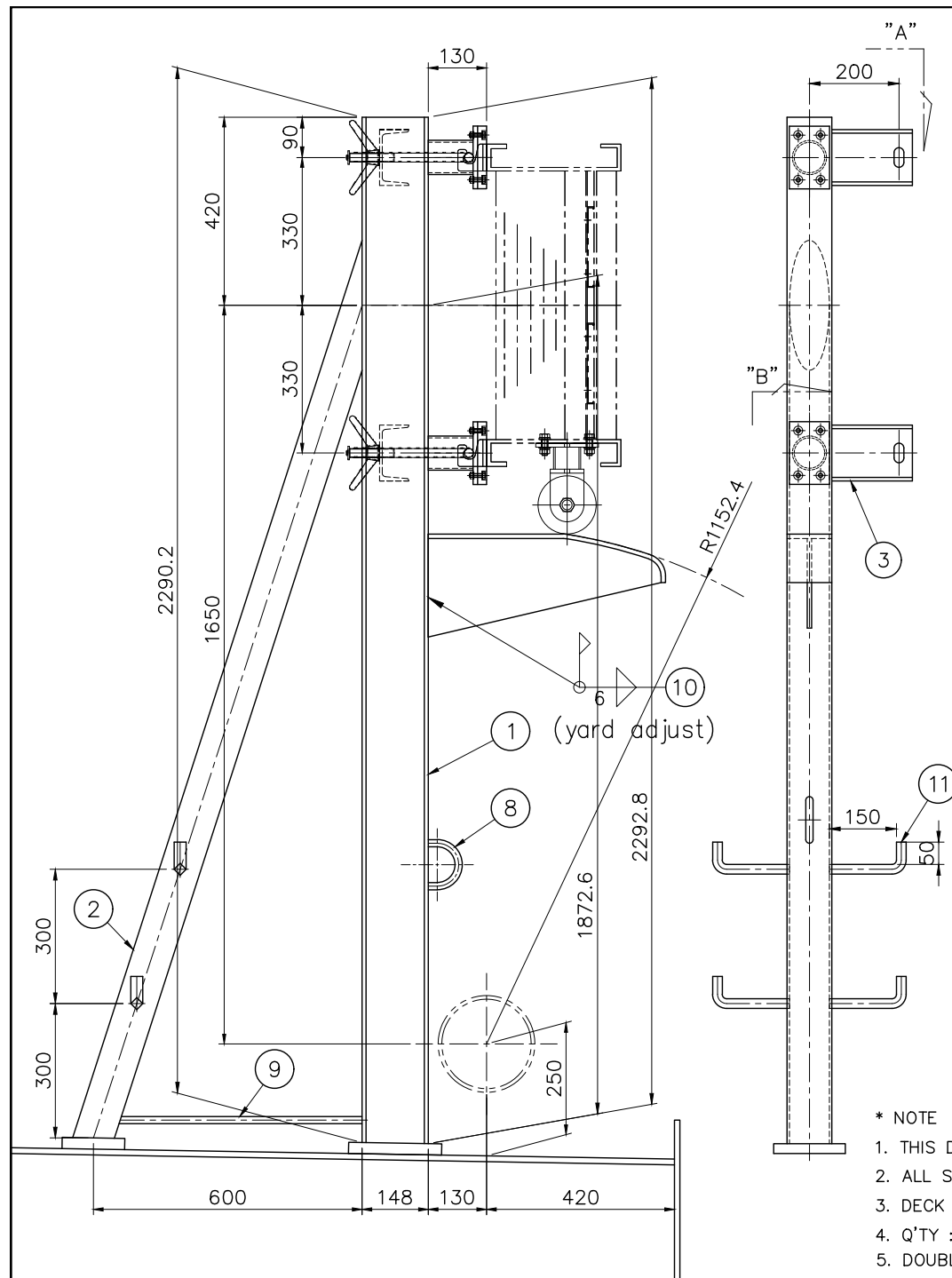
(FOR ONE SET)

7	STIFFENER	PL. 10t	SS400	1	-
6	BOLT/N/S.W	M12x55L	SUS304	2	-
5	STOPPER	PL. 10t	SS400	1	-
4	STOPPER	PL. 10t	SS400	2	-
3	BASE PLATE	PL. 10t	SS400	1	-
2	HOUSING	125A	SGP	1	-
1	SPRING	ø120x280L	SUP	1	-
NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS



Project Title. ACCOMMODATION/PILOT SLANT LADDER					
Drawing Title. POST - SPRING SET					
Dwg No. 18J010-SS-VDPSP-01	SCALE 1/6	SHIP NO.	4193/94		
		Approved	S.H.HONG		
JUNG-A MARINE co.,LTD.	DATE 2019.01.31	Checked	S.W.SHIN		
		Designed	C.W.PARK		

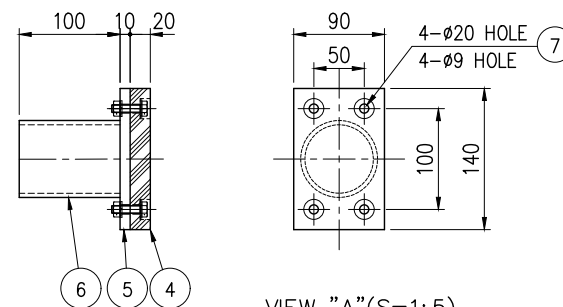




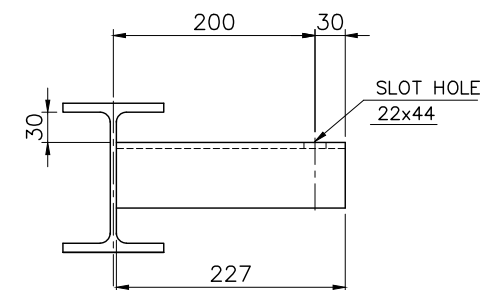
* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. DECK CAMBER : 50 / 2900
4. Q'TY : 2SETS/SHIP(1SET SYMMETRY)
5. DOUBLING PAD : 25t (YARD SUPPLY)

VIEW "B"(S=1:5)



VIEW "A"(S=1:5)



(FOR ONE SET)

11	STEP	SS400	4	19x19 S.Q.B
10	LADDER SUPPORT	SS400	1	10t
9	SUPPORT BAR	SS400	1	ø16 R.B
8	EYE FOR LIFE NET	SS400	1	ø16 R.B
7	BOLT/NUT/W/S.W	SUS304	8	M8x35L
6	SUPPORT	SGP	2	65A PIPE
5	PAD	SS400	2	10t
4	PAD	RUBBER	2	20t
3	LASHING SOCKET	SS400	2	[125x65x6/8t
2	SUPPORT	SGP	1	80A
1	FRAME	SS400	1	H148x100x6/9t
NO	DESCRIPTION	MAT'L	Q'TY	REMARK

Project Title.

ACCOMMODATION/PILOT SLANT LADDER

Drawing Title.

REST 11A

Dwg No.

AC-VET-RAAY-01

SCALE

SHIP NO.	4193/94
----------	---------

Approved

S.W.SHIN

DATE

Checked	S.W.SHIN
---------	----------

Designed	C.W.PARK
----------	----------



JUNG-A MARINE CO., LTD.

0. 2021.11.10

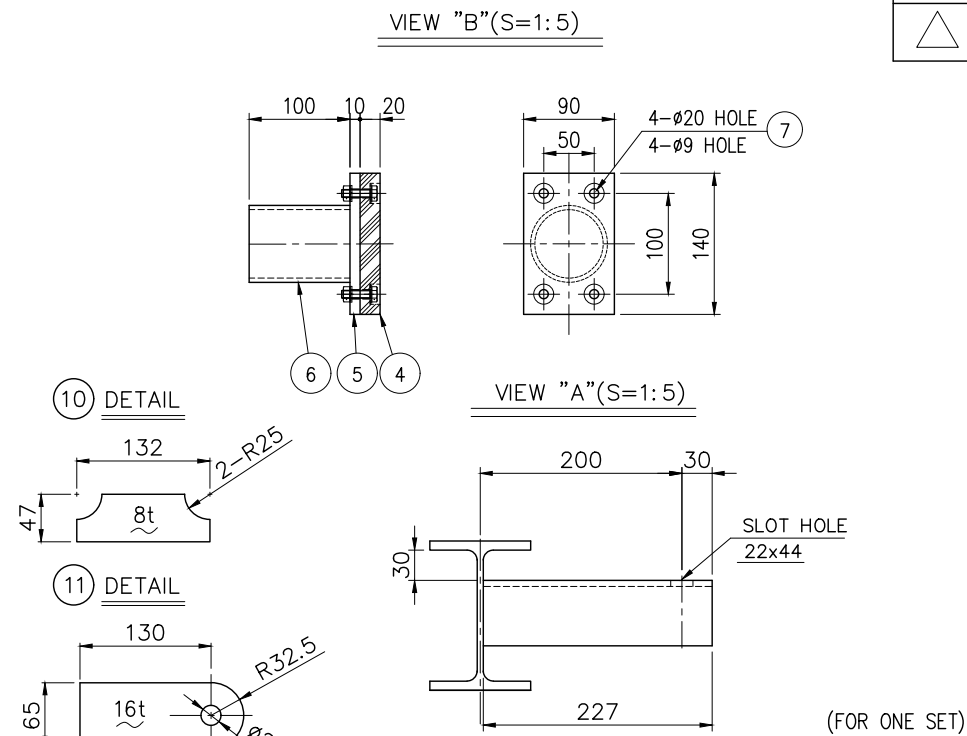
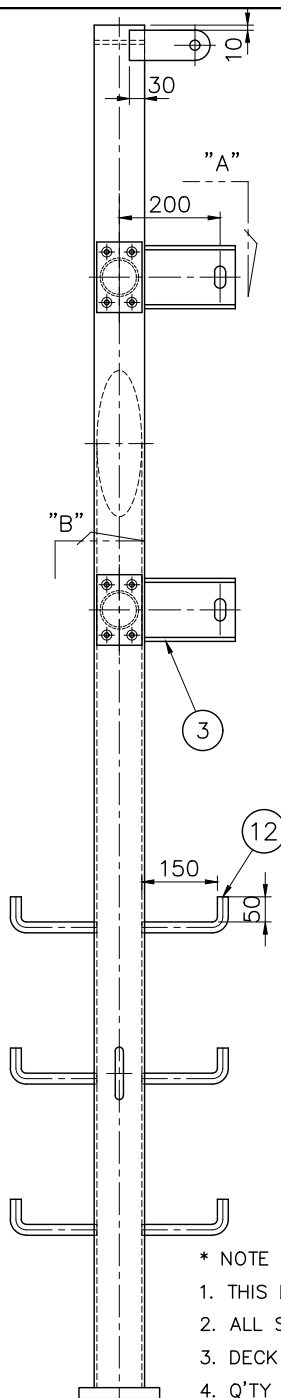
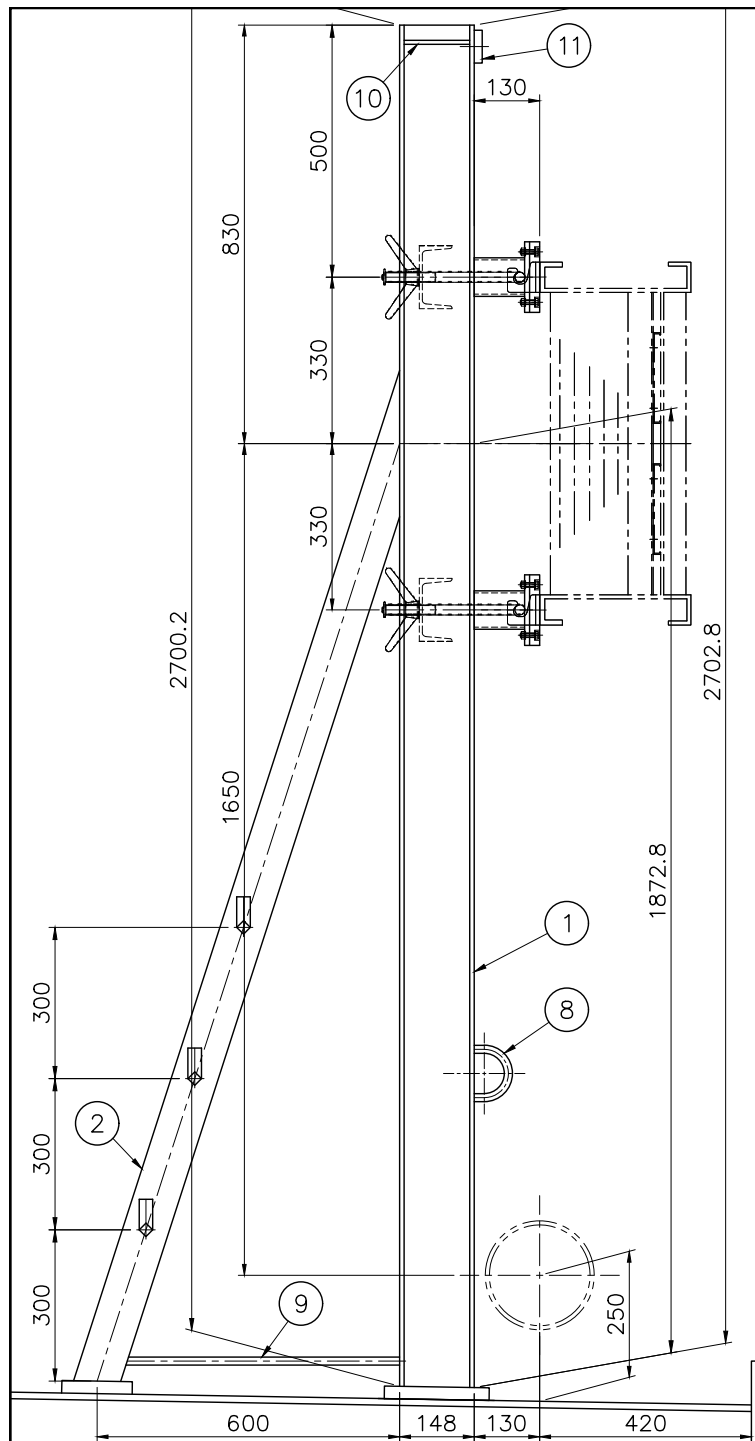
Designed	C.W.PARK
----------	----------

REV.No.



PAGE No.	
----------	--

P - 23A



NO	DESCRIPTION	MAT'L	Q'TY	REMARK
12	STEP	SS400	6	19x19 S.Q.B
11	FALL ARREST EYE	SS400	2	16t x 65 F.BAR
10	STIFFENER	SS400	2	8t
9	SUPPORT BAR	SS400	1	ø16 R.B
8	EYE FOR LIFE NET	SS400	1	ø16 R.B
7	BOLT/NUT/W.S.W	SUS304	8	M8x35L
6	SUPPORT	SGP	2	65A PIPE
5	PAD	SS400	2	10t
4	PAD	RUBBER	2	20t
3	LASHING SOCKET	SS400	2	[125x65x6/8t
2	SUPPORT	SGP	1	80A
1	FRAME	SS400	1	H148x100x6/9t

Project Title. ACCOMMODATION/PILOT SLANT LADDER			
Drawing Title. REST (11B)			
Dwg No. AC-VET-RAAY-01	SCALE 1/15	SHIP NO.	4193/94
		Approved	S.W.SHIN
JUNG-A MARINE co.,LTD.	DATE 2021.11.10	Checked	S.W.SHIN
		Designed	C.W.PARK

* NOTE

- THIS DWG. SHOWS ST'BD SIDE.
- ALL STEEL PARTS TO BE GALVANIZED.
- DECK CAMBER : 50 / 2900
- Q'TY : 2SETS/SHIP(1SET SYMMETRY)
- DOUBLING PAD : 25t (YARD SUPPLY)

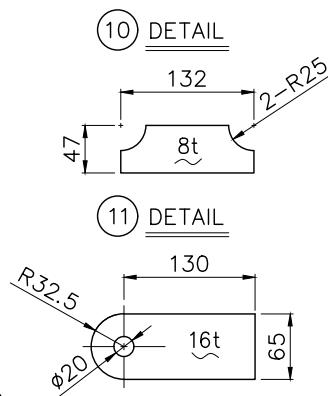
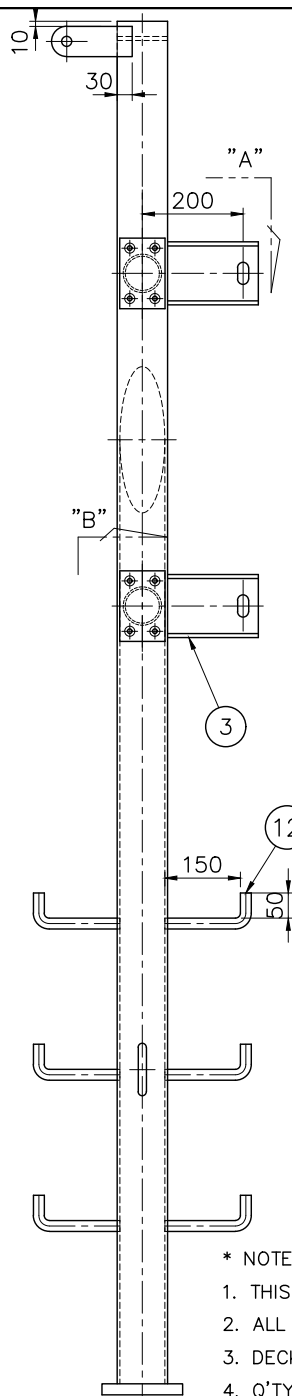
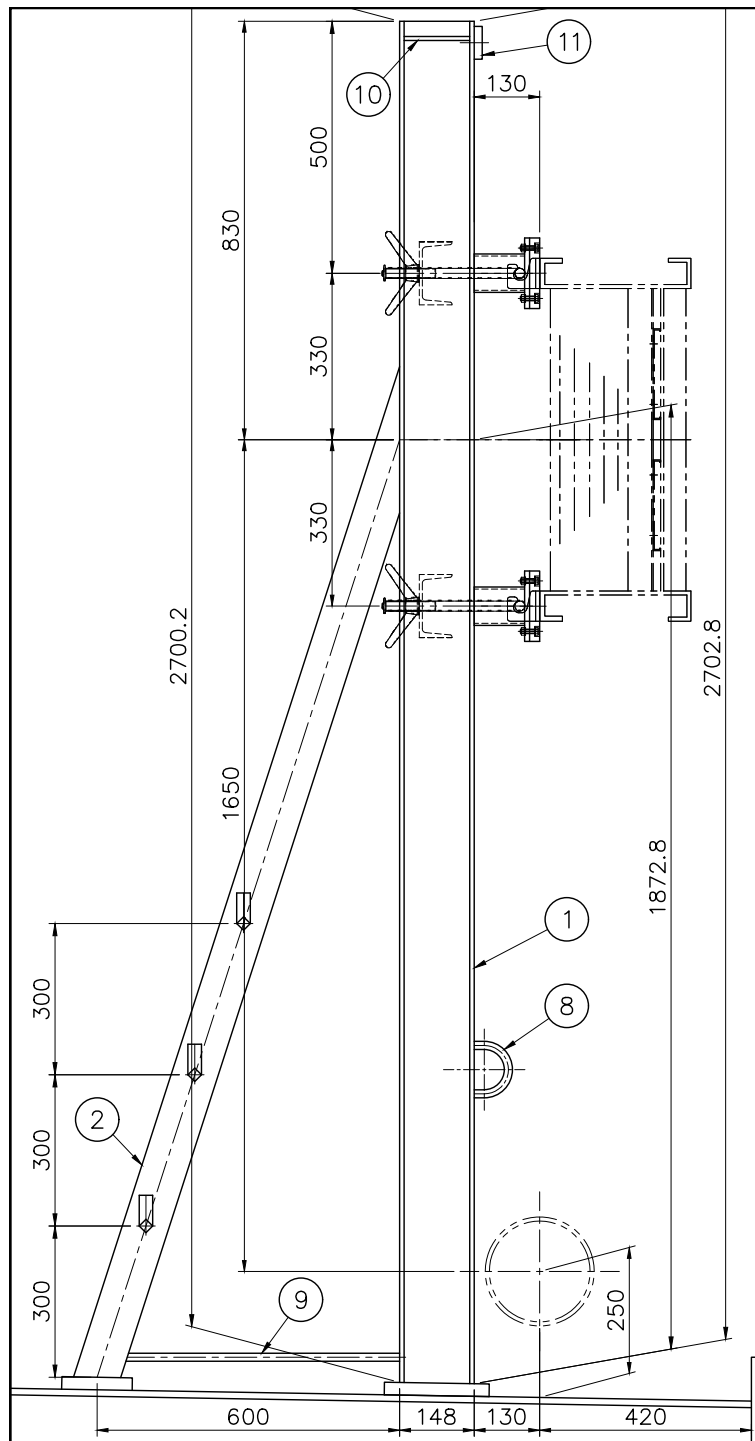
REV.No.



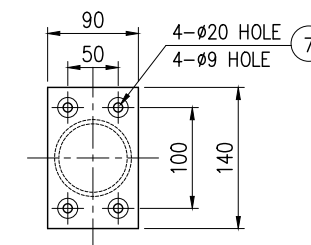
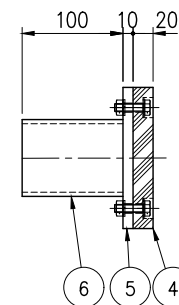
PAGE No.

P

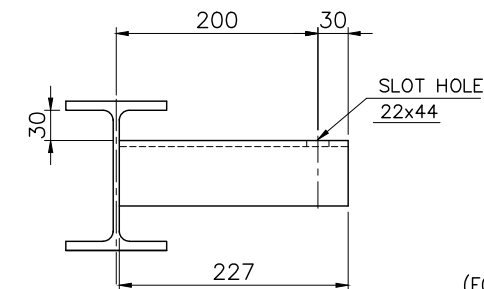
- 24



VIEW "B"(S=1:5)



VIEW "A"(S=1:5)



(FOR ONE SET)

12	STEP	SS400	6	19x19 S.Q.B
11	FALL ARREST EYE	SS400	2	16tx65 F.BAR
10	STIFFENER	SS400	2	8t
9	SUPPORT BAR	SS400	1	ø16 R.B
8	EYE FOR LIFE NET	SS400	1	ø16 R.B
7	BOLT/NUT/W.S.W	SUS304	8	M8x35L
6	SUPPORT	SGP	2	65A PIPE
5	PAD	SS400	2	10t
4	PAD	RUBBER	2	20t
3	LASHING SOCKET	SS400	2	[125x65x6/8t
2	SUPPORT	SGP	1	80A
1	FRAME	SS400	1	H148x100x6/9t
NO	DESCRIPTION	MAT'L	Q'TY	REMARK

Project Title. **ACCOMMODATION/PILOT SLANT LADDER**

Drawing Title. **REST (11C)**

Dwg No.	AC-VET-RAAY-01	SCALE	SHIP NO.	4193/94
		1/15	Approved	S.W.SHIN
 JUNG-A MARINE CO.,LTD.		DATE	Checked	S.W.SHIN
		2021.11.10	Desianed	C.W.PARK

* NOTE

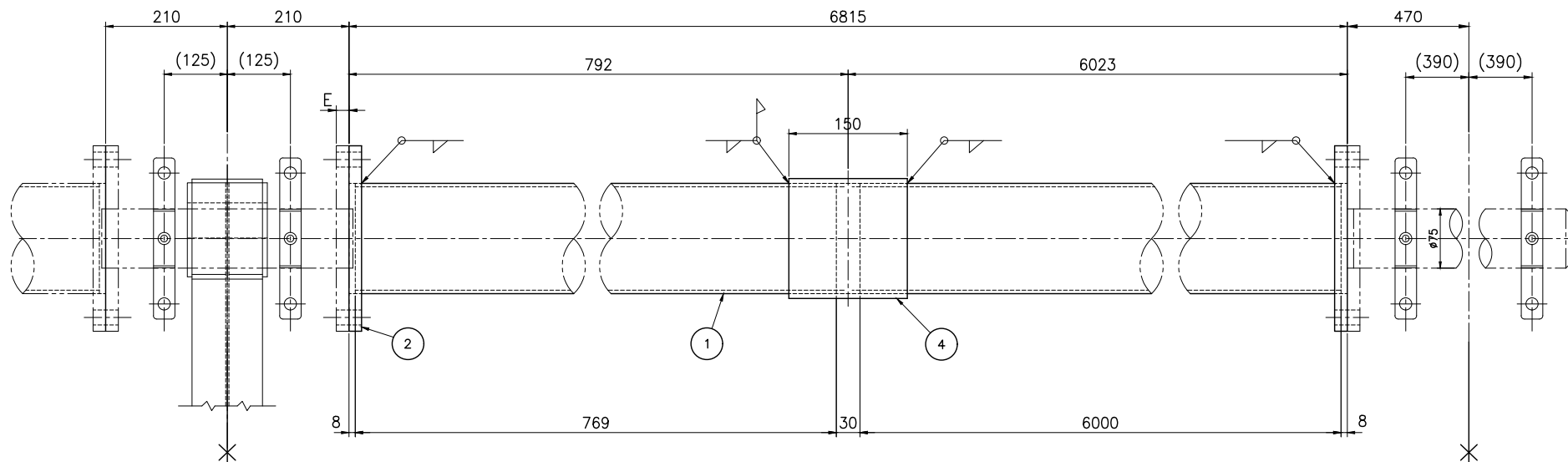
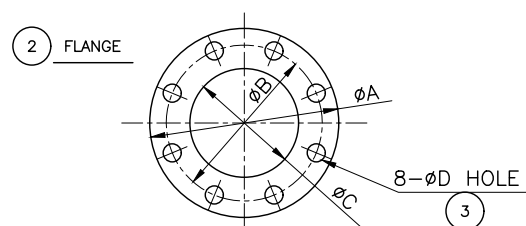
1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. DECK CAMBER : 50 / 2900
4. Q'TY : 2SETS/SHIP(1SET SYMMETRY)
5. DOUBLING PAD : 25t (YARD SUPPLY)

REV.No.



PAGE No.

P - 24A

FWD. STOWING DAVIT ∇ UPPER P/F ∇ 

* NOTE

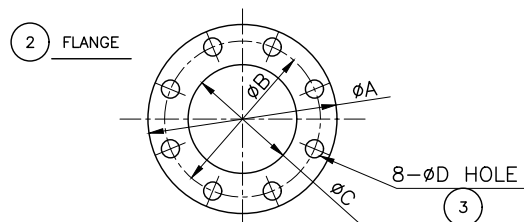
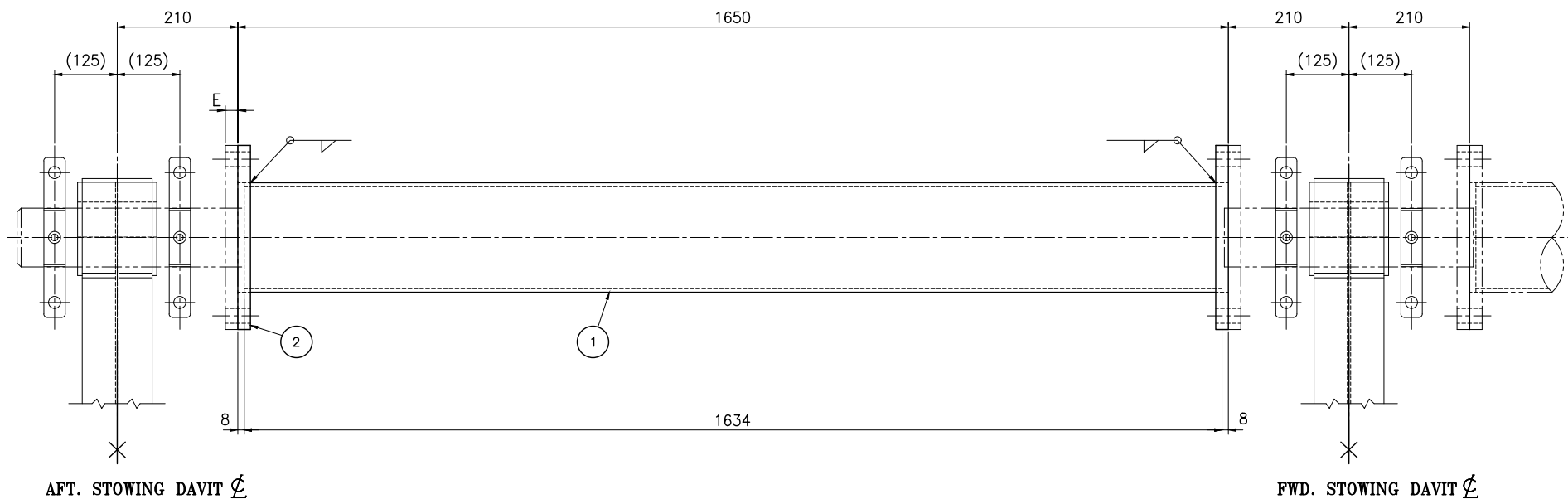
1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

(FOR ONE SET)

NO.	SPINDLE	SPINDLE O.D.	FLANGE	A	B	C	D	E	BOLT/NUT/W/S.W	MARK
1	100A	ø114.3	100A	200	165	115.4	19	16	M16x60L	
2	125A	ø139.8	125A	235	200	141.2	19	16	M16x60L	
3	150A	ø165.2	150A	265	230	166.6	19	18	M16x60L	
4	200A	ø216.5	200A	320	280	218.0	23	20	M20x70L	

NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS
4	SLEEVE	200A TYPE	SS400	1	—
3	BOLT/NUT/W/S.W	—	SUS304	16	—
2	FLANGE	5Kg/CmxFF TYPE	SS400	2	—
1	PIPE	200A	SGP	2	—

Project Title. ACCOMMODATION/PILOT SLANT LADDER					
Drawing Title. SPINDLE PIPE(FWD)					
Dwg No. 18J010-SS-VSPAY-01		SCALE N/S	SHIP NO. 4193/94		
 JUNG-A MARINE co.,LTD.		DATE 2021.11.10	Approved S.H.HONG		
			Checked S.H.HONG		
			Designed C.W.PARK		



NO.	SPINDLE	SPINDLE O.D.	FLANGE	A	B	C	D	E	BOLT/NUT/W/S.W	MARK
1	100A	Ø114.3	100A	200	165	115.4	19	16	M16x60L	
2	125A	Ø139.8	125A	235	200	141.2	19	16	M16x60L	
3	150A	Ø165.2	150A	265	230	166.6	19	18	M16x60L	
4	200A	Ø216.5	200A	320	280	218.0	23	20	M20x70L	

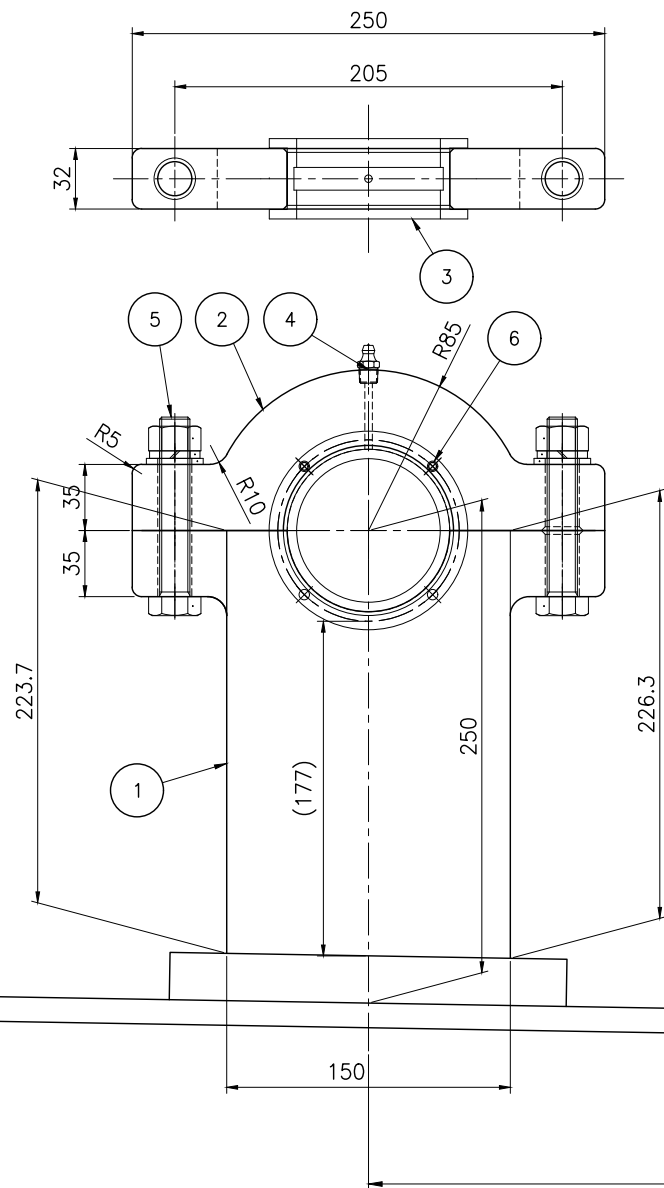
* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

(FOR ONE SET)

NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS
3	BOLT/NUT/W/S.W	-	SUS304	16	-
2	FLANGE	5Kg/CmxFF TYPE	SS400	2	-
1	PIPE	200A	SGP	2	-

Project Title. ACCOMMODATION/PILOT SLANT LADDER					
Drawing Title. SPINDLE PIPE(AFT)					
Dwg No. 18J010-SS-VSPAY-01		SCALE N/S	SHIP NO. 4193/94		
JUNG-A MARINE co.,LTD.		DATE 2021.11.10	Approved S.H.HONG		
			Checked S.H.HONG		
			Designed C.W.PARK		



* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. DECK CAMBER : 50 / 2900
4. DOUBLING PAD(25t) TO BE SUPPLIED BY YARD.
3. Q'TY : 12SETS/SHIP (6SETS SYMMETRY)

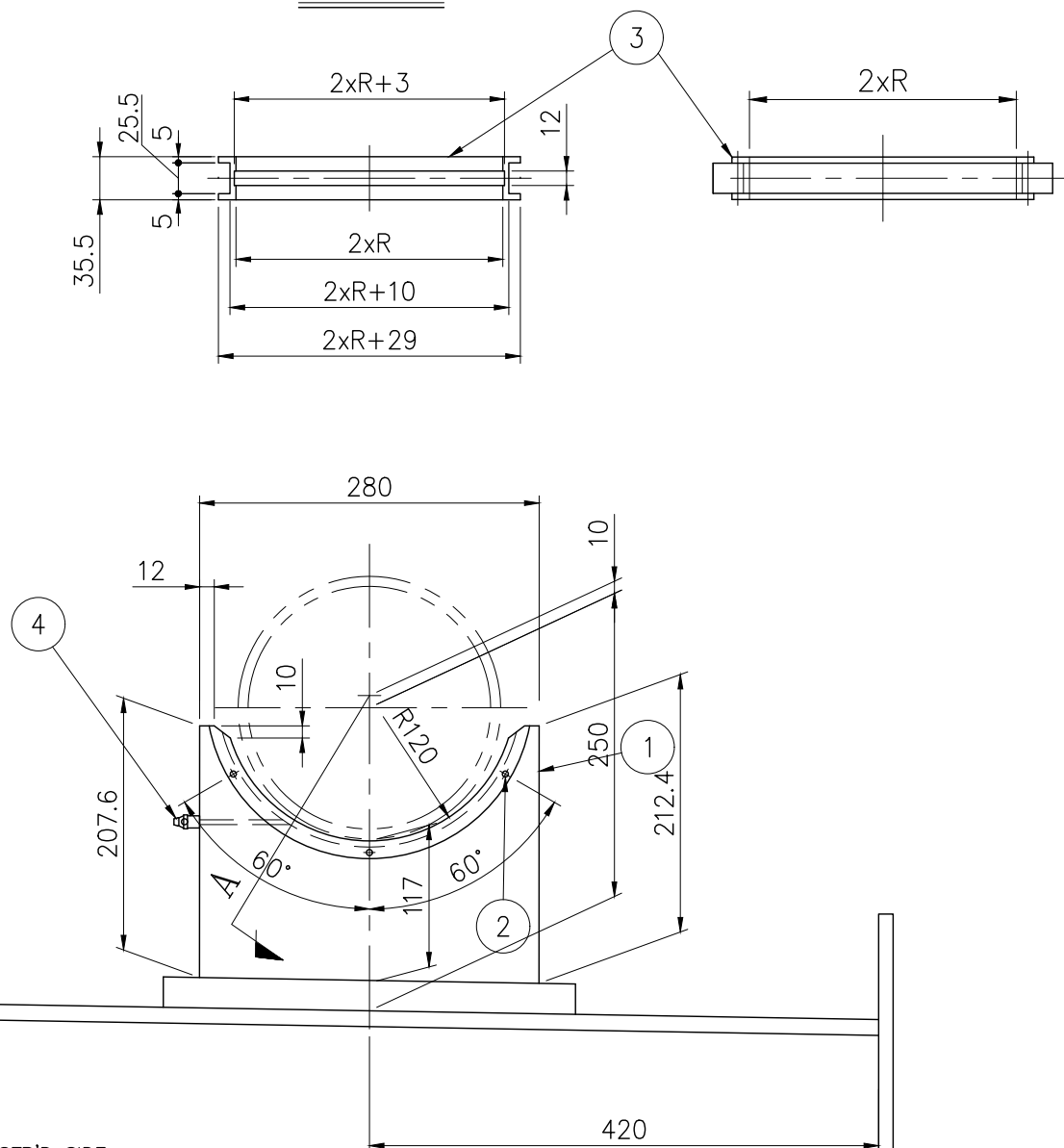
(FOR ONE SET)

6	PIN	ø5x12L	SUP	8	-
5	BOLT/N/P.W./S.W.	M16x100L	SUS304	2	-
4	GREASE NIPPLE	A-PT 1/8"	BRASS	1	-
3	BUSH	-	BRASS	2	-
2	HOLDER	PL. 32t	SS400	1	-
1	BODY	PL. 32t	SS400	1	-
NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS

Project Title. ACCOMMODATION/PILOT SLANT LADDER					
Drawing Title. SHAFT HOLDER					
Dwg No. 18J010-SS-VSHAY-01		SCALE 1/4	SHIP NO. 4193/94		
 JUNG-A MARINE co.,LTD.		DATE 2021.11.10	Approved S.H.HONG		
			Checked S.H.HONG		
			Designed C.W.PARK		



SECTION A



(FOR ONE SET)

NOTE

1. THIS DWG. SHOWS STB'D SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. DECK CAMBER : 50 / 2900
4. DOUBLING PLATE : 25t (YARD SUPPLY)
5. Q'TY : 2SETS/SHIP(1SET SYMMETRY)

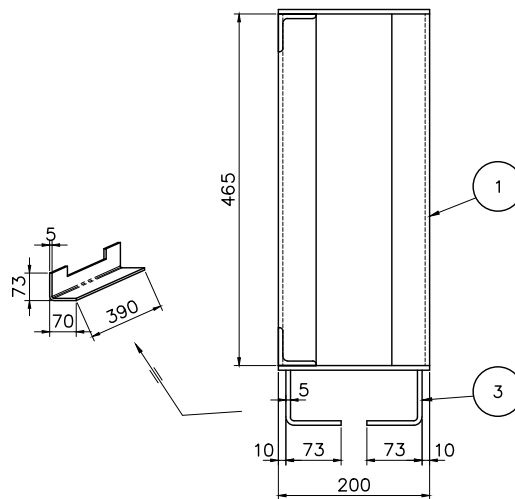
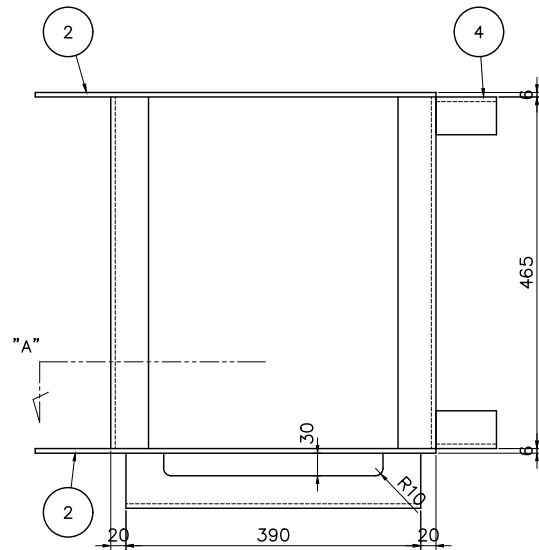
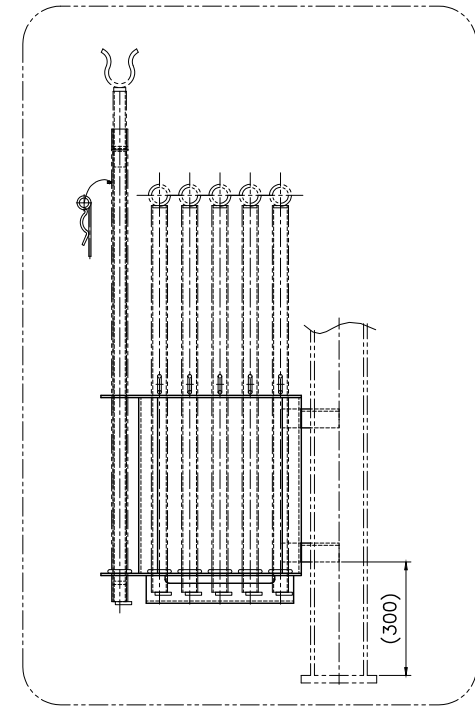
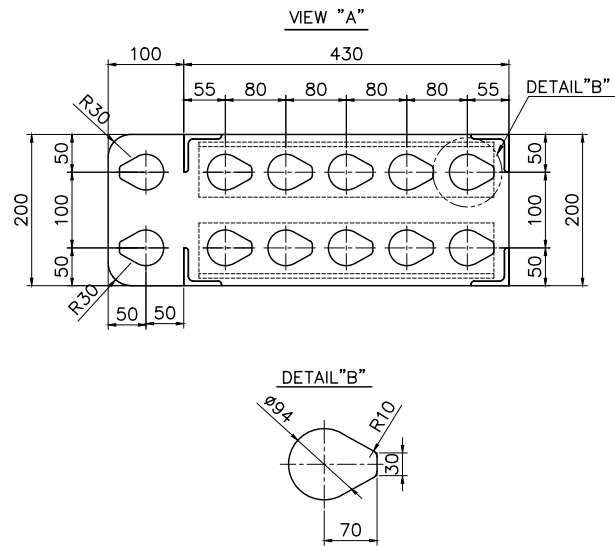
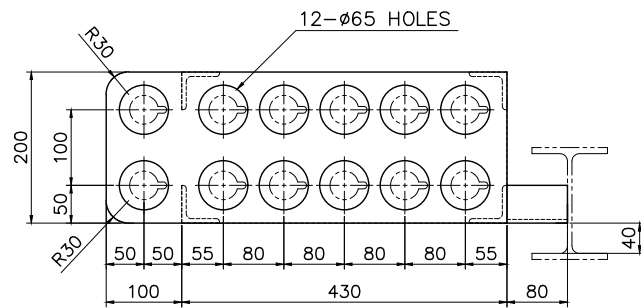
4	GREASE NIPPLE	BRASS	2	A-PT 1/8"
3	BUSH	BRASS	2	BODY
2	SPRING PIN	SUP	1	ø3x35L
1	BODY	SS400	1	25t
No.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved	S.H.HONG	Scale	1/5 A4	Dwg No.	AC-VET-SPRT-04
Checked	S.W.SHIN	Date	2021.11.10	Title	
Designed	C.W.PARK	Ship No.	4193/94		



JUNG-A MARINE CO.,LTD.

SPINDLE REST



* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. THIS ITEM SHALL BE INSTALLED ON THE FIRST REST FROM UPPER P/F.
4. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

(FOR ONE SET)

5	BRACKET	EA.50x6t	SS400	1	—
4	STOPPER	PL. 5t	SS400	1	—
3	LOWER PLATE	PL. 6t	SS400	1	—
2	UPPER PLATE	PL. 6t	SS400	1	—
1	MAIN FRAME	EA.50x6t	SS400	1	—
NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS

Project Title.

ACCOMMODATION/PILOT SLANT LADDER

Drawing Title.

HANDRAIL STANCHION STAND

Dwg No.

18J010-SS-VSDBL-01

SCALE

1/6

SHIP NO.

4193/94

Approved S.H.HONG



JUNG-A MARINE co.,LTD.

DATE

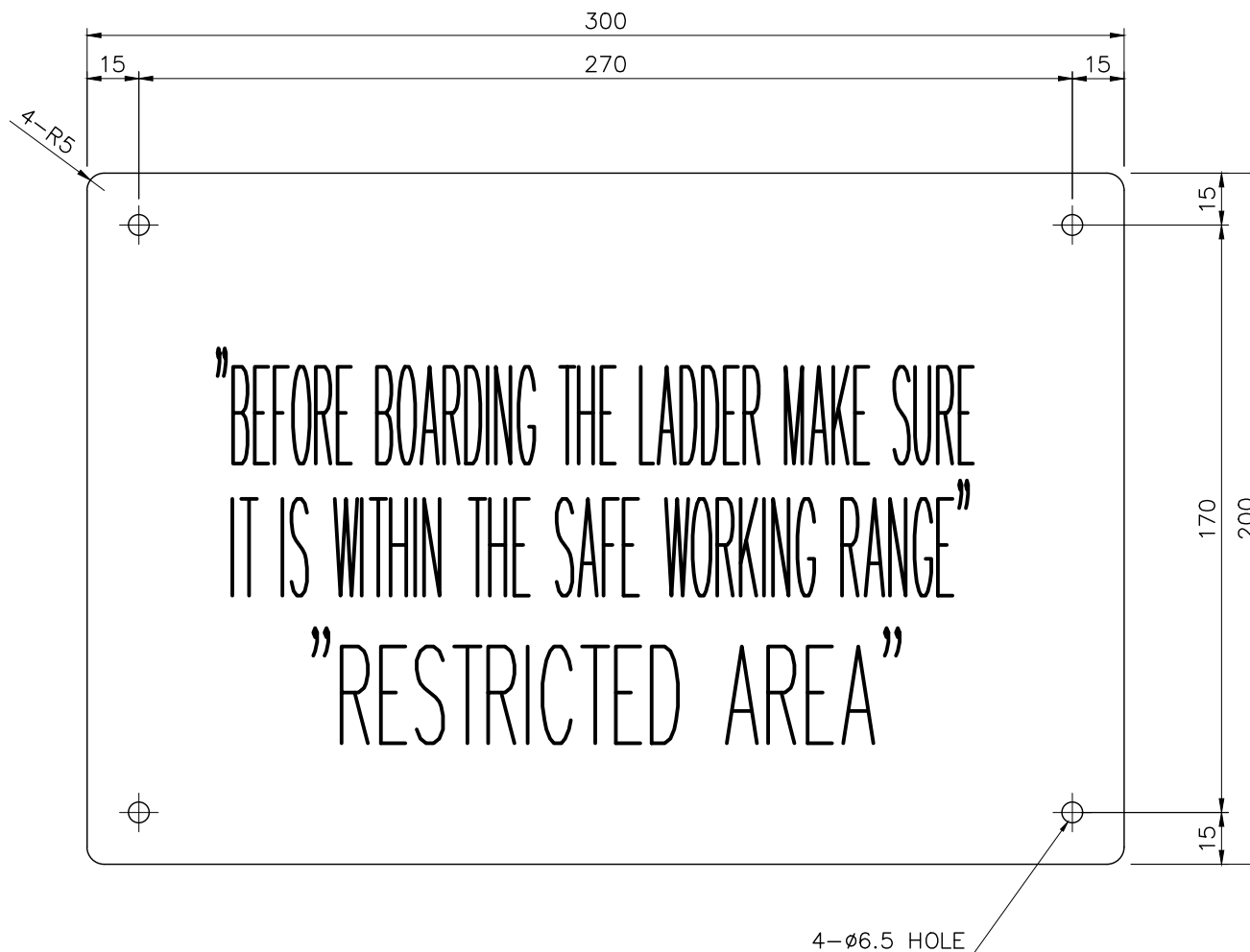
2019.01.31

Checked

S.W.SHIN

Designed

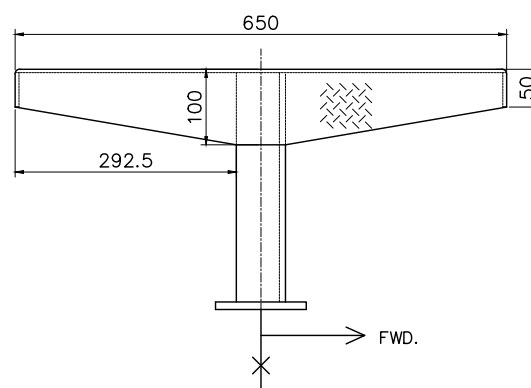
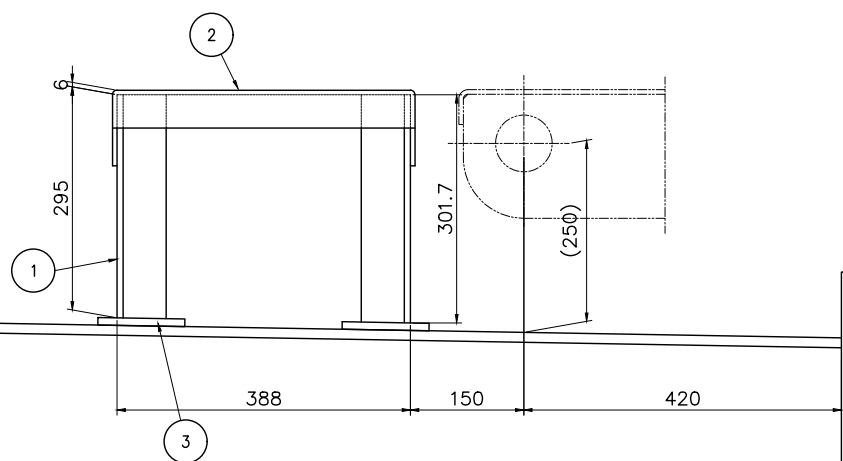
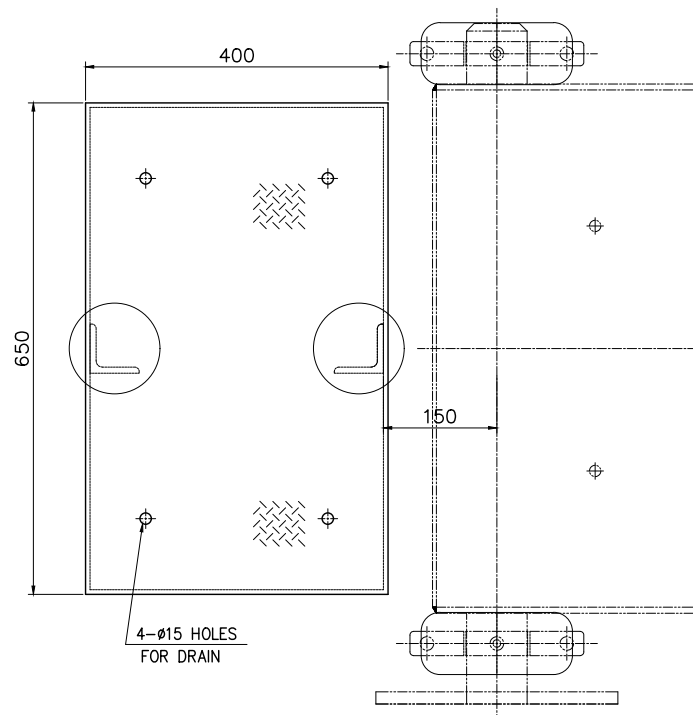
C.W.PARK



NOTE

1. Q'TY : 2 SETS/SHIP
2. MAT'L : SUS316 x 1t
3. LETTER COLOR : RED
4. LETTER DEPTH : 0.2mm
5. THIS ITEM SHOULD BE INSTALLED ON YARD ADJUST.

Project Title.		ACCOMMODATION/PILOT SLANT LADDER	
Drawing Title.		CAUTION PLATE	
Dwg No.	AC-STD-CTPL-01	SCALE	SHIP NO.
		1/2	4193/94
JUNG-A MARINE co.,LTD.	DATE	Approved	S.H.HONG
		Checked	S.W.SHIN
JUNG-A MARINE co.,LTD.	2021.11.10	Designed	C.W.PARK



* NOTE

1. THIS DWG. SHOWS ST'BD SIDE.
2. ALL STEEL PARTS TO BE GALVANIZED.
3. DECK CAMBER : 50 / 2900
4. DOUBLING PAD(10t) TO BE SUPPLIED BY MAKER.
3. Q'TY : 2SETS/SHIP (1SET SYMMETRY)

(FOR ONE SET)

NO.	DESCRIPTION	SPEC.	MAT'L	Q'TY	REMARKS
3	DOUBLING PAD	10t	SS400	2	—
2	TOP PLATE	PL. 6t(CH'KED)	SS400	1	—
1	FRAME	EA.65x8t	SS400	2	—

Project Title. ACCOMMODATION/PILOT SLANT LADDER					
Drawing Title. ACCESS PLATFORM					
Dwg No. 18J010-SS-VAPAY-01		SCALE 1/6	SHIP NO. 4193/94		
			Approved	S.H.HONG	
 JUNG-A MARINE co.,LTD.		DATE 2021.11.10	Checked	S.H.HONG	
			Designed	C.W.PARK	

LOAD TEST METHOD

PAGE No.

P - 31

SHIP No.

4193/94

TEST DATE

MESSRS.

HYUNDAI MIPO

TEST PLACE

JUNG-A MARINE

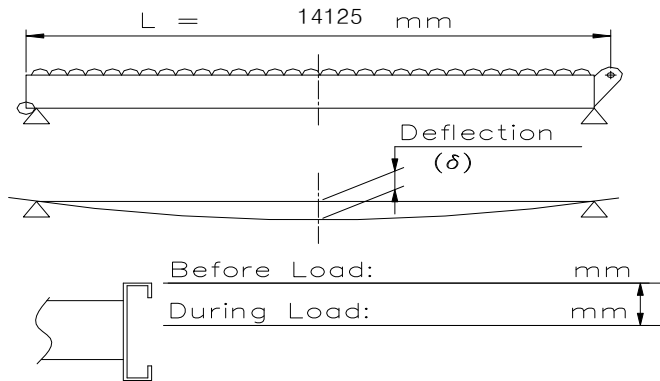
CLASS

ABS

1. Particulars

- 1) Ladder Length : 1412.5cm
- 2) Step Q'ty : 40ea/set
- 3) Load Test Weight 75kg/Step x 40Step = 3000.0 kg
- 4) The Method of Supporting : Horizontally Both End Support

2. Checking for Deflection

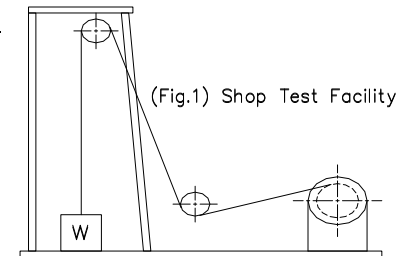


Allowable Deflection (δ)	$L/75 =$	18.8cm
Actual (δ) for St'bd		
Actual (δ) for Port		

3. Load Test for winch

1) Summary of Winch Spec.

Hoisting Load	632kg
Supporting Load	780kg
Rope speed(3Layer)	abt 17.0 m/min
Wire rope type	Ø12(6x37) wire rope
Motor type	AIR 4.5HP x 600RPM

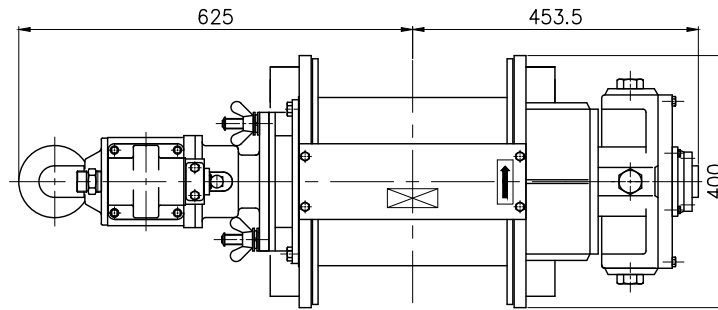
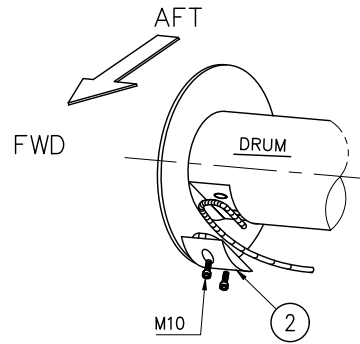


2) Rope Speed Test

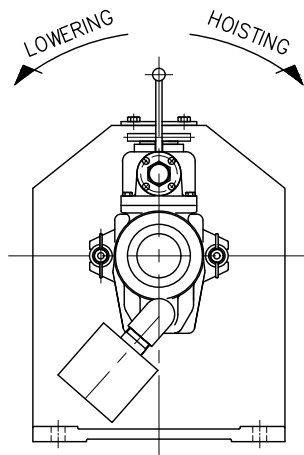
Hoisting Load	Position	Rope Speed(3rd Layer)				Manual Operation
			Height	Time	Speed	
632kg	St'bd Side	Hoisting	3m	Sec	m/min	
		Lowering	3m	Sec	m/min	
	Port Side	Hoisting	3m	Sec	m/min	
		Lowering	3m	Sec	m/min	

3) Static Load Test (Fig.1)

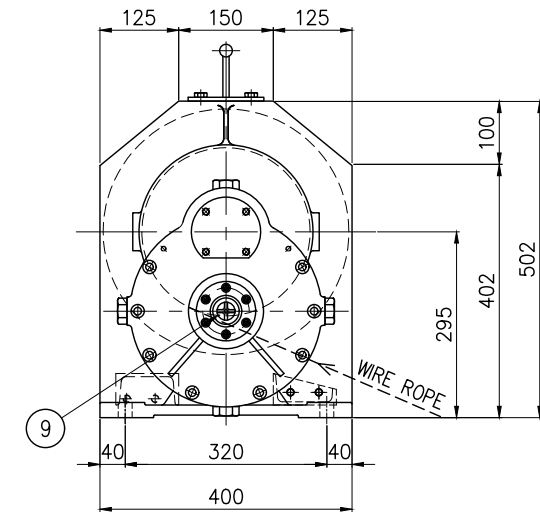
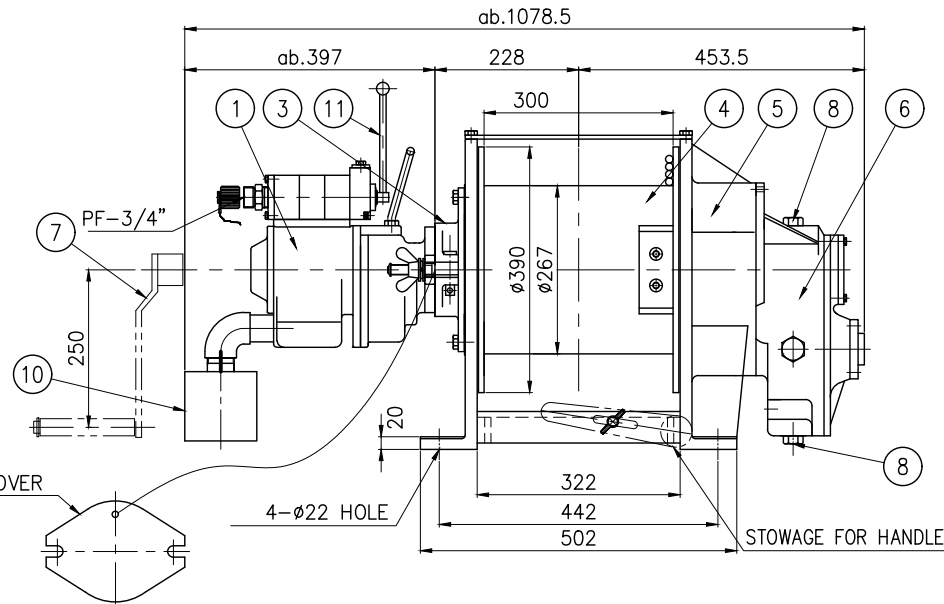
Static load =	1170kg
St'bd Side	
Port Side	



FWD.
SIDE SHELL(STB'D SIDE)
AFT.



BLIND COVER



SPECIFICATION		
HOISTING LOAD	632 KG	
SUPPORTING LOAD	780 KG	
ROPE SPEED	at. 3 Layer ab 17 m/min	
DRUM CAPACITY	ø12 x 71M (3 Layer)	
AIR PRESSURE	6 KG/CM ²	
AIR MOTOR	4.5PS x 600RPM	
WEIGHT(ABOUT)	185 KG	
OIL	QUANTITY	2.0 L
	ISO GRADE	150

Q'TY : 2SET/SHIP (HALF SYMMETRY)

GEAR RATIO : 1/36.3 (13/59x14/112)

HANDLE FORCE : 15.2 kgf (900x(27.6/2)/25x36.3x0.9)

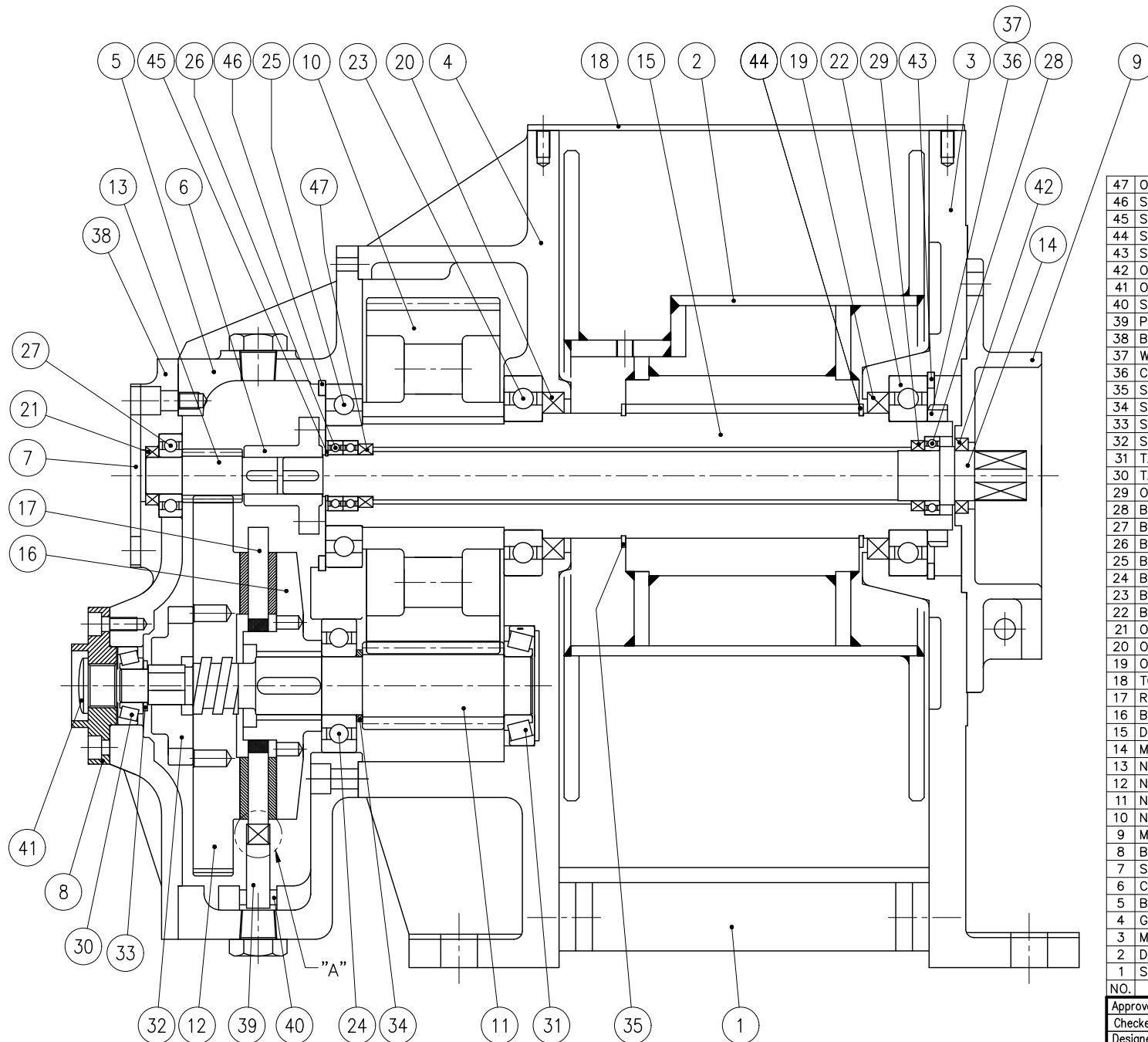
THIS DWG. SHOWS STB'D SIDE, PORT SIDE SYMMETRY.

DESIGNATION : ISO7364-P-5-R(L)-D

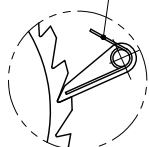
11	CONTROL LEVER	SS400	1
10	SILENCER	SS400	1
9	LEVEL GAUGE	P.P	1
8	OIL PLUG/DRAIN PLUG	BS	1/1
7	HANDLE	SS400	1

6	GEAR CASE	FCD450	1	
5	GEAR CASE	FCD450	1	
4	DRUM	STPG370	1	
3	MOTOR STAND	FCD450	1	
2	CLAMP	SS400	1	
1	AIR MOTOR	ASS'Y	1	
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Approved	S.H.HONG	Scale	1/12	A4	Dwg No.	OTW-1320-A2673004.5
Checked	S.W.SHIN	Date	2021.11.10		Title	AIR WINCH (OTW-1320-A)
Designed	C.W.PARK	Ship No.	4193/94			



47	OIL SEAL	NBR	1	25388
46	STOP RING	SK5	1	H110
45	STOP RING	SK5	1	S25
44	STOP RING	SK5	1	S75
43	STOP RING	SK5	1	H120
42	OIL SEAL	NBR	1	30458
41	OIL LEVEL GAUGE	ZnDC2	1	
40	SPRING	SWPB	2	
39	PAWL	S45C	2	
38	B/R CASE COVER	FCD200	1	
37	WASHER	S45C	1	
36	CROWN NUT	S45C	1	AN13(M65x2)
35	STOP RING	SK5	2	S75
34	SPACER	S45C	1	
33	SPACER	S45C	1	
32	STOP COLLAR	S45C	1	
31	TAPER ROLLER B/R	SUJ	1	30207K
30	TAPER ROLLER B/R	SUJ	1	30204K
29	OIL SEAL	NBR	1	30428
28	BALL BEARING	SUJ	1	6906ZZ
27	BALL BEARING	SUJ	1	62/22ZZ
26	BALL BEARING	SUJ	2	6905ZZ
25	BALL BEARING	SUJ	1	6212
24	BALL BEARING	SUJ	1	6307
23	BALL BEARING	SUJ	1	6213ZZ
22	BALL BEARING	SUJ	1	6213ZZ
21	OIL SEAL	NBR	1	22358
20	OIL SEAL	NBR	1	7510013
19	OIL SEAL	NBR	1	7510013
18	TOP PLATE	SS400	1	
17	RATCHET WHEEL	S45C	1	
16	BRAKE DISC	S45C	1	
15	DRUM SHAFT	S45C	1	
14	MAIN SHAFT	SCM440H	1	
13	NO.4 PINION GEAR	SCM440H	1	2Mx14T
12	NO.3 GEAR	SCM440H	1	2Mx112T
11	NO.2 PINION GEAR	SCM440H	1	3.5Mx13T
10	NO.1 GEAR	SCM440H	1	3.5Mx59T
9	MOTOR STAND	FCD200	1	
8	BEARING COVER	SS400	1	
7	SHAFT END COVER	SS400	1	
6	COUPLING	SCM440H	1	
5	BRAKE CASE	FCD450	1	
4	GEAR CASE	FCD450	1	
3	MOTOR FRAME	FCD450	1	
2	DRUM	SS400 STPG370	1	
1	STAY	SS400	1	

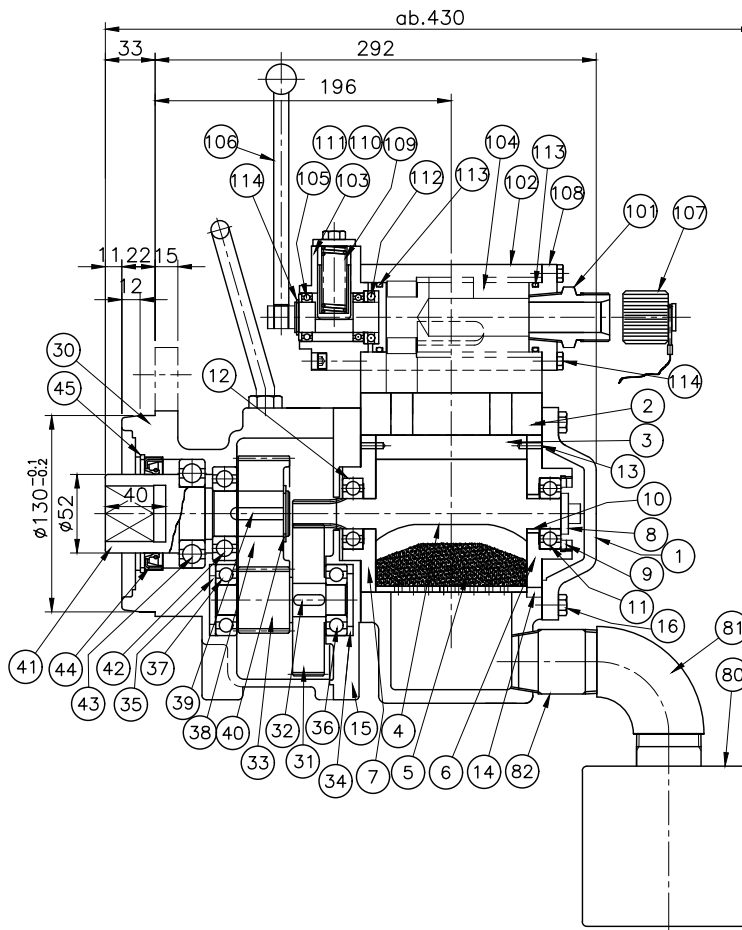
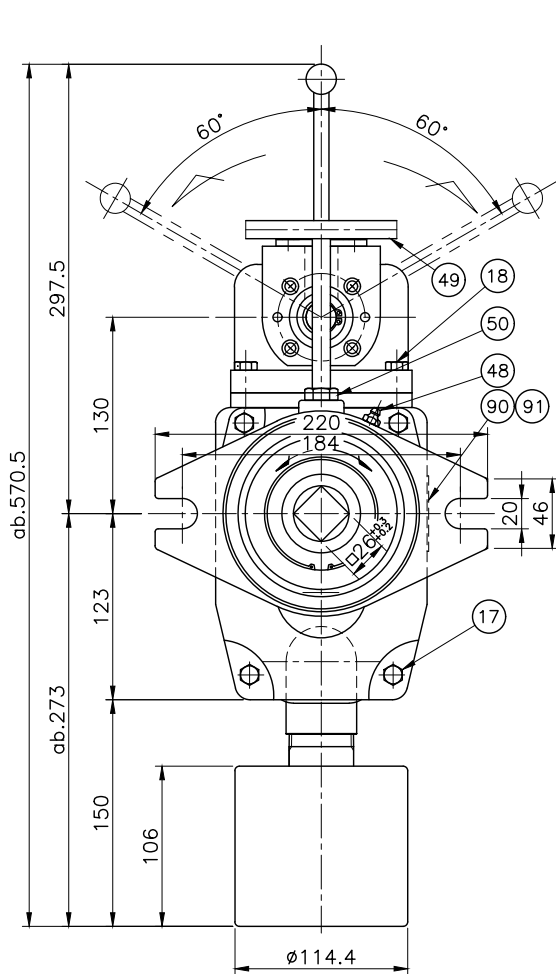


DETAIL "A"

NO.	DESCRIPTION	MAT'L	Q'TY	REMARK
Approved	S.H.HONG	Scale	N/S A4	Dwg No. OTW-1320-ASD(36.3)
Checked	S.W.SHIN	Date	2021.11.10	Title
Designed	C.W.PARK	Ship No.	4193/94	
				WINCH SECTION (OTW-1320-ASD)



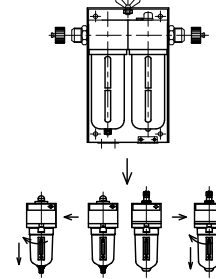
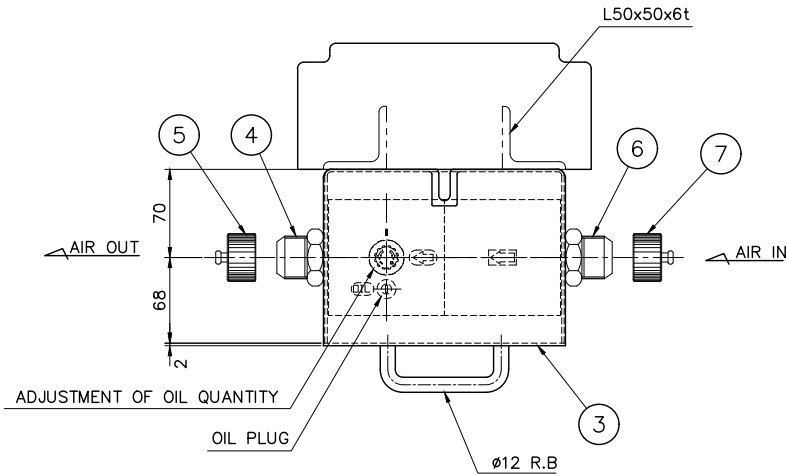
JUNG-A MARINE co.,ltd.



SPECIFICATION

T Y P E	HORSE POWER (MAX)	MAX HORSE TORQUE	SPEED	TORQUE		GEAR RATIO	WEIGHT	AIR CONSUMPTION (FREE AIR)	HOSE DIA	AIR PRESSURE
	PS	Kgf.m	rpm	MAX.	STARTING					
				Kgf.m	Kgf.m		Kg	m³/min	IN	Kg/cm²
OTM-4.5P-200DA	4.5	17.0	190	34.0	27.2	1/19.5	39	4.1	3/4"	6
OTM-4.5P-300DA	4.5	10.7	300	21.4	17.1	1/12.5				
OTM-4.5P-450DA	4.5	7.2	450	14.4	11.5	1/8.3				
OTM-4.5P-600DA	4.5	5.5	600	11.0	8.8	1/6.4				

(FOR ONE SET)				
114	STOP RING	SUP	1	S20
113	"O"-RING	NBR	2	P40
112	THRUST BEARING	SUJ	1	#51104
111	SPRING COVER	SS400	1	
110	SPRING RECEIVE	SUS	1	
109	SPRING	SWO-A	1	
108	VALVE INLET	SS400	1	
107	INLET CAP & WIRE	BsBm& SUS304	1	
106	VALVE LEVER	S45C	1	
105	BALL BEARING	SUJ	2	#6804ZZ
104	VALVE&SHAFT	S45C	1	
103	VALVE COVER	FC200	1	
102	VALVE BODY	FC200	1	
101	INLET NIPPLE	BsBm	1	
91	DRIVE SCREW	SUS304	4	
90	NAME PLATE	SUS304	1	
82	PIPE	SGP	1	32A x 50L
81	ELBOW	FCMB	1	1-1/4B
80	SILENCER	SGP	1	100A
50	HEX. NUT	SUS304	1	M12
49	HANDLE	SS400	1	
48	GREASE NIPPLE	BS	1	A-PT1/8"
45	STOP RING	SUP	1	H75
44	OIL SEAL	NBR	1	SB527512
43	BALL BEARING	SUJ	1	#6207Z
42	BALL BEARING	SUJ	1	#6206Z
41	MAIN SHAFT	SCM21	1	
40	STOP RING	SUP	1	S30
39	KEY	S45C	2	7 x 7
38	REDUCTION GEAR NO.3	SCM21	1	
37	BALL BEARING	SUJ	1	#6204Z
36	BALL BEARING	SUJ	1	#6204Z
35	BEARING EXTRACT PLATE	SS400	1	
34	BEARING EXTRACT PLATE	SS400	1	
33	REDUCTION GEAR NO.2	SCM21	1	
32	KEY	S45C	1	7 x 7
31	REDUCTION GEAR NO.1	SCM21	1	
30	GEAR CASING	FC200	1	
18	HEX. HEAD BOLT/WASHER	SUS304	4	M8 x 25L
17	HEX. HEAD BOLT/WASHER	SUS304	6	M8 x 25L
16	HEX. HEAD BOLT/WASHER	SUS304	4	M8 x 25L
15	ADAPTER PLATE	FC200	1	
14	DOWEL PIN	S45C	1	6 x 8
13	DOWEL PIN	S45C	2	7 x 12
12	BALL BEARING	SUJ	1	#6205ZZ
11	BALL BEARING	SUJ	1	#6204ZZ
10	SPACER	S45C	1	
9	NUT FOR BEARING	SS400	1	
8	END PLATE/BOLT	SS400 SCM440	1	M6 x 30L
7	FRONT END PLATE	FC200	1	
6	REAR END PLATE	FC200	1	
5	VANE	BAKELITE	5	
4	ROTOR	SCM21	1	
3	CYLINDER	FC200	1	
2	MOTOR HOUSING	FCD450	1	
1	MOTOR HOUSING COVER	FCD450	1	
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK
Approved	S.H.HONG	Scale	1/3	1/3
Checked	S.W.SHIN	Date	2021.11.10	1/3
Designed	C.W.PARK	Ship No.	4193/94	1/3
JUNG-A MARINE CO.,LTD.				AIR MOTOR (AUTO RETURN TYPE)



REMARKS

1. KEEP OIL LEVEL 80 % IN OILER.
2. ON ACTING, DRIPPING OIL HAS TO BE CONTROLLED ON 1~2 DRIPS EVERY SECOND.
3. SOMETIMES TAKE OFF THE DRAINING BY DRAIN VALVE OF FILTER
4. IF THE MOTOR POWER COMING SLOW DOWN. CLEAN THE ELEMENT IN FILTER BOTTLE BY USING NEUTRAL DETERGENT

MAKER : PARKER

FILTER : P3NFA26

OILER : P3NLA26

NOTE

1. THIS DWG. SHOWS ST'BD SIDE, PORT SIDE IS SYMMETRY.
2. THIS ITEM SHOULD BE INSTALLED ON WINCH SEAT.
(REF. ARR'T DWG P-5)
3. Q'TY: 2SETS/SHIP(1SET SYMM)

(FOR ONE SET)

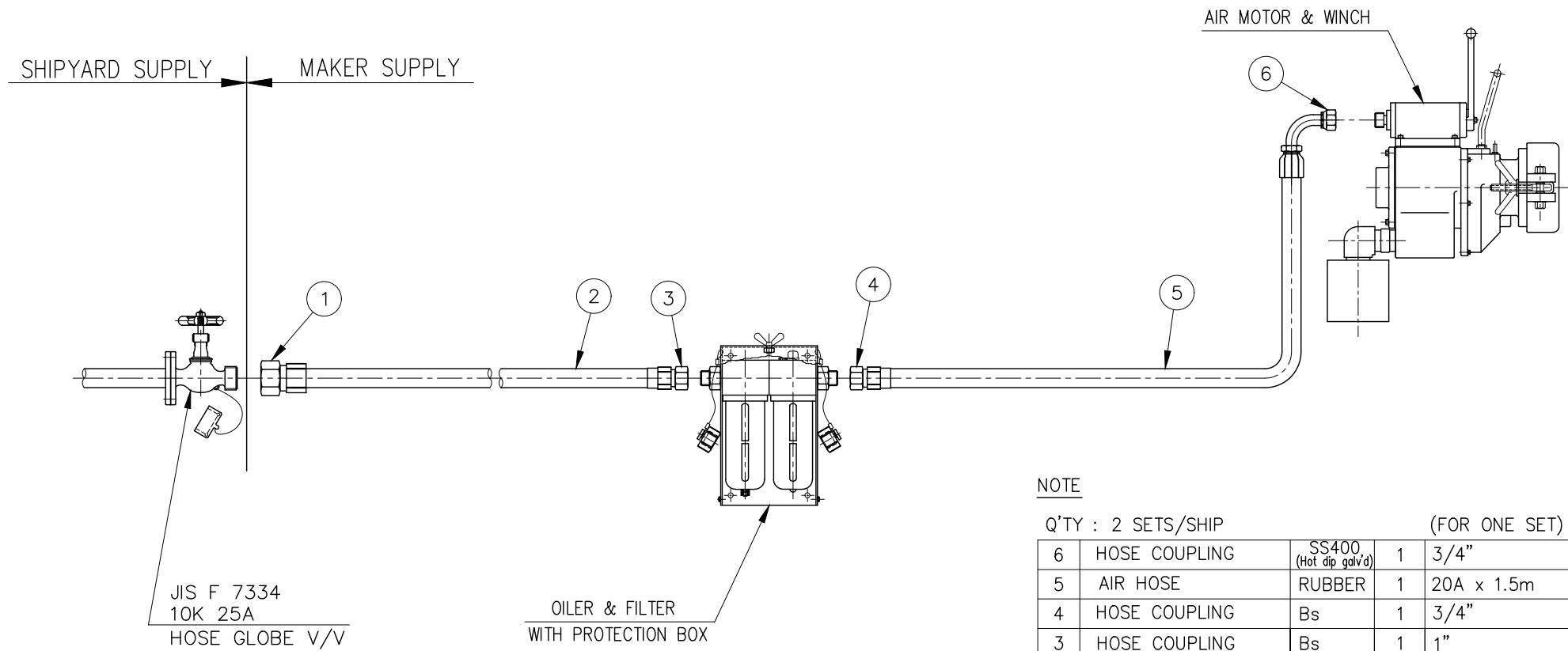
8	B/F BOLT	SUS304	2	M8x20L
7	CAP	BS	1	PF 1"
6	HOSE NIPPLE	BS	1	PF 1"
5	CAP	BS	1	PF 3/4"
4	HOSE NIPPLE	BS	1	PF 3/4"
3	CASE	SUS304	1	PLATE-2T
2	OILER	PUR	1	3/4"
1	FILTER	PUR	1	3/4"

NO.	DESCRIPTION	MAT'L	Q'TY	REMARK
Approved	S.H.HONG	Scale	1/6-A4	Dwg No. AC-STD-OIFI-01
Checked	S.W.SHIN	Date	2021.11.10	Title
Designed	C.W.PARK	Ship No.	4193/94	



JUNG-A MARINE CO., LTD.

OILER & FILTER



NOTE

Q'TY : 2 SETS/SHIP

(FOR ONE SET)

6	HOSE COUPLING	SS400 (Hot dip galvd)	1	3/4"
5	AIR HOSE	RUBBER	1	20A x 1.5m
4	HOSE COUPLING	Bs	1	3/4"
3	HOSE COUPLING	Bs	1	1"
2	AIR HOSE	RUBBER	1	25A x 4.5m
1	VALVE COUPLING	Bs	1	M42 x P2
NO.	DESCRIPTION	MAT'L	Q'TY	REMARK

Project Title.

ACCOMMODATION/PILOT SLANT LADDER

Drawing Title.

PIPING PLAN

Dwg No.

AC-VET-PIPL-01

SCALE

N/S

SHIP NO.

4193/94

Approved

S.H.HONG

DATE

2021.11.10

Checked

S.W.SHIN

Designed

C.W.PARK



JUNG-A MARINE

co.,LTD.

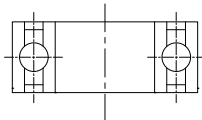
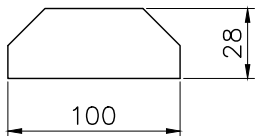
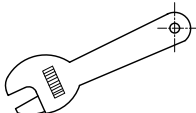
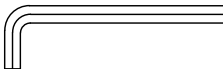
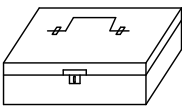
LUBRICATION OIL CHART

NO.	REVISION	DATE	CHECKED
	UPDATE NEW-DATA	2016.01.21.	K.W.KIM

EQUIP'T/MODEL/ MAKER/SET	OIL BRAND	ISO	OIL Q'TY /SET	APPLICATION POINT	REMARK(REFILL CHANGE INTERVAL)	PROVIDE INITIAL FILL'G YES/NO
1) WINCH	1) 1. OMALA S2 G220 -----SHELL 2. SPARTAN EP 220-----ESSO 3. MOBIL SHC 630-----MOBIL MOBILGEAR 600 XP 220-----MOBIL 4. MEROPA 220-----CHEVRON 5. ENERGOL GR-XP 220-----BP 6. ALPHA SP 220-----CASTROL 7. SUPER GEAR EP 220-----S.K 8. EPONA SA 220-----TOTAL EPONA Z 220-----TOTAL 9. GULFSEA GEAR OIL 220-----GULF 10. RENOLIN CLP 220-----LUKOIL	ISO VG 220	6 LTR	1) GEAR BOX	1),4) INITIAL FILLING JUST BEFORE THE DELIVERY. AFTER THE FIRST OIL FILLING REPLACE OIL EVERY SIX MONTHS.	1)-4) NO
2) FALL	2) 1. GADUS S2 A320 2 -----SHELL 2. SURRET FLUID 4K-----ESSO 3. MOBILARMA 798-----MOBIL 4. TEXCLAD 2-----CHEVRON 5. ENERGREASE MP-MG2-----BP 6. SPHEEROL SX2-----CASTROL BIOTEC MP-----CASTROL 7. GEAR COMPOUND NO.1-----S.K 8. CERAN AD PLUS-----TOTAL 9. GULFSEA HYPERBAR SUPER CS-----GULF 10. RENOLIT CXI 2-----LUKOIL	-	1.0 LTR	2) WIRE ROPE	2),3) OCCASIONALLY	
3) AIR MOTOR	3) 1. GADUS S2 V220 2-----SHELL 2. CAZAR K2-----ESSO 3. MOBILITH SHC 460-----MOBIL MOBILGREASE XHP 222-----MOBIL 4. MULTIFAK EP 2-----CHEVRON 5. ENERGREASE MP-MG2-----BP 6. SPHEEROL SX2-----CASTROL 7. CROWN GREASE EP 2-----S.K CROWN GREASE NO. 2 8. CERAN XM220-----TOTAL 9. GULFSEA HYPERBAR CS2--GULF 10. RENOLIT CXI 2-----LUKOIL	-	0.2 LTR	3) BEARING		
4) OILER SET	4) 1. TURBO OIL T32-----SHELL TELLUS S2 V 32-----SHELL 2. UNIVIS N 32-----ESSO 3. DTE 10 EXCEL 32-----MOBIL 4. RANDO HDZ 32 -----CHEVRON 5. ENERGOL SHF-HV 32 -----BP 6. HYSPIN AWH-M32-----CASTROL 7. ZIC SUPERVIS X 32-----S.K ZIC SUPERVIS AW 32-----S.K 8. VISGA 32-----TOTAL 9. GULFSEA HYDRAULIC HVI PLUS 32-GULF 10. RENOLIN B32 HVI-----LUKOIL 11. TURBINE OIL 32 -----EPPCO	ISO VG 32	0.7 LTR	4) OILER		

TITLE	SPARE PARTS & TOOL LIST (OTW-1320 / 4.5HP)	PAGE NO.	P - 38
		SHIP NO.	4193/94

(Q'TY : 1 SET / SHIP)

NO.	NAME	SKETCH	MAT'L	SUPPLY PER SHIP	REMARKS
				SPARE	
1	BALL BEARING (#6204ZZ)		SUJ	1	FOR AIR MOTOR (4.5HP)
2	BALL BEARING (#6205ZZ)			1	FOR AIR MOTOR (4.5HP)
3	VANE		BAKELITE	2	FOR AIR MOTOR (4.5HP)
4	ADJUSTABLE SPANNER (8 INCH)		FC	1	FOR WINCH
5	HEXAGONAL BAR (M10) (FOR BOLT M12)		SCM3	1	FOR WINCH
6	SPARE BOX	 PAINTING : 7.5BG 7/2	SS400	1	FOR SPARE





STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

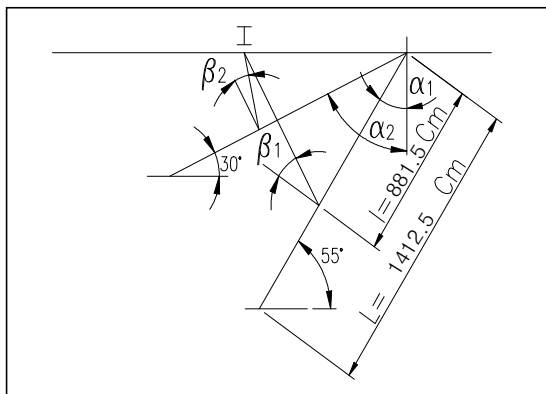
1. CONSTRUCTION

- 1) LADDER ----- [300 x 8/8t
- 2) STOWING DAVIT----- H194x150x6/9t
- 3) UPPER PLATFORM ----- STEEL PLATE WELDING

2. PRINCIPAL DIMENSION

- 1) MAIN MATERIAL OF LADDER ----- AL-ALLOY
- 2) LADDER LENGTH ----- 1412.5 Cm
- 3) LADDER WEIGHT ----- $W_L = 895$ Kg
- 4) STOWING DAVIT WEIGHT ----- $W_D = 325$ Kg
- 5) UPPER PLATFORM WEIGHT ----- $W_U = 350$ Kg
- 6) PERSON LOAD ----- $W_P = \frac{75 \text{ Kg/ STEP} \times 40}{3000} \text{ Kg}$ EA

3. EQUIPMENT CONDITION OF LADDER



$$\begin{aligned} \alpha_1 &= 35 & \sin \alpha_1 &= 0.57358 \\ \alpha_2 &= 60 & \sin \alpha_2 &= 0.86602 \\ \beta_1 &= 27.5 & \cos \beta_1 &= 0.88701 \\ \beta_2 &= 15 & \cos \beta_2 &= 0.96593 \\ l &= 881.5 \text{ Cm} \end{aligned}$$

$$L = 1412.5 \text{ Cm}$$

4. BASIC LENGTH

- 1) l_1 = FROM SPINDLE CENTER TO THE GRAVITY CENTER OF W_L = 165 Cm
- 2) l_2 = FROM SPINDLE CENTER TO THE GRAVITY CENTER OF W_D = 108.3 Cm
- 3) l_3 = FROM SPINDLE CENTER TO THE GRAVITY CENTER OF W_U = 98.5 Cm



STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

5. LADDER LOAD

1) LADDER

$$W_t = W_L + W_P = 3895.0 \text{ Kg}$$

2) LADDER LOAD IN CASE OF LIGHT BALLAST WATER LINE ARRIVAL CONDITION

$$P_{U1} = \frac{W_t \times L \times \sin \alpha_1}{2 \times l \times \cos \beta_1} = \frac{3895 \times 1413 \times 0.5736}{2 \times 882 \times 0.887} = 2017.9 \text{ Kg}$$

3) LADDER LOAD IN CASE OF SOME WATER LINE CONDITION

$$P_{U2} = \frac{W_t \times L \times \sin \alpha_2}{2 \times l \times \cos \beta_2} = \frac{3895 \times 1413 \times 0.866}{2 \times 882 \times 0.9659} = 2797.9 \text{ Kg}$$

4) MAX. LADDER LOAD IN CASE OF HORIZONTAL CONDITION

$$P_{U3} = \frac{W_t \times L}{2 \times l} = 3121 \text{ Kg (STOWING DAVIT POINT)}$$

$$P_{U4} = W_t - P_{U3} = 774 \text{ Kg (UPPER PIN POINT)}$$

5) EXAMINE OF WIRE ROPE : $\Phi 12(6 \times 37)$ -- According to "iso2408" Annex C Table C.4

$$(1) \text{ BREAKING LOAD OF WIRE ROPE (} \sigma_a \text{) : } 7673 \text{ Kg}$$

(2) COMBINATION LOAD OF WIRE ROPE

$$\sigma_0 = \frac{P_{U3}}{4} = 780 \text{ Kg}$$

(3) SAFETY FACTOR

$$S.F = \frac{\sigma_a}{\sigma_0} = 9.8 > 5 \text{ ----- O.K}$$



STRENGTH CALCULATION

Date : 2021-11-10

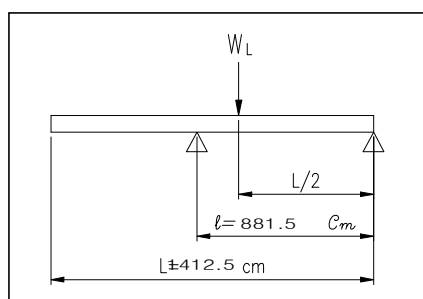
Hull.No : 4193/94

6. LOAD ON WINCH DRUM

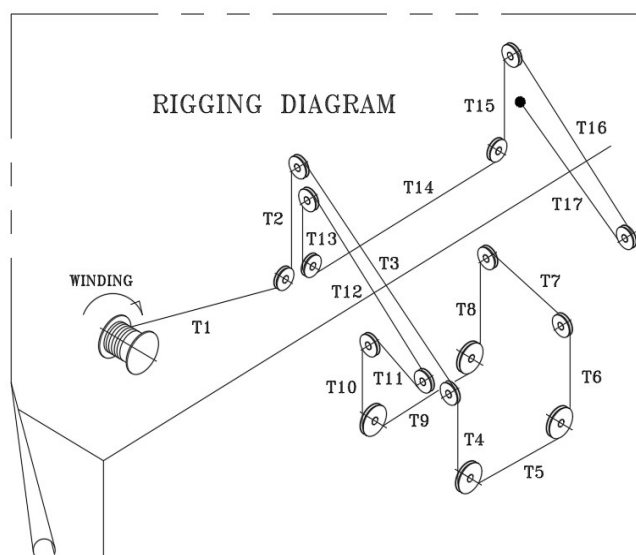
1) SUPPORTING LOAD (T_{r1})

$$T_{r1} = \frac{P_{U3}}{4} = 780 \text{ Kg}$$

2) MAX. HOISTING LOAD (WHEN HORIZONTAL CONDITION) (T_{r2})



$$P_U = \frac{W_L \times L}{2 \times l} = 717 \text{ Kg}$$



μ : FRICTION LOSS OF SHEAVE (BUSH BEARING) = 1.06 BEARING SHEAVE = 1.02

n_1 : NUMBER OF SHEAVE ROTATED ON LADDER AND DAVIT =

7

n_2 : NUMBER OF SHEAVE ROTATED ON POST AND DECK =

2

$$T_{r2} = T_4 = \frac{P_U}{4} \times \mu^{n_1} \times \mu^{n_2} = 179.3 \times 1.149 \times 1.124 = 231 \text{ Kg}$$



STRENGTH CALCULATION

Date : 2021-11-10

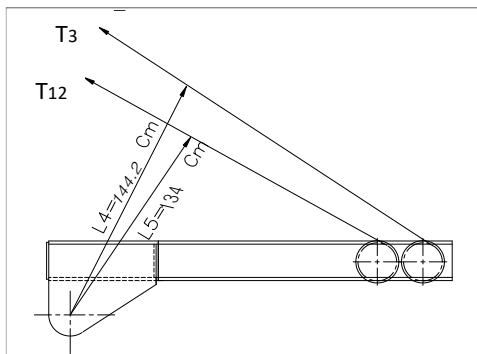
Hull.No : 4193/94

3) STOWING LOAD (Tr3)

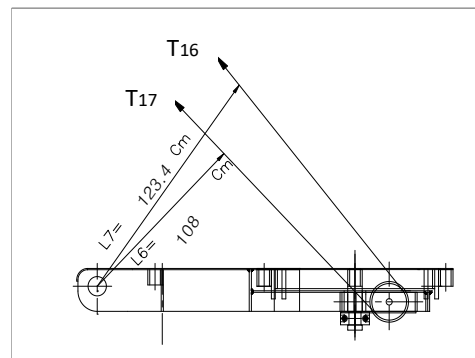
$$\sum (W \times L^0) = (W_L \times l_1) + (W_D \times l_2) + (W_U \times l_3)$$

$$= 217347.5 \quad \text{Kg Cm}$$

STOWING DAVIT



UPPER PLATFORM



$$T_{16} = T_{17} \times \mu$$

$$T_{12} = T_{17} \times \mu^5$$

$$T_3 = T_{17} \times \mu^{14}$$

$$(T_{17} \times l_6) + (T_{16} \times l_7) + (T_3 \times l_4) + (T_{12} \times l_5) = W L^0$$

$$(T_{17} \times l_6) + (T_{17} \times \mu \times l_7) + (T_{17} \times \mu^{14} \times l_4) + (T_{17} \times \mu^5 \times l_5) = W L^0$$

$$T_{17} \times \{ (l_5 \times \mu^5) + (l_4 \times \mu^{14}) + (l_7 \times \mu) + l_6 \} = W L^0$$

$$T_{15} = \frac{W L^0}{(l_5 \times \mu^5) + (l_4 \times \mu^{14}) + (l_7 \times \mu) + l_6}$$

$$= 326 \quad \text{Kg}$$

$$\therefore Tr3 = T_{15} \times \mu^{16} = 632 \quad \text{Kg}$$

$$l_4 = 144.2$$

$$l_5 = 134$$

$$l_6 = 108$$

$$l_7 = 123.4$$

$$\mu_1 = 1.02 \quad (\text{BEARING})$$

$$\mu_2 = 1.06 \quad (\text{BUSH})$$

$$\mu_1^7 = 1.15 \quad (\text{BEARING})$$

$$\mu_2^5 = 1.34 \quad (\text{BUSH})$$

$$\mu_2^7 = 1.50 \quad (\text{BUSH})$$

$$\mu_2^9 = 1.69 \quad (\text{BUSH})$$



STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

7. WITH REGARD TO TORQUE,HP OF MOTOR

1) MAX. TORQUE ON MOTOR SHAFT WHEN ROPE IS 3TH LAYER

$$T_M = T_{r3} \times \frac{D}{2RY} = 632 \times \frac{0.327}{2 \times 36.3 \times 0.8} = 3.56 \text{ Kg M}$$

2) REQUIRED HP OF MOTOR

$$Hp = \frac{T_M \times RPM}{716.2} = \frac{3.56 \times 600}{716.2} = 2.98 \text{ Hp}$$

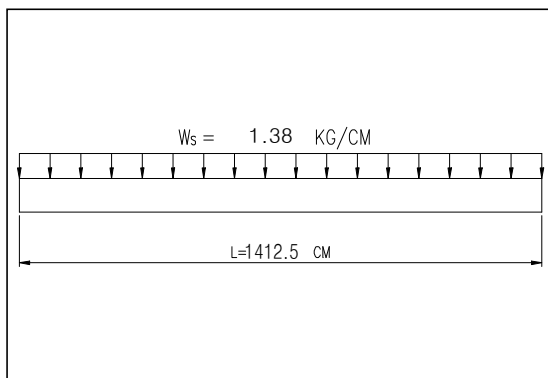
8. BENDING MOMENT FOR LADDER

1) WITH REGARD TO ONE SIDE FRAME OF LADDER(W_s)

$$W_t = 3895 \text{ Kg}$$

$$W_s = \frac{W_t}{2L} = 1.38 \text{ Kg/Cm}$$

2) LADDER CONDITION AT HORIZONTAL (BOTH ENDS SUPPORTING)



3) MAX. BENDING MOMENT (M_t)

$$M_t = \frac{W_s L^2}{8} = 343855 \text{ Kg Cm}$$

4) SECTION MODULUS OF LADDER ONE SIDE FRAME (Z)

$$Z = 298.91 \text{ Cm}^3$$

5) MAX. BENDING STRESS ON LADDER ONE SIDE FRAME

$$\sigma_o = \frac{M_t}{Z} = 1150 \text{ Kg/Cm}^2$$

6) 0.2 % PROOF STRESS OF A6061-T6 :

$$\sigma_a = 2500 \text{ Kg/Cm}^2$$

7) SAFETY FACTOR (S)

$$S = \frac{\sigma_a}{\sigma_o} = 2.2 > 2 \text{ ----- O.K}$$



STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

9. WITH REGARD TO MAX. DEFLECTION OF LADDER (δ)

$$W_s = \frac{WP}{2L} = 1.06$$

$$\delta = \frac{5 \times W_s \times L^4}{384 \times E \times I} = \frac{5 \times 1.06 \times 1412.5^4}{384 \times 7.0 \times 10^5 \times 4484}$$

E : YOUNG MODULUS CAL-ALLOY
= $7.0 \times 10^5 \text{ Kg/Cm}^2$
I = 4483.62 Cm^4

$$= 17.5 \text{ cm} < \frac{L}{75} = 18.8 \text{ ---- O.K}$$

10. WITH REGARD TO STRENGTH OF STOWING DAVIT

$$l_{DC} = 124 \text{ Cm}$$

$$P = (P_{U3} + 2W_D)/2 = 1560 + 325$$

$$= 1885 \text{ Kg}$$

1) BENDING MOMENT OF "X-X" SECTION

$$M = P \times l_{DC} = 233780 \text{ Kg Cm}$$

2) SECTION MODULUS OF FRAME OF "X-X" SECTION

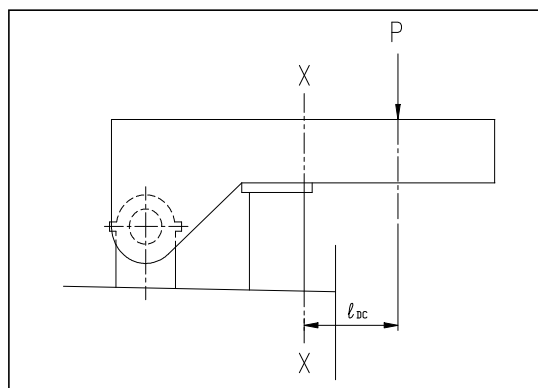
$$Z = 467.2 \text{ Cm}^3$$

3) BENDING STRESS (σ_0)

$$\sigma_0 = \frac{M}{Z} = 500 \text{ Kg/Cm}^2$$

4) BREAKING STRESS OF SS400 (σ_a) : 4100 Kg/Cm^2

$$5) \text{ SAFETY FACTOR(S) } S = \frac{\sigma_a}{\sigma_0} = 8.2 > 5 \text{ ---- O.K}$$



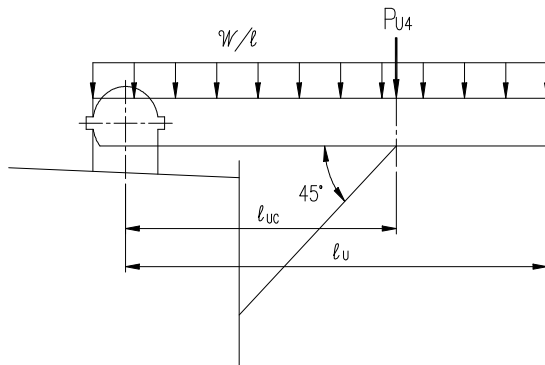


STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

11. WITH REGARD TO STRENGTH OF UPPER PLATFORM SUPPORTING ROD



$$\begin{aligned} l_{UC} &= 165 \text{ Cm} \\ l_U &= 195 \text{ Cm} \\ P &= P_{U4} + Wg = 1706 \text{ Kg} \\ P_{U4} &= 774 \text{ Kg} \\ Wg &= W + W_U = 931 \text{ Kg} \\ W &= 408 \text{ Kg/m}^2 ; 1.4245 \text{ M}^2 \\ &= 581.2 \text{ Kg} \\ W_U &= 350 \text{ Kg} \end{aligned}$$

1) COMPRESS LOAD ON ONE SIDE UPPER PLATFORM SUPPORTING ROD

$$\begin{aligned} P_0 &= \left(P_{U4} + \frac{Wg \times l_U}{2 \times l_{UC}} \times \frac{1}{2} \times \frac{1}{\cos 30^\circ \times \cos 13.3^\circ} \right) \\ &= \left(774 + \frac{931.196 \times 195}{2 \times 165} \right) \times \frac{1}{2 \times 0.97} \\ &= 683 \text{ Kg} \end{aligned}$$

2) SECTION AREA OF $\phi 32$ ROUND BAR

$$A = \frac{\pi \times D^2}{4} = 8.04 \text{ Cm}^2$$

3) COMPRESS STRESS

$$\sigma_p = \frac{P_0}{A} = 84.90 \text{ Kg/Cm}^2$$

4) BREAKING STRESS OF SS400 = 4100 Kg/Cm (σ_a)

5) SAFETY FACTOR (S)

$$S = \frac{\sigma_a}{\sigma_p} = 48.3 > 5 \text{ ----- O.K}$$

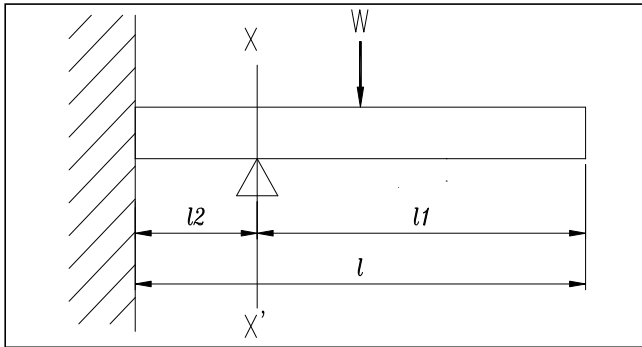


STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

12. WITH REGARD TO STRENGTH OF LOWER PLATFORM



$$\left[\begin{array}{lcl} l & = & 83 \text{ Cm} \\ l_1 & = & 68 \text{ Cm} \\ W & = & W_W + W_L = 216.6 \text{ Kg} \\ W_W & = & 408 \text{ Kg/m}^2 ; 0.4648 \text{ m}^2 \\ & = & 189.6 \text{ Kg} \\ W_L & = & 27 \text{ Kg} \\ W_S & = & \frac{W_W}{l} = 2.6 \text{ Kg} \end{array} \right.$$

1) MAX. BENDING MOMENT (M)

$$M_{\max.} = \frac{W_S \times l_1^2}{2} = \frac{2.6 \times 68^2}{2} = 6035 \text{ Kg} \cdot \text{Cm}$$

2) SECTION MODULUS OF UPPER PLATFORM FRAME (Z)

$$Z = 49.6 \text{ Cm}^3$$

3) MAX. BENDING STRESS ON LOWER PLATFORM (σ)

$$\sigma = \frac{M}{Z} = 121.7 \text{ Kg / Cm}^2$$

4) 0.2% PROOF STRESS OF A6063-T5 (σ_o)

$$\sigma_o = 1100 \text{ Kg / Cm}^2$$

5) SAFETY FACTOR (S)

$$S = \frac{\sigma_o}{\sigma} = \frac{1100}{121.7} = 9.0 > 2 \text{ ----- O.K.}$$



STRENGTH CALCULATION

Date : 2021-11-10

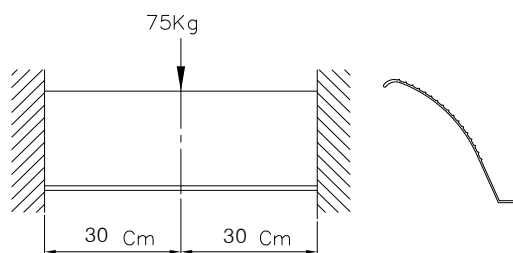
Hull.No : 4193/94

13. WITH REGARD TO STRENGTH OF LADDER STEP

1) TEST LOAD

$$Wt = 75 \text{ Kg/step}$$

2) STEP CONDITION AT HORIZONTAL



$$\left\{ \begin{array}{l} \text{AREA OF STEP} : 8.47 \text{ Cm}^2 \\ \text{Inertia moment of step} : 13.05 \text{ Cm}^4 \end{array} \right.$$

3) MAX. BENDING MOMENT (M)

$$M_{\max.} = \frac{Wt \times L}{8} = \frac{75.0 \times 60}{8} = 562.5 \text{ Kg} \cdot \text{Cm}$$

4) SECTION MODULUS OF ONE STEP (Z)

$$Z = 1.67 \text{ Cm}^3$$

5) MAX. BENDING STRESS ON LOWER PLATFORM (σ)

$$\sigma = \frac{M}{Z} = 336.8 \text{ Kg / Cm}^2$$

6) 0.2 % PROOF STRESS OF A6063-T5 :

$$\sigma_a = 1100 \text{ Kg/Cm}^2$$

7) SAFETY FACTOR (S)

$$S = \frac{\sigma_a}{\sigma_0} = 3.3 > 2 \text{ ----- O.K}$$

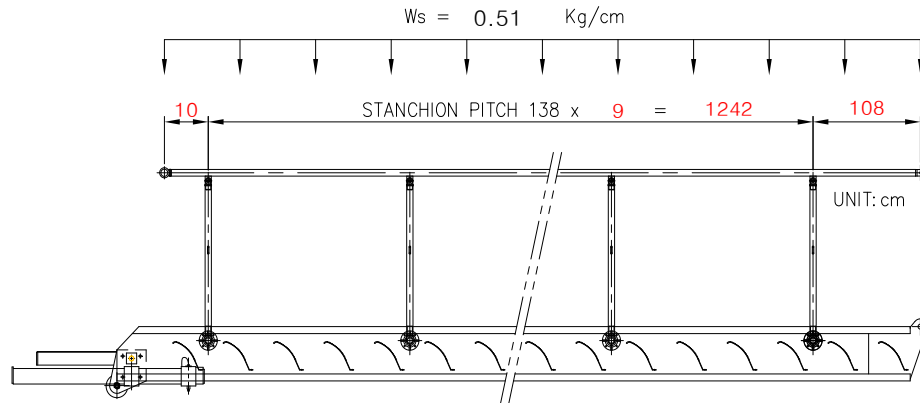


STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

14. WITH REGARD TO STRENGTH OF LADDER HANDRAIL



☛ END PART OF HANDRAIL

1) DISTRIBUTED LOAD ON HANDRAIL (W_s) : $W_s = 0.51 \text{ Kg / Cm}$

2) BENDING MOMENT (M)

$$M = \frac{W_s \times \ell^2}{2} = \frac{0.51 \times 108^2}{2} = 2974.3 \text{ Kg} \cdot \text{Cm}$$

3) SECTION MODULUS OF SECTION (Z)

$$Z = \frac{\pi \times (D^4 - d^4)}{32 \times D} = \frac{\pi \times (4.3^4 - 3.6^4)}{32 \times 4.3} = 3.97 \text{ Cm}^3$$

4) BENDING STRESS (σ)

$$\sigma = \frac{M}{Z} = 749.4 \text{ Kg / Cm}^2$$

5) BREAKING STRESS OF A6061-T6 (σ_o) : $\sigma_o = 2500 \text{ Kg / Cm}^2$

6) SAFETY FACTOR (S)

$$S = \frac{\sigma_o}{\sigma} = \frac{2500}{749.4} = 3.3 > 2 \text{ ----- O.K.}$$

☛ MID PART OF HANDRAIL

1) BENDING MOMENT (M)

$$M = \frac{W_s \times \ell^2}{8} = \frac{0.51 \times 138^2}{8} = 1214.1 \text{ Kg} \cdot \text{Cm}$$

2) BENDING STRESS (σ)

$$\sigma = \frac{M}{Z} = 306 \text{ Kg / Cm}^2$$

3) SAFETY FACTOR (S) $S = \frac{\sigma_a}{\sigma_o} = 8.2 > 2 \text{ ----- O.K.}$

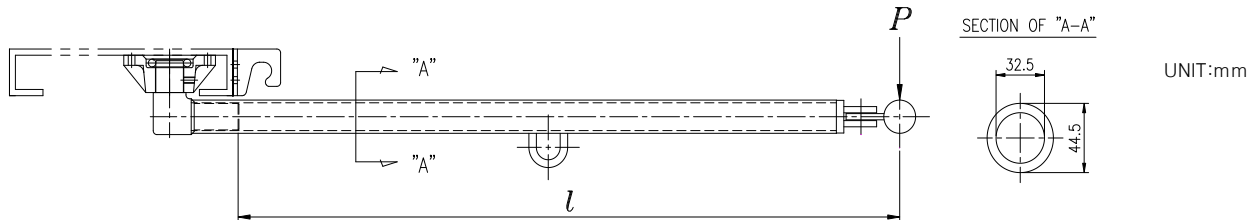


STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

15. WITH REGARD TO STRENGTH OF LADDER STANCHION



$$\left\{ \begin{array}{l} \text{LENGTH OF HANDRAIL (L)} : 13.60 \text{ m} \\ \text{QUANTITY OF STANCHION (N)} : 10 \text{ EA} \end{array} \right.$$

1) TOTAL LOAD ON HANDRAIL (W)

$$W = W_s \times L = 51 \text{ Kg/m}^2 \times 13.60 \text{ m} = 694 \text{ Kg}$$

2) LOAD ON ONE STANCHION (P)

$$P = \frac{W}{N} = \frac{694}{10} = 69.36 \text{ Kg}$$

3) SECTION MODULUS OF SECTION "A-A" (Z)

$$Z = \frac{\pi(D^4 - d^4)}{32 D} = \frac{\pi(4.45^4 - 3.25^4)}{32 \times 4.45} = 6.19 \text{ Cm}^3$$

4) BENDING MOMENT (M)

$$M = P \times l = 69.36 \times 101.7 = 7053.9 \text{ Kg}\cdot\text{cm}$$

5) BENDING STRESS (σ)

$$\sigma = \frac{M}{Z} = 1140 \text{ Kg / Cm}^2$$

6) BREAKING STRESS OF A6061-T6 (σ_o) : $\sigma_o = 2500 \text{ Kg / Cm}^2$

7) SAFETY FACTOR (S)

$$S = \frac{\sigma_o}{\sigma} = \frac{2500}{1140.2} = 2.2 > 2 \text{ ----- O.K.}$$



STRENGTH CALCULATION

Date : 2021-11-10

Hull.No : 4193/94

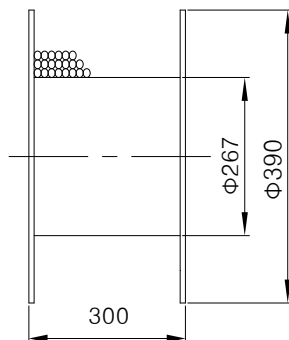
16. WITH REGARD TO MOTOR CAPACITY, TORQUE & ROPE SPEED ON EACH LAYER

1) TORQUE AT EACH LAYER

$$\begin{aligned}
 &T_{r3} = 632 \text{ Kg} \\
 &\text{1st LAYER : } T_{r3} \times \frac{0.279}{2 \times 36.3 \times 0.8} = 3.04 \text{ Kg M} \\
 &\text{2nd LAYER : } T_{r3} \times \frac{0.303}{2 \times 36.3 \times 0.8} = 3.30 \text{ Kg M} \\
 &\text{3rd LAYER : } T_{r3} \times \frac{0.327}{2 \times 36.3 \times 0.8} = 3.56 \text{ Kg M}
 \end{aligned}$$

2) ROPE SPEED AT EACH LAYER

$$\begin{aligned}
 &\text{1st LAYER : } \pi \times 0.279 \times \frac{600}{36.3} = 14.5 \text{ M/MIN} \\
 &\text{2nd LAYER : } \pi \times 0.303 \times \frac{600}{36.3} = 15.7 \text{ M/MIN} \\
 &\text{3rd LAYER : } \pi \times 0.327 \times \frac{600}{36.3} = 17.0 \text{ M/MIN}
 \end{aligned}$$



$$\begin{aligned}
 D1 &= d + (\phi \times 1) = 0.279 & \phi &= 12 \\
 D2 &= d + (\phi \times 3) = 0.303 & d &= 267 \\
 D3 &= d + (\phi \times 5) = 0.327 & w &= 300 \\
 n &= W / \phi \\
 &= 25
 \end{aligned}$$

$$\begin{aligned}
 \text{DRUM CAPACITY} &= n\pi(D1+D2+D3) \\
 &= 71 \text{ M}
 \end{aligned}$$